



US012414332B2

(12) **United States Patent**
Liaw

(10) **Patent No.:** **US 12,414,332 B2**
(45) **Date of Patent:** ***Sep. 9, 2025**

(54) **STRUCTURE AND METHOD FOR SEMICONDUCTOR DEVICES**

H10D 62/121 (2025.01); **H10D 62/151** (2025.01); **H10D 64/017** (2025.01);
(Continued)

(71) Applicant: **Parabellum Strategic Opportunities Fund LLC**, Wilmington, DE (US)

(58) **Field of Classification Search**
CPC H10D 30/501–509; H10D 30/6735; H10D 30/6212; H01L 29/775; H01L 29/778; H01L 29/785; H01L 29/7851; H01L 29/7856; H01L 29/42392; H01L 29/7853
See application file for complete search history.

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(73) Assignee: **Parabellum Strategic Opportunities Fund LLC**, Austin, TX (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 138 days.

This patent is subject to a terminal disclaimer.

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Primary Examiner — Joseph C. Nicely
(74) *Attorney, Agent, or Firm* — Almanac IP Advisors LLP

(21) Appl. No.: **18/360,508**

(22) Filed: **Jul. 27, 2023**

(65) **Prior Publication Data**
US 2023/0378301 A1 Nov. 23, 2023

Related U.S. Application Data

(60) Continuation of application No. 17/590,409, filed on Feb. 1, 2022, now Pat. No. 11,728,402, which is a
(Continued)

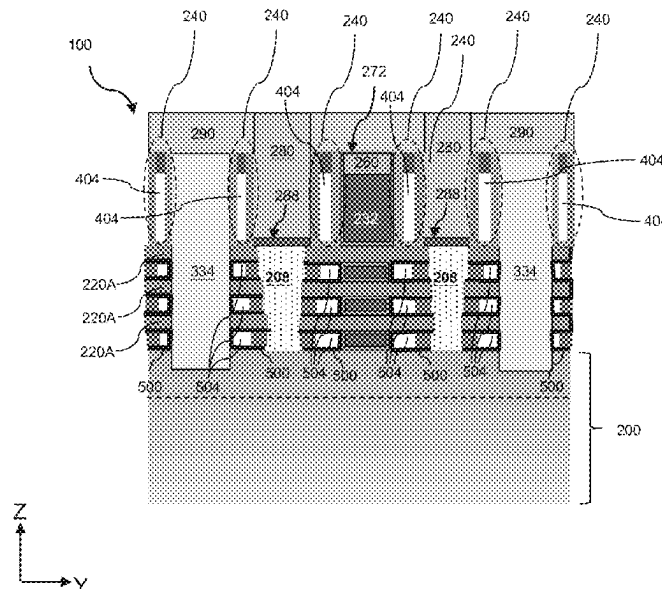
(51) **Int. Cl.**
H10D 30/67 (2025.01)
H10D 30/01 (2025.01)
(Continued)

(52) **U.S. Cl.**
CPC **H10D 30/6735** (2025.01); **H10D 30/0243** (2025.01); **H10D 30/6212** (2025.01); **H10D 30/6757** (2025.01); **H10D 62/116** (2025.01);

(57) **ABSTRACT**

The present disclosure provides an integrated circuit (IC) device, including: a semiconductor substrate having a top surface; a first source/drain feature and a second source/drain feature disposed on the semiconductor substrate; and a plurality of semiconductor layers including a first semiconductor layer and a second semiconductor layer. Each of the first semiconductor layer and the second semiconductor layer extends longitudinally in a first direction and connects the first source/drain feature and the second source/drain feature. The first semiconductor layer is stacked over the second semiconductor layer in a second direction perpendicular to the first direction. A length of the first semiconductor layer along the first direction is less than a length of the second semiconductor layer along the first direction. The IC device further includes a gate structure engaging center portions of the first semiconductor layer and the second semiconductor layer.

20 Claims, 49 Drawing Sheets



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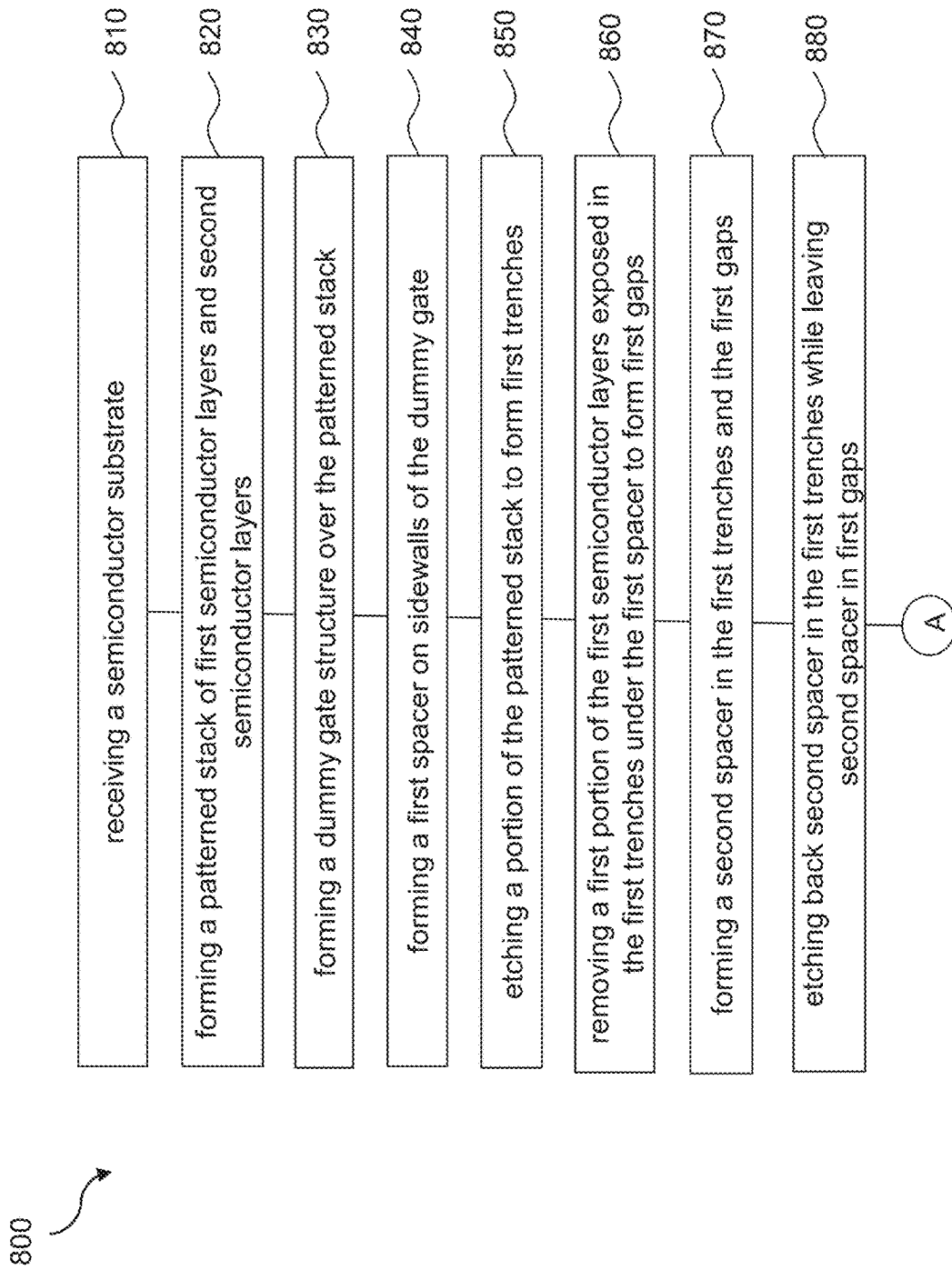


FIG. 1A

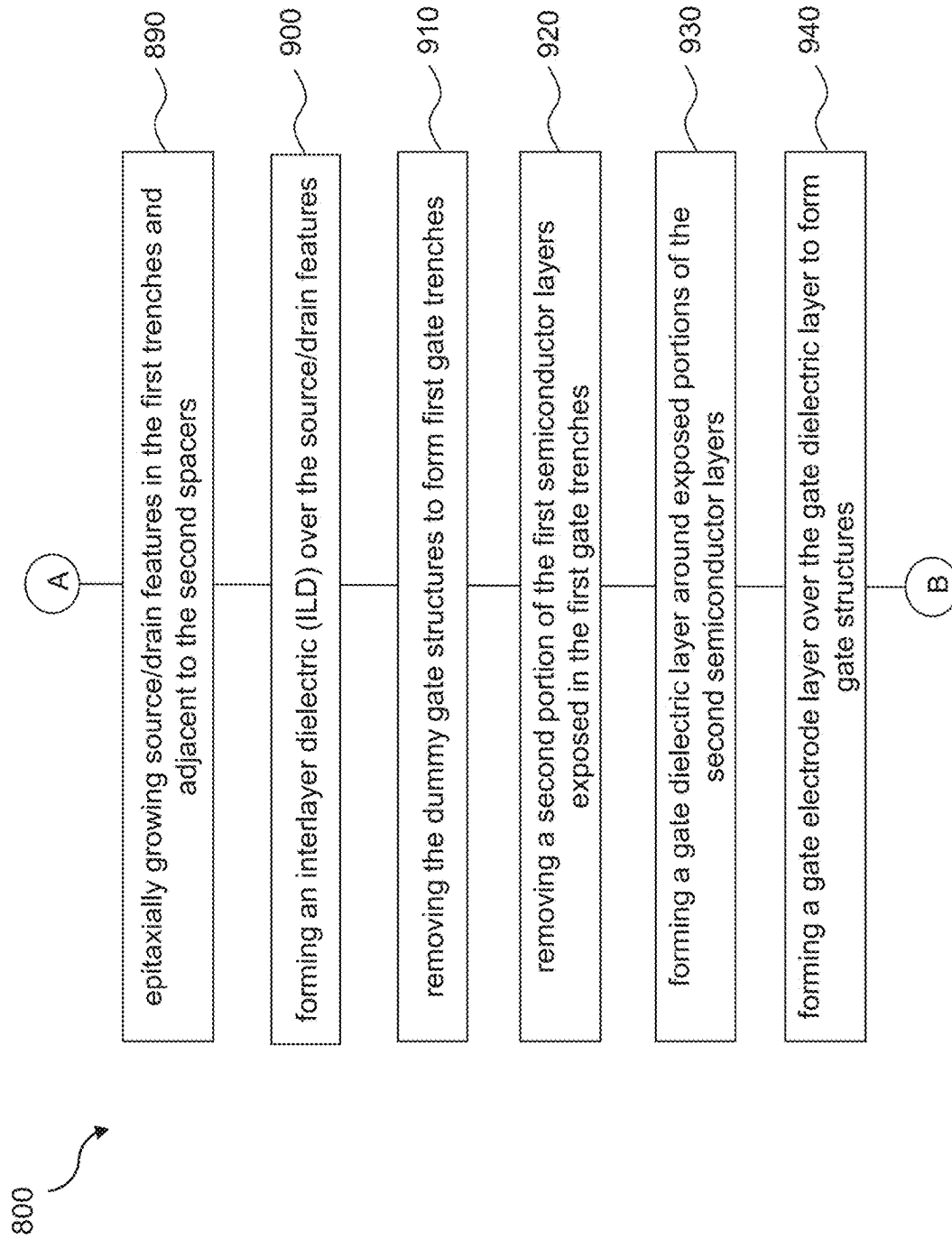


FIG. 1B

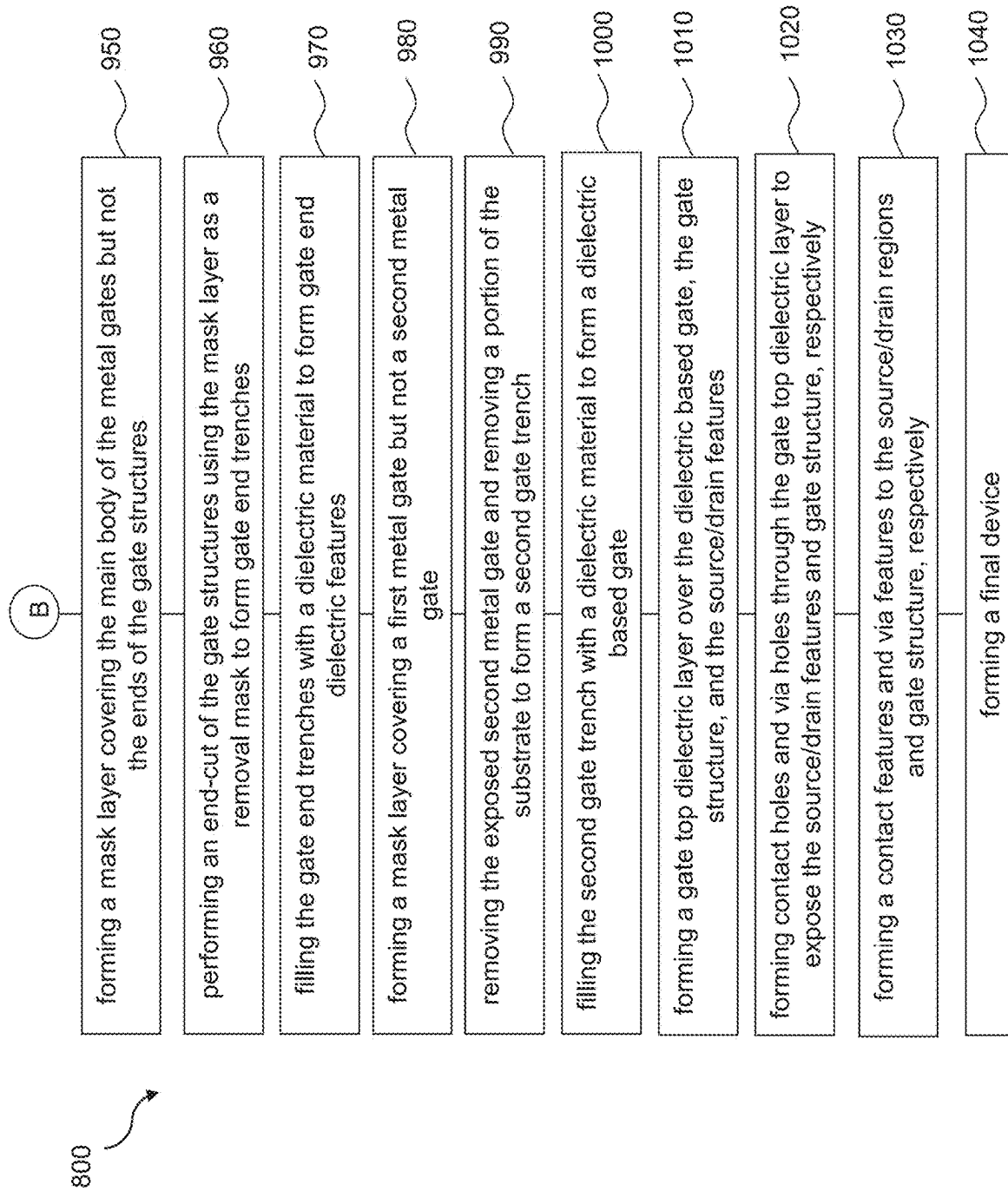


FIG. 1C

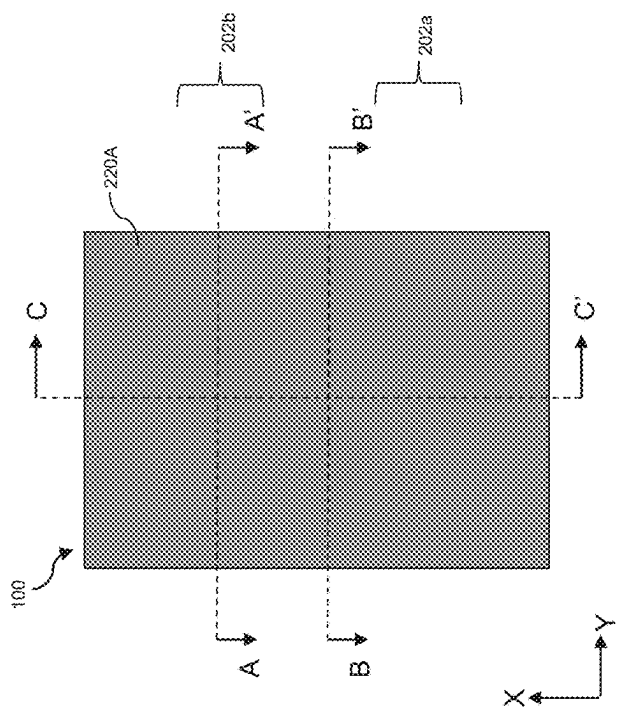


FIG. 2A

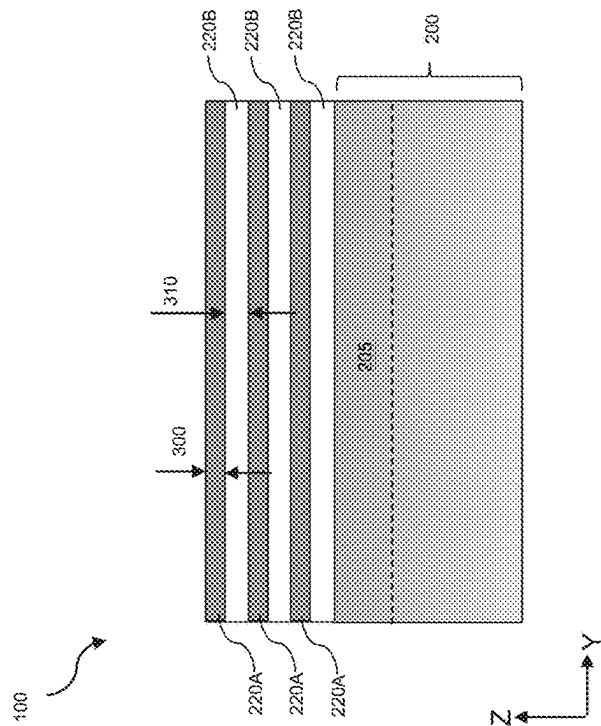


FIG. 2B

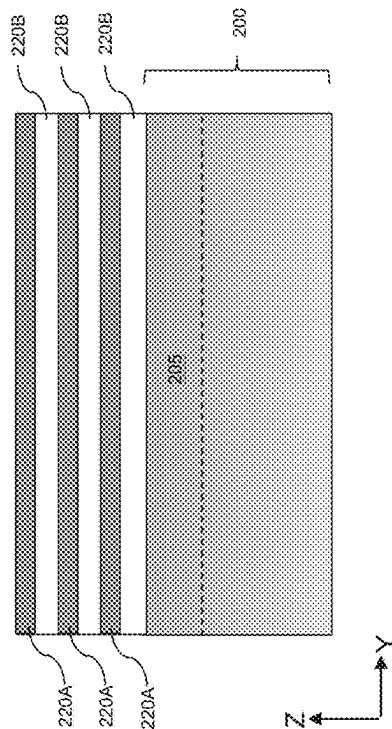


FIG. 2C

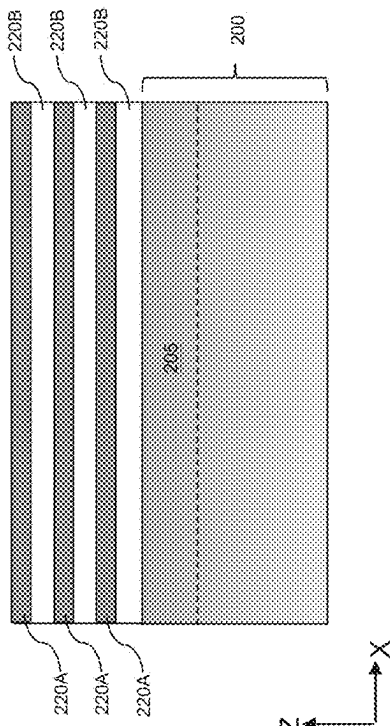


FIG. 2D

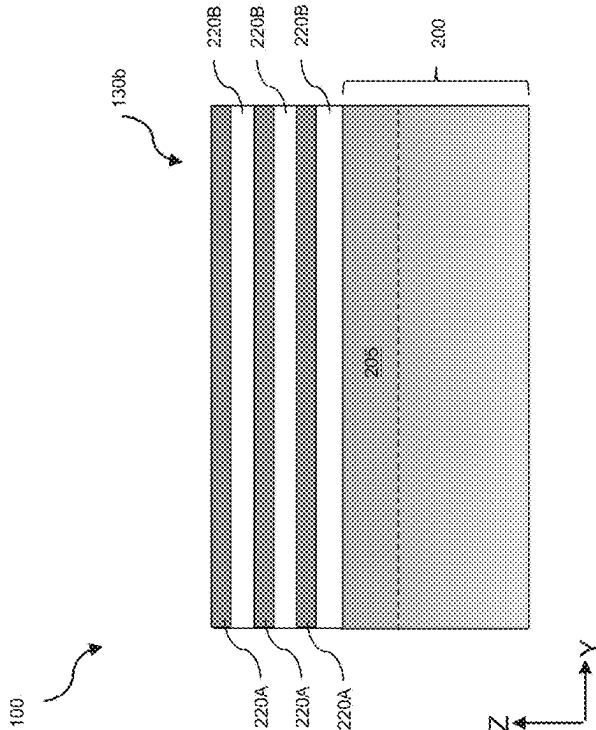


FIG. 3A

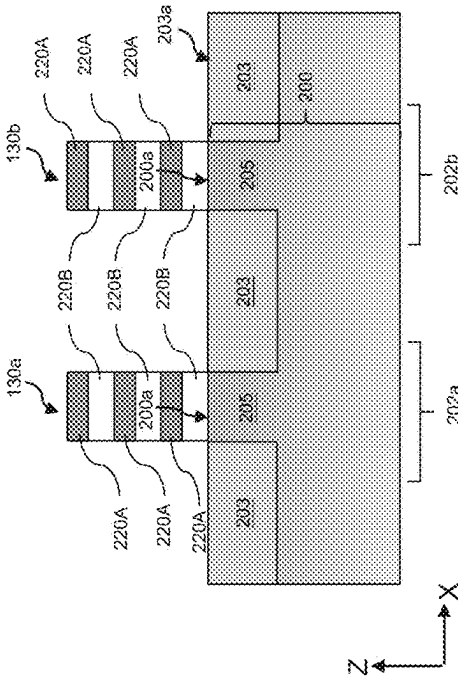


FIG. 3B

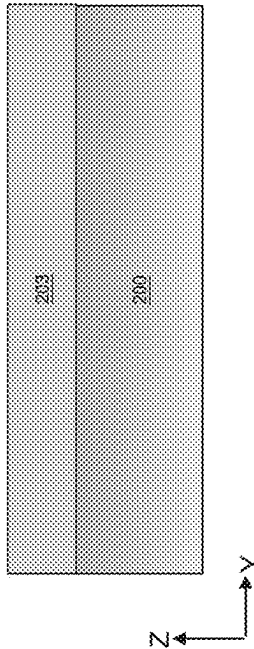


FIG. 3C

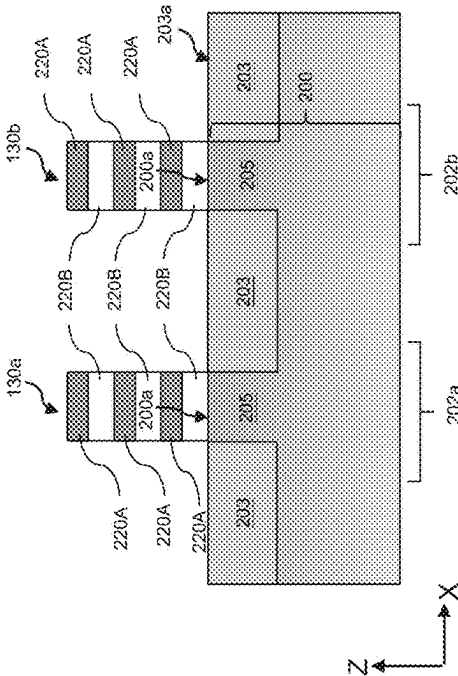


FIG. 3D

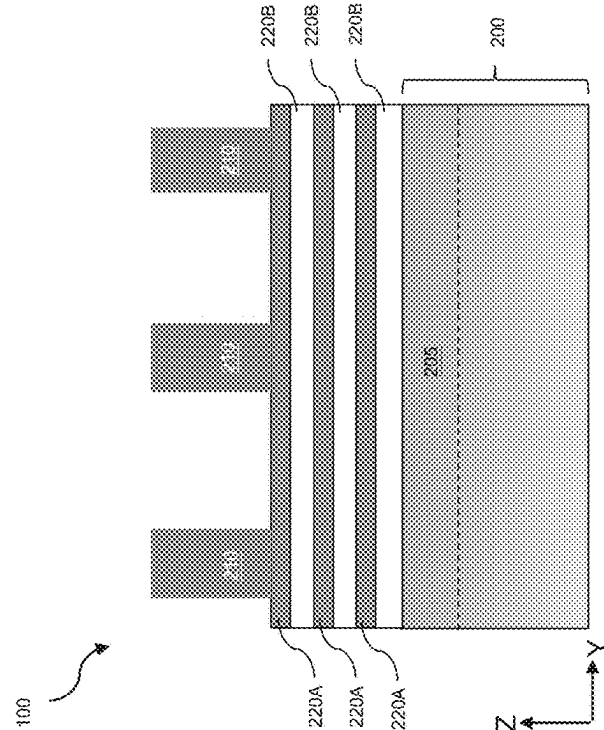


FIG. 4B

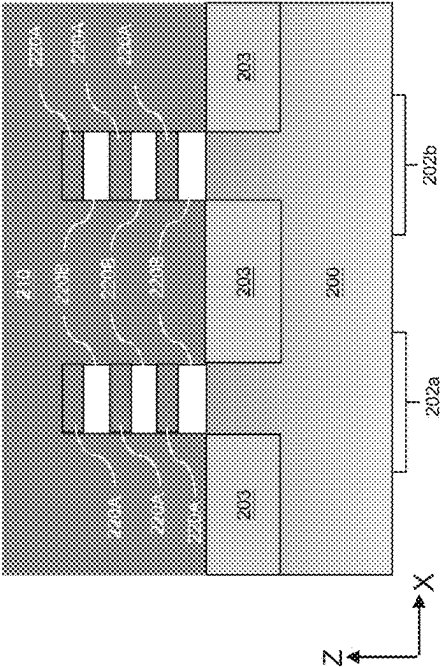


FIG. 4D

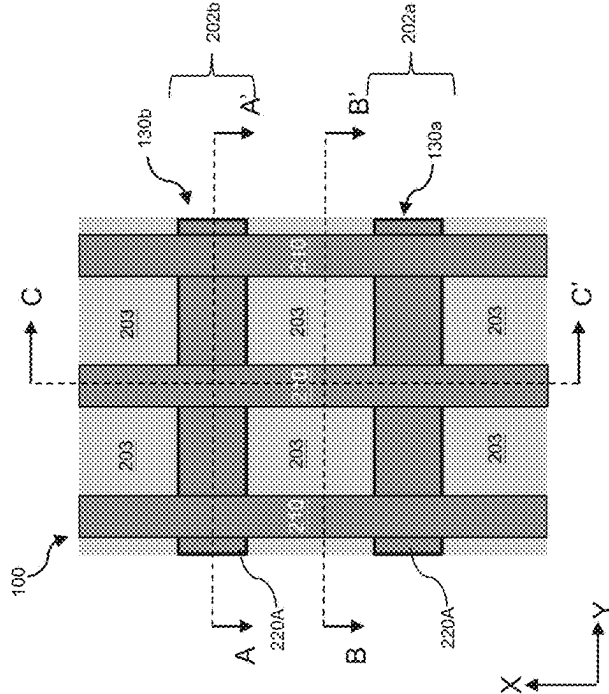


FIG. 4A

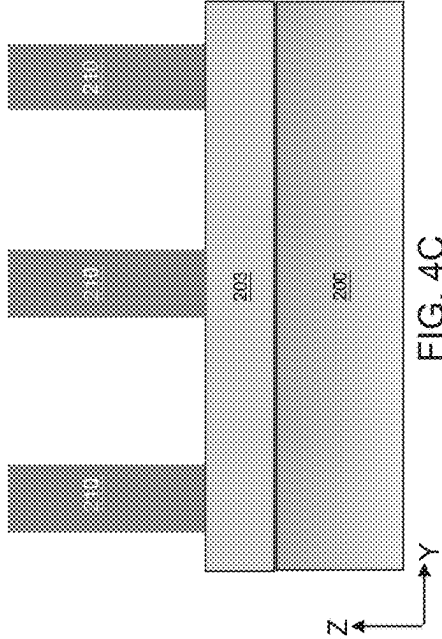
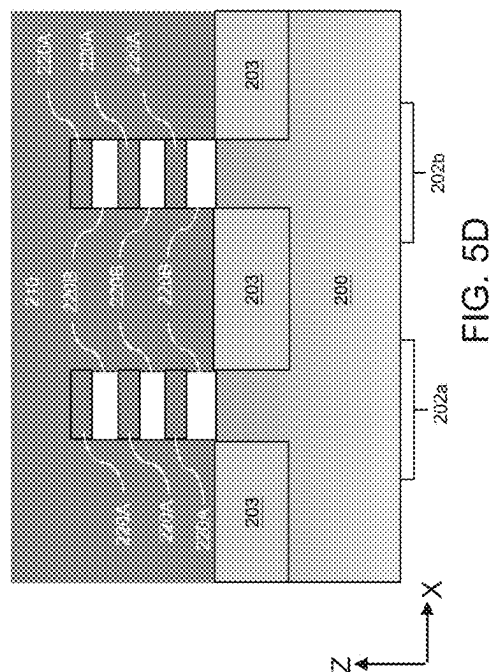
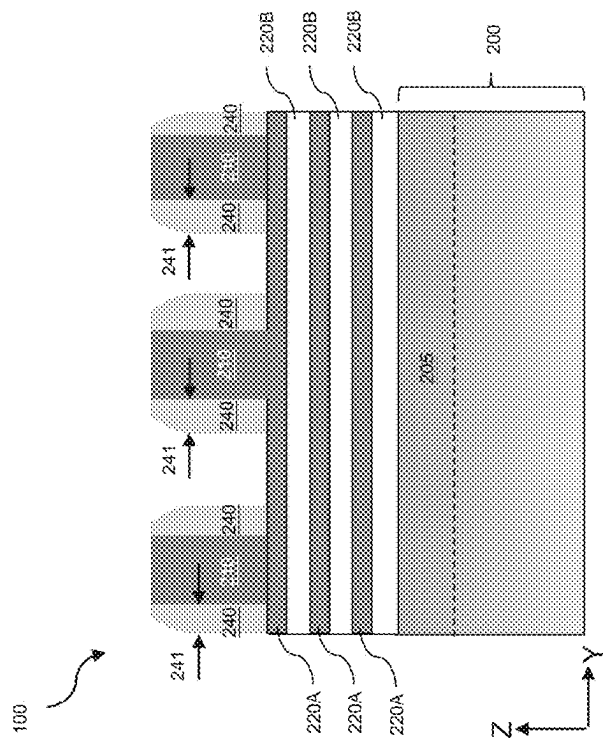
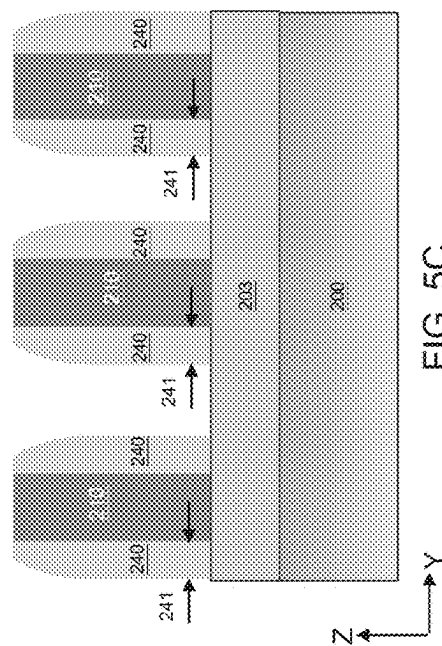
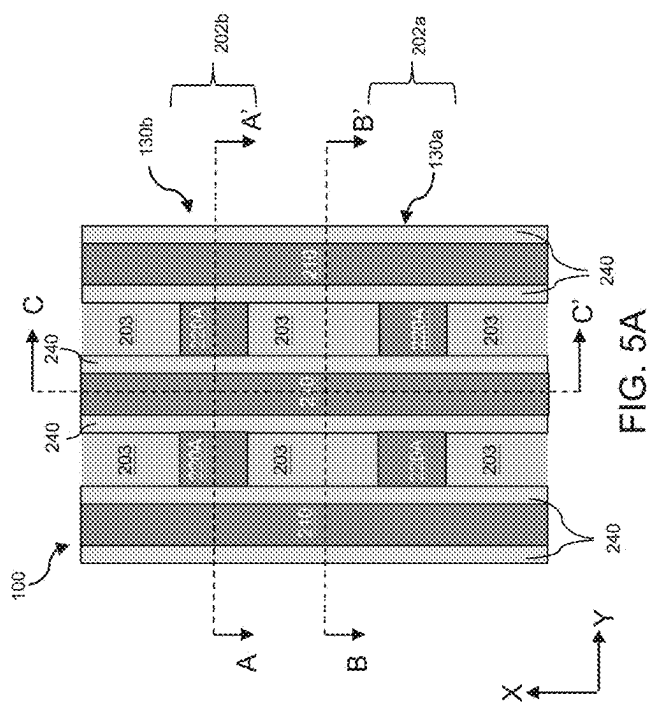


FIG. 4C



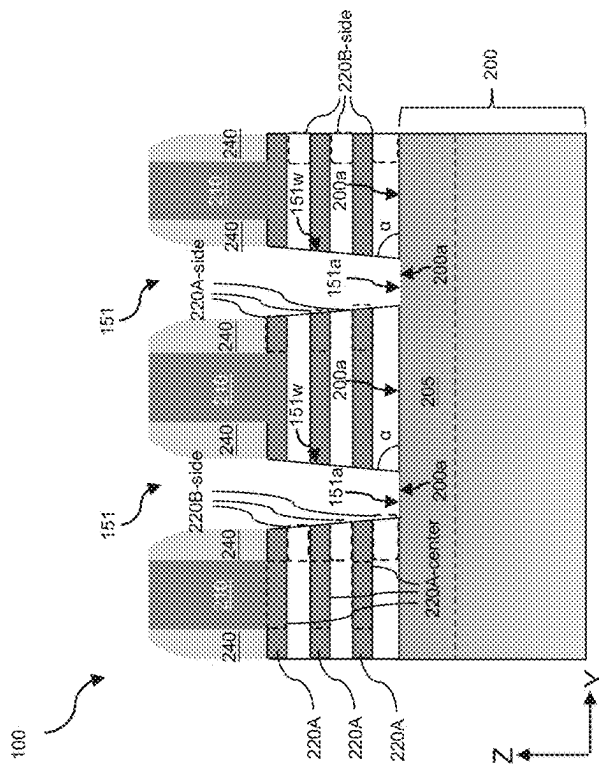


FIG. 6B

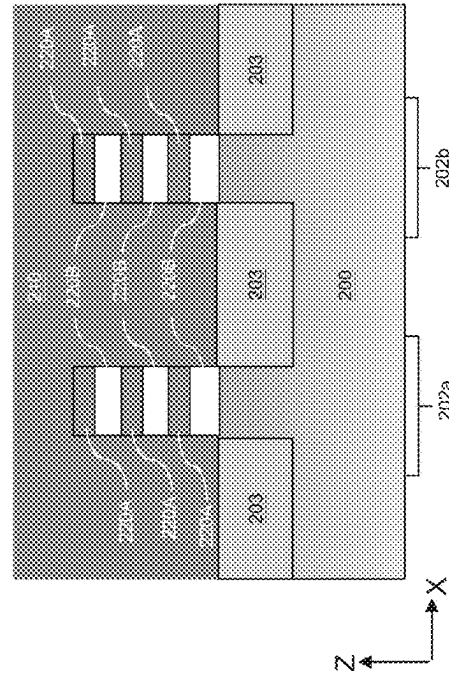


FIG. 6D

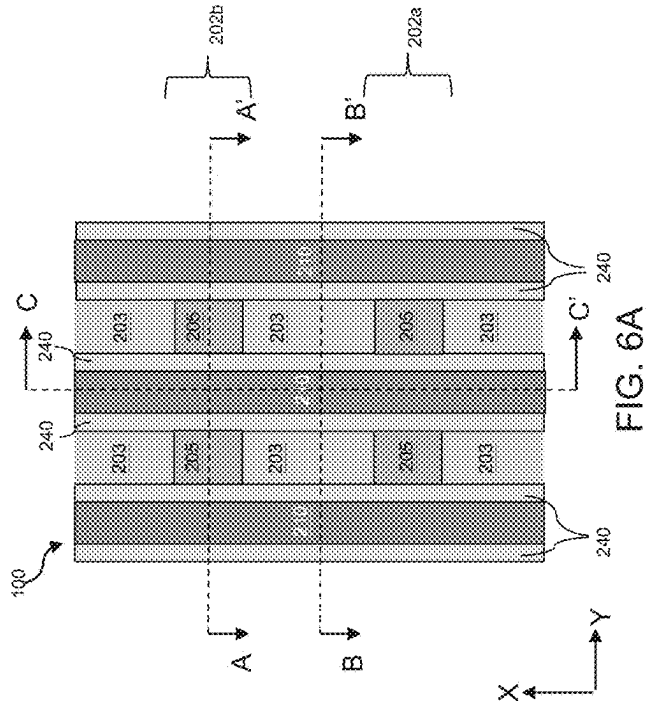


FIG. 6A

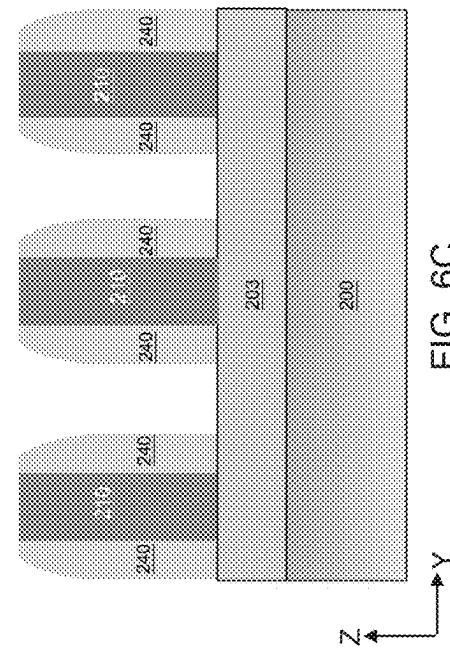
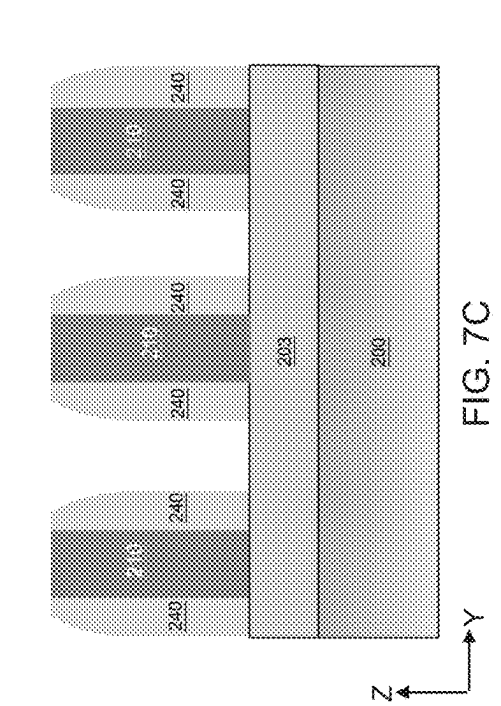
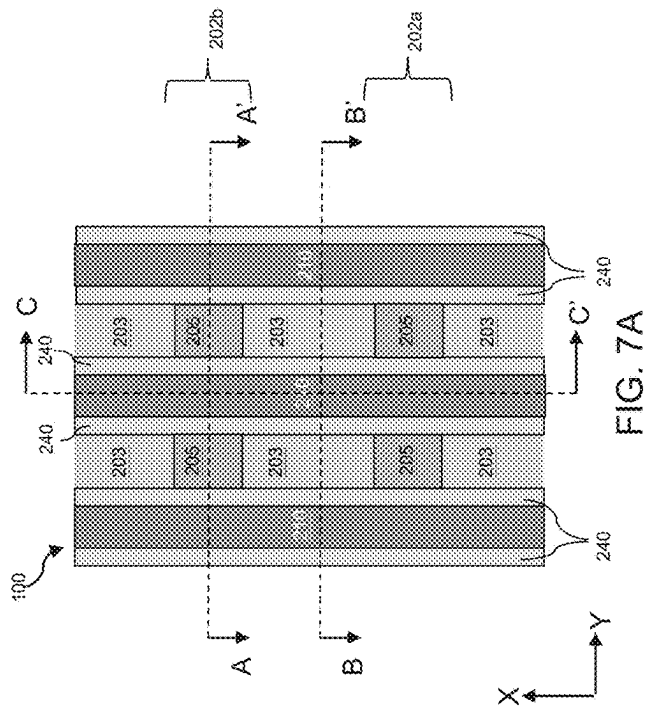
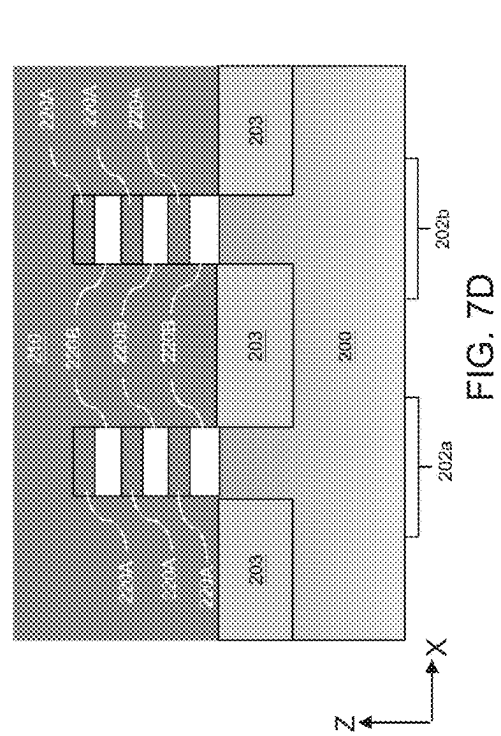
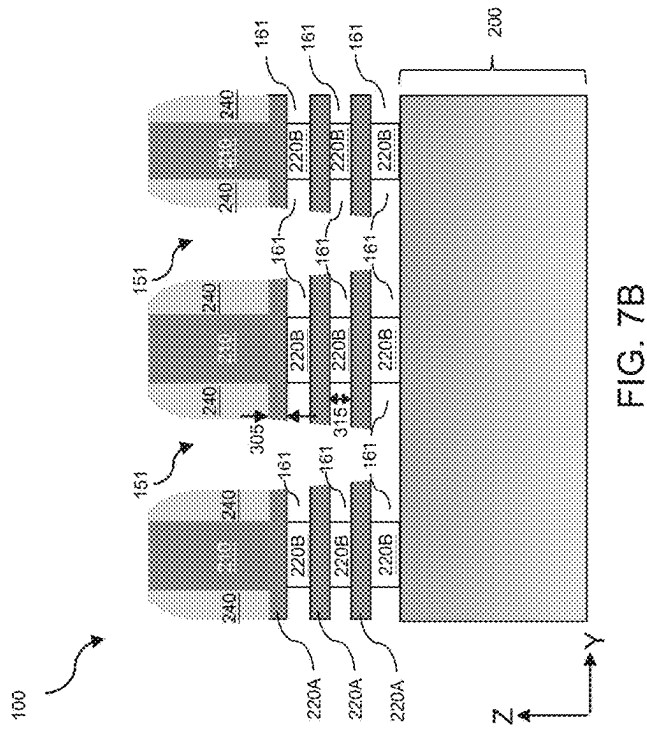


FIG. 6C



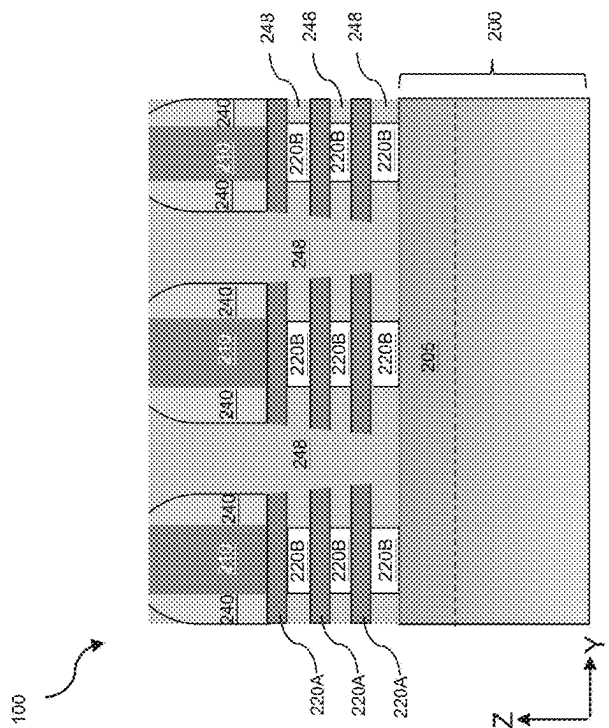


FIG. 8A

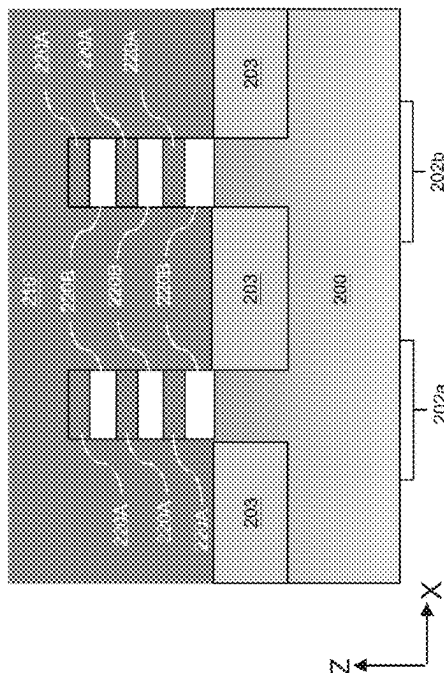


FIG. 8B

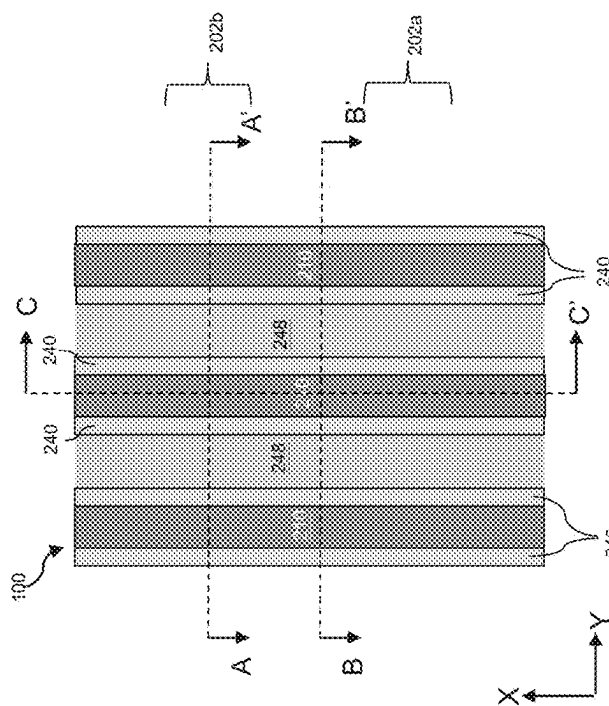


FIG. 8C

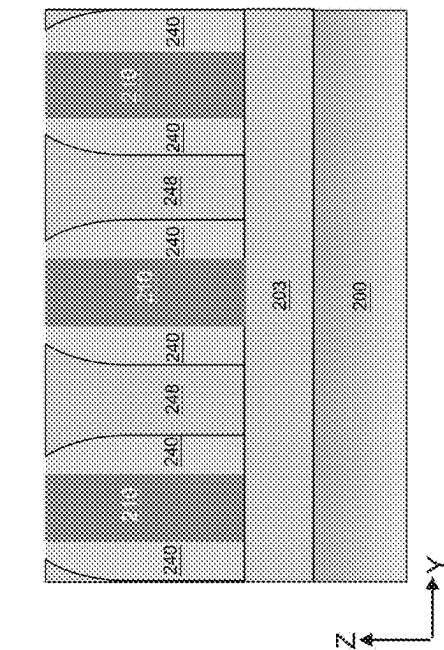


FIG. 8D

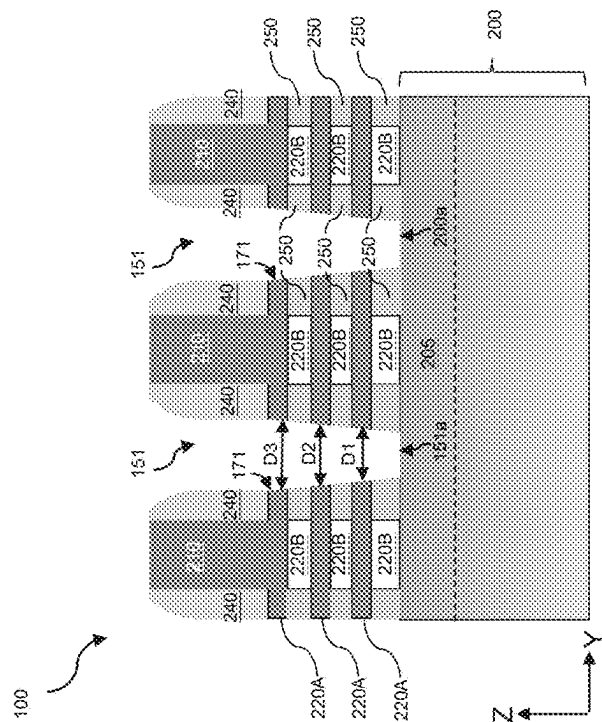


FIG. 9B

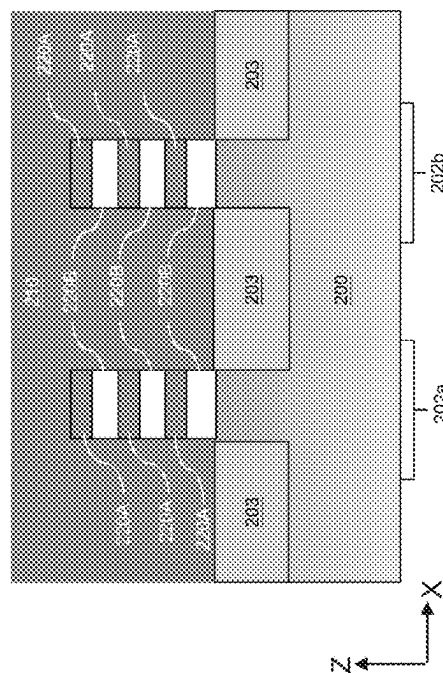


FIG. 9D

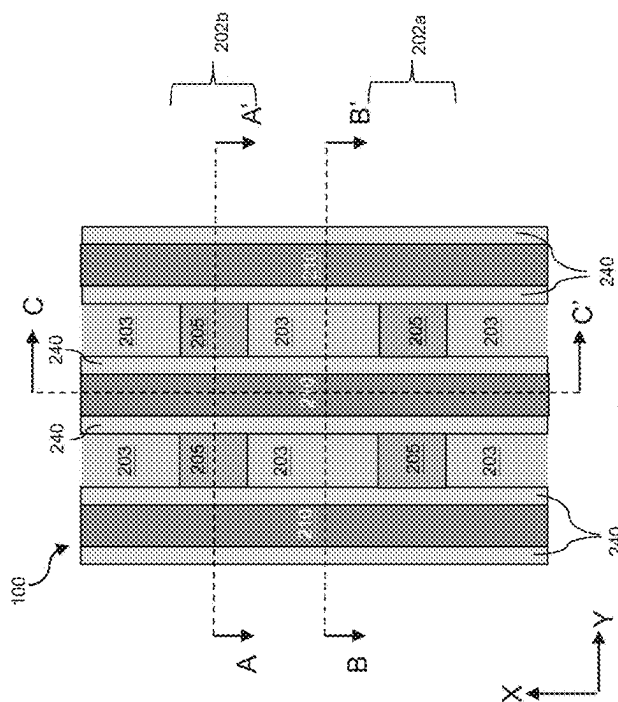


FIG. 9A

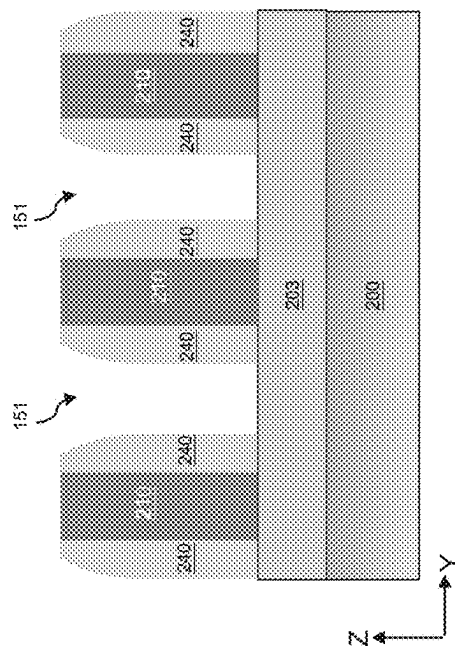


FIG. 9C

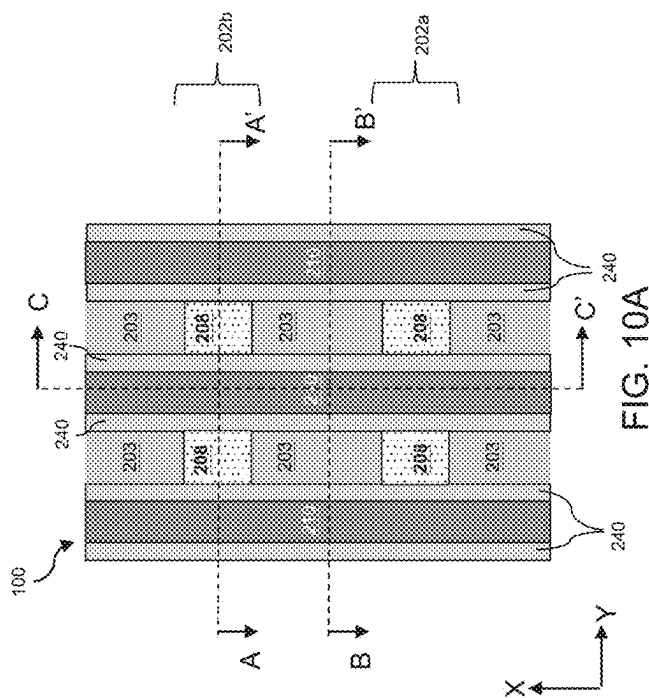


FIG. 10A

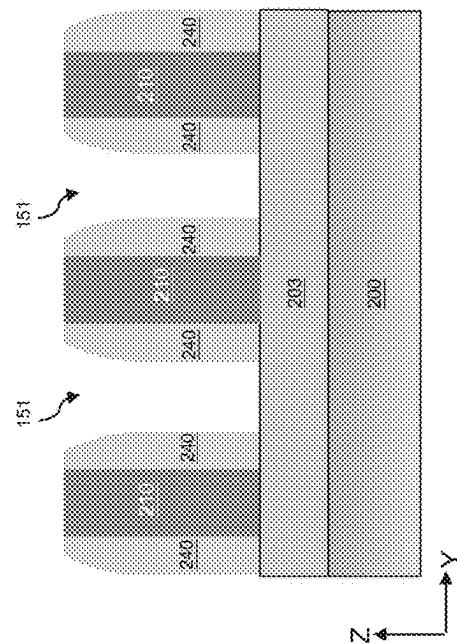


FIG. 10C

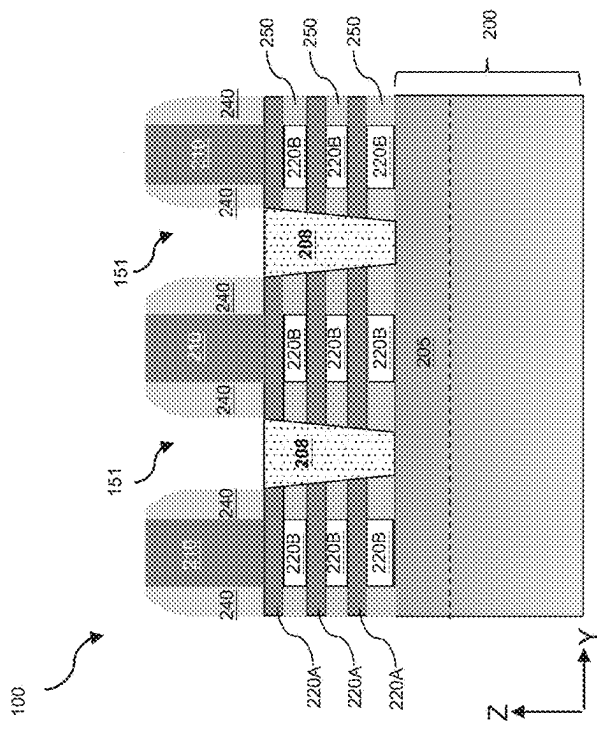


FIG. 10B

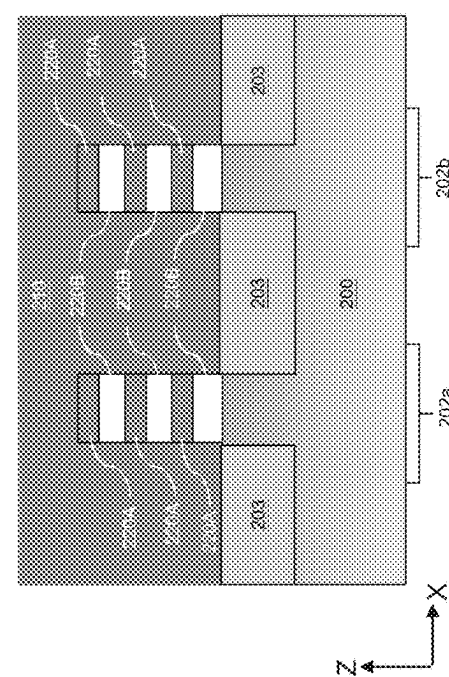
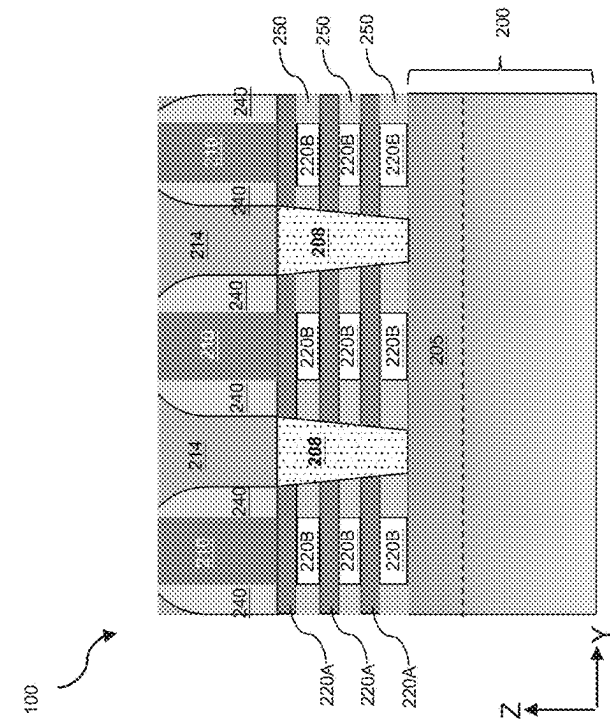
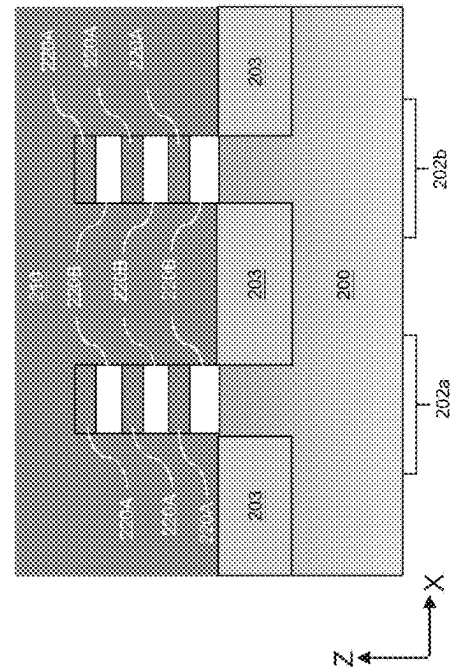


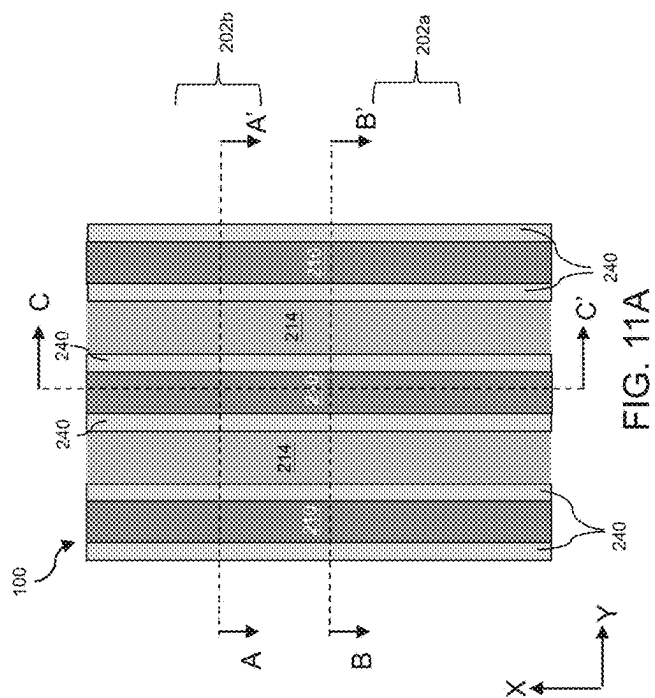
FIG. 10D



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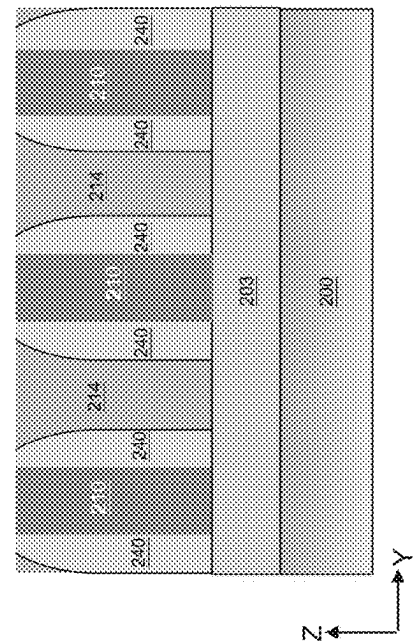


FIG. 11C

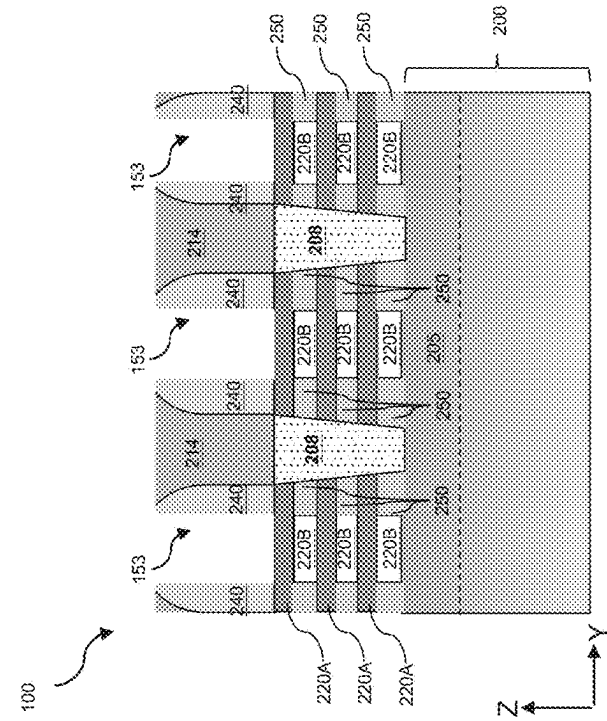


FIG. 12B

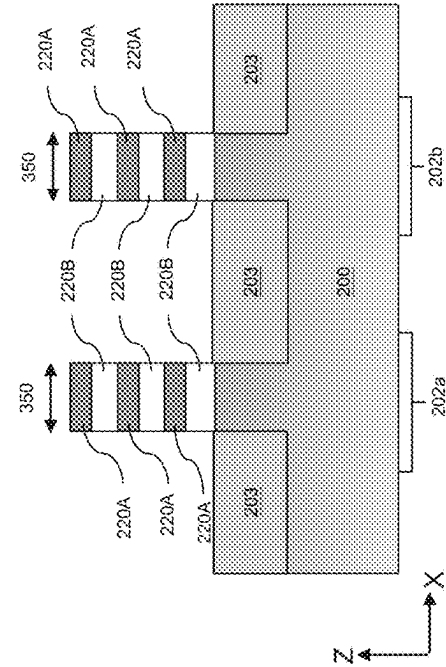


FIG. 12D

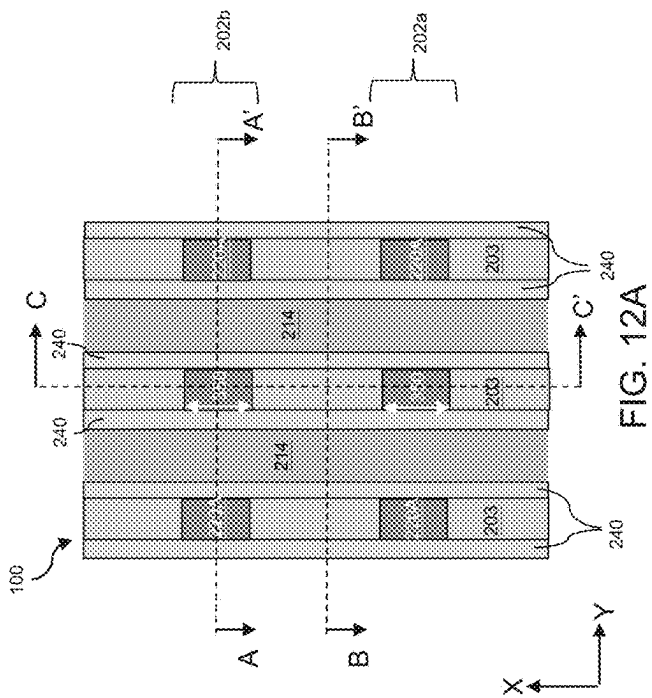


FIG. 12A

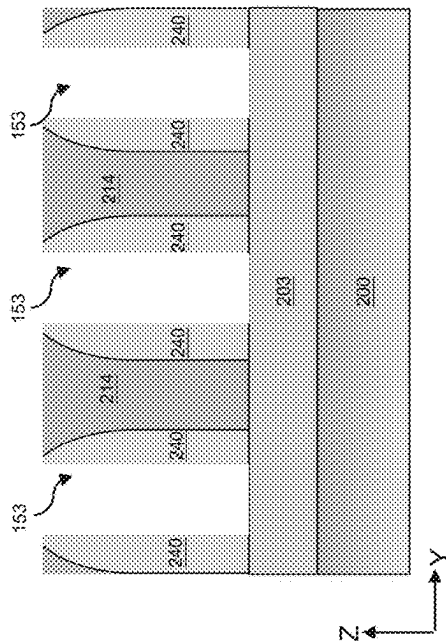


FIG. 12C

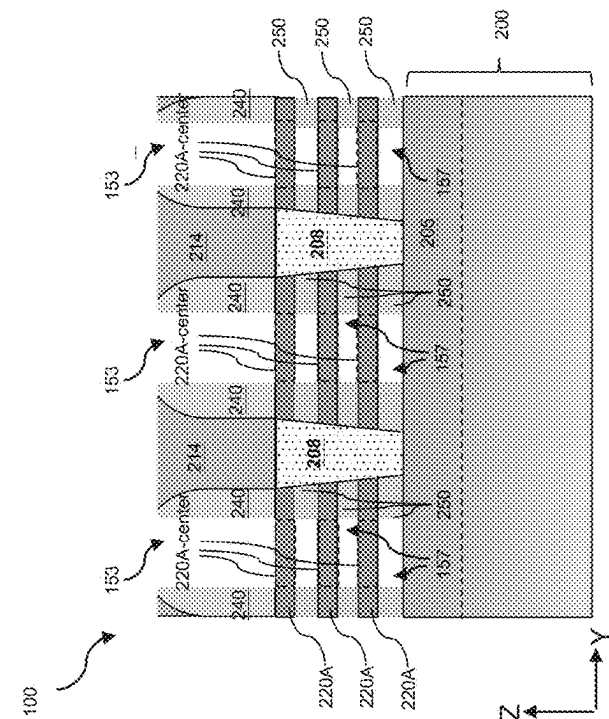


FIG. 13B

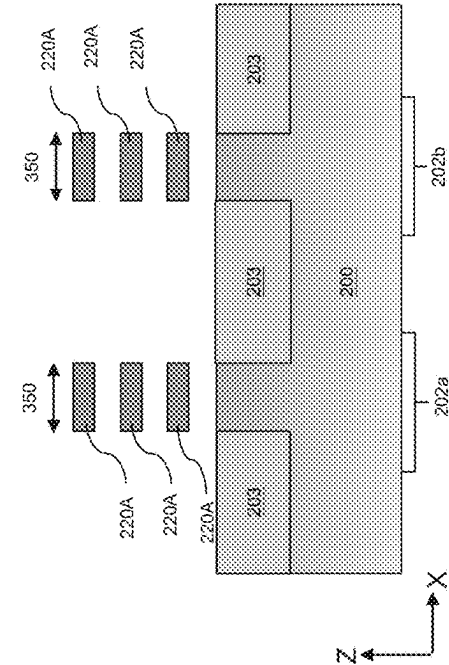


FIG. 13D

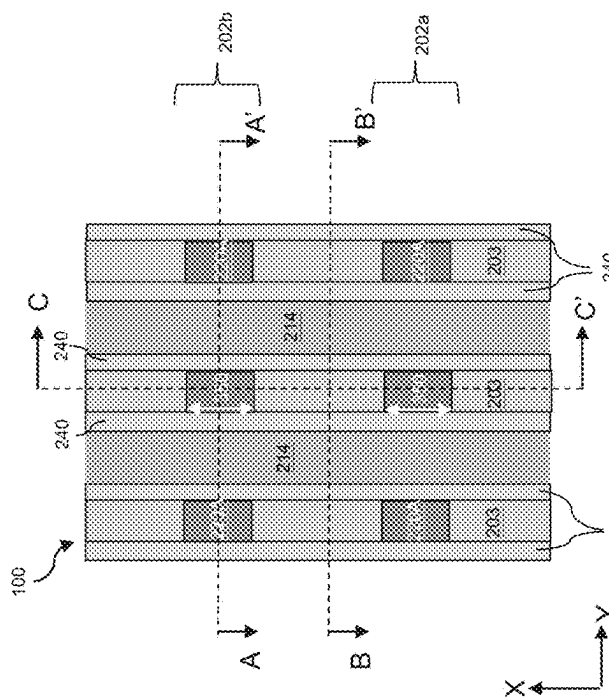


FIG. 13A

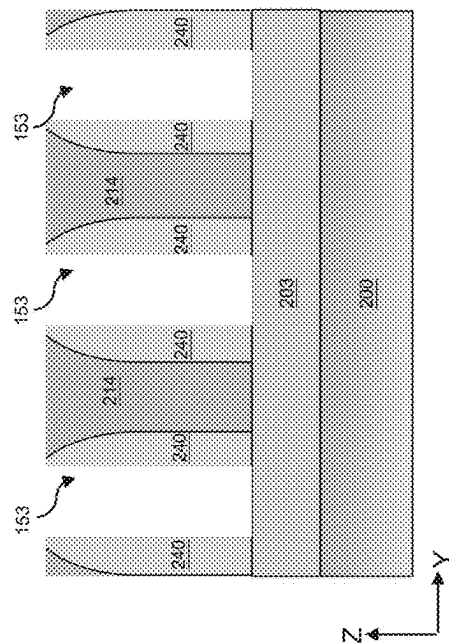


FIG. 13C

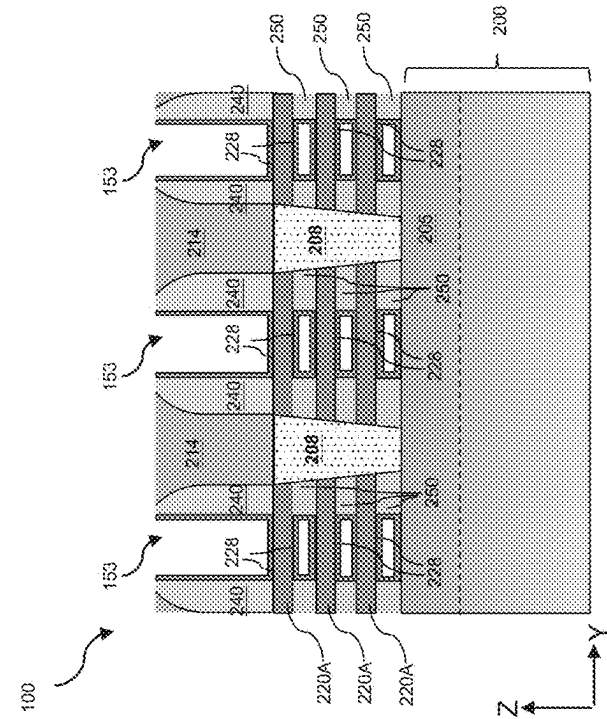


FIG. 14B

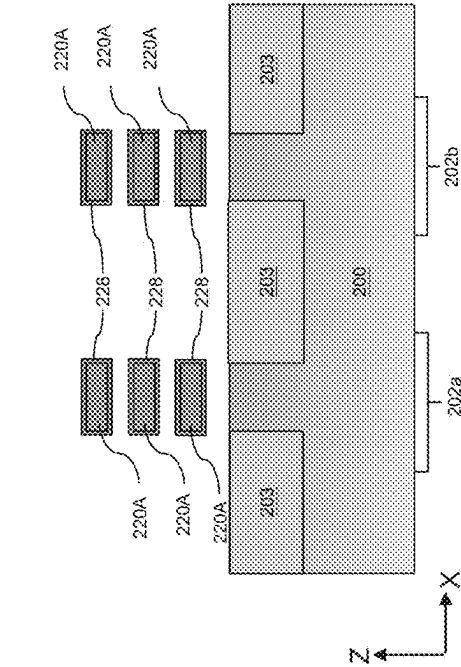


FIG. 14D

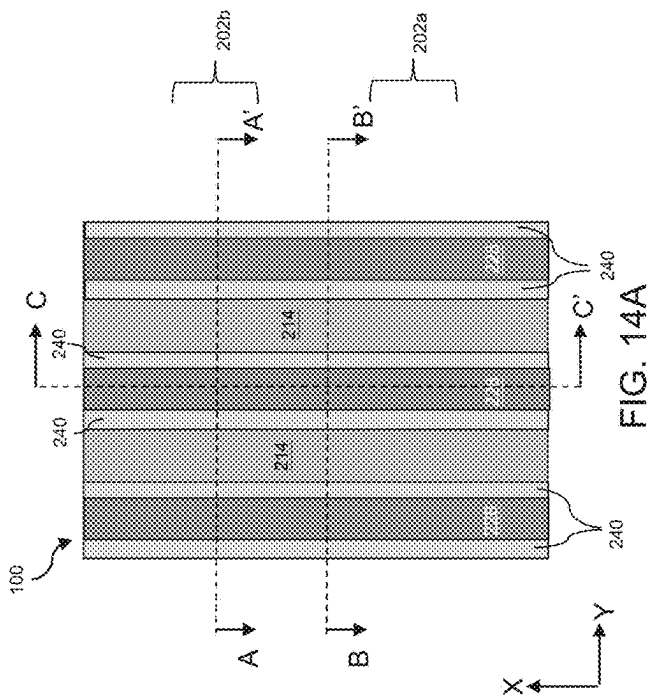


FIG. 14A

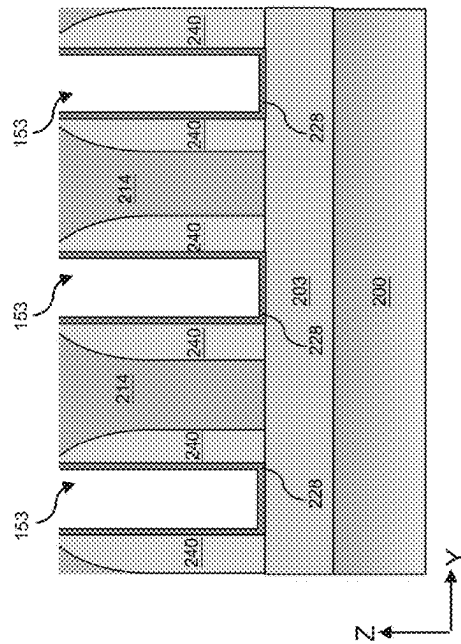


FIG. 14C

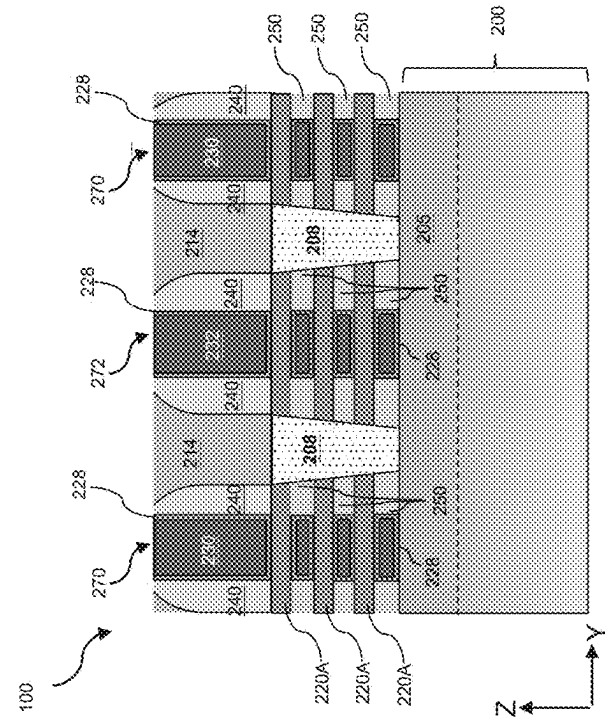


FIG. 15B

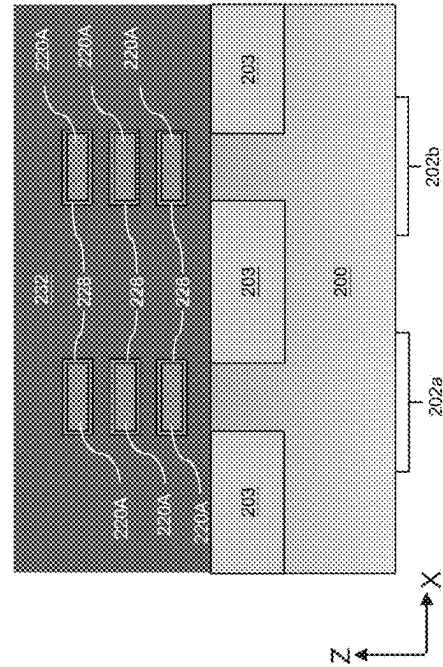


FIG. 15D

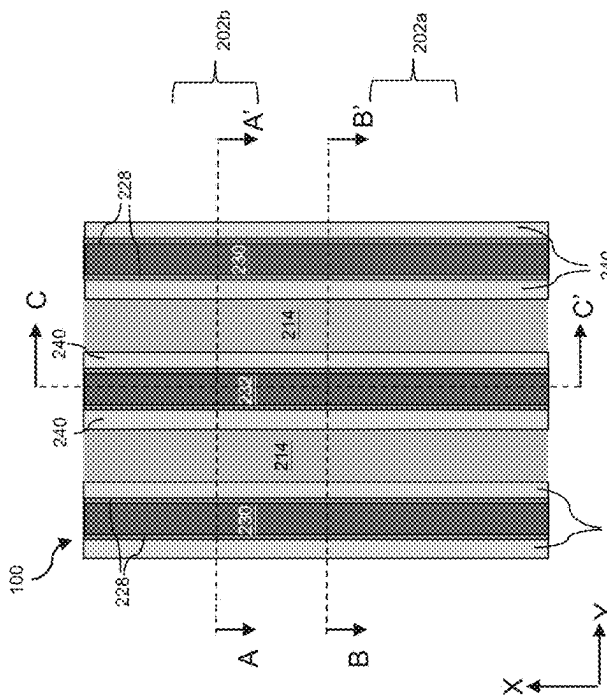


FIG. 15A

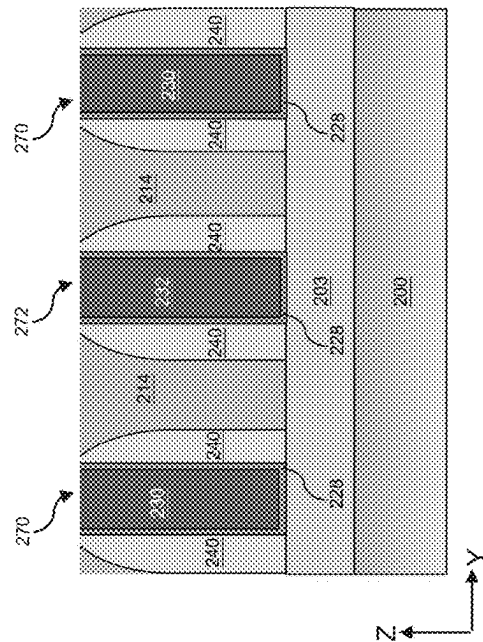


FIG. 15C

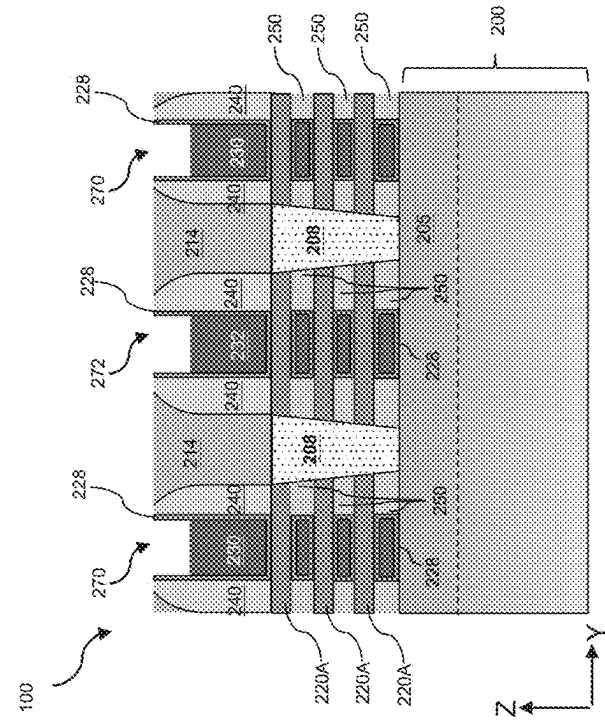


FIG. 16B

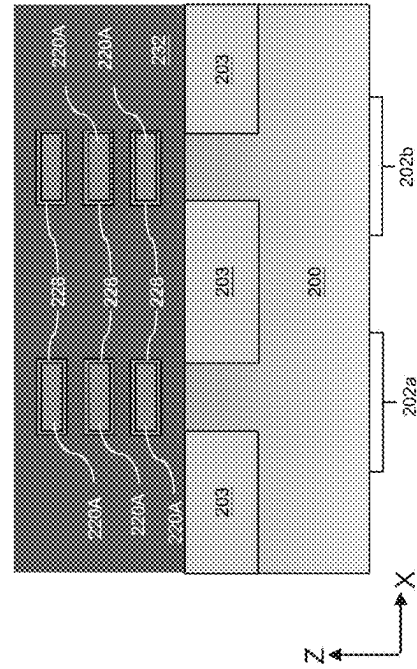


FIG. 16D

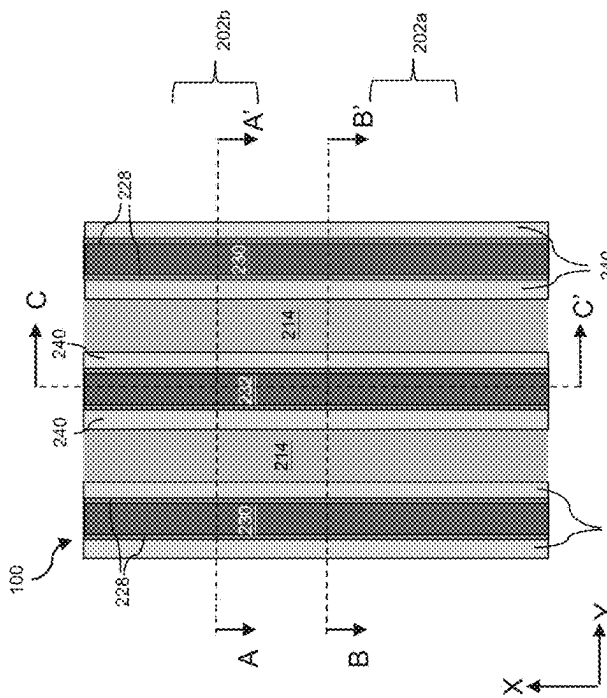


FIG. 16A

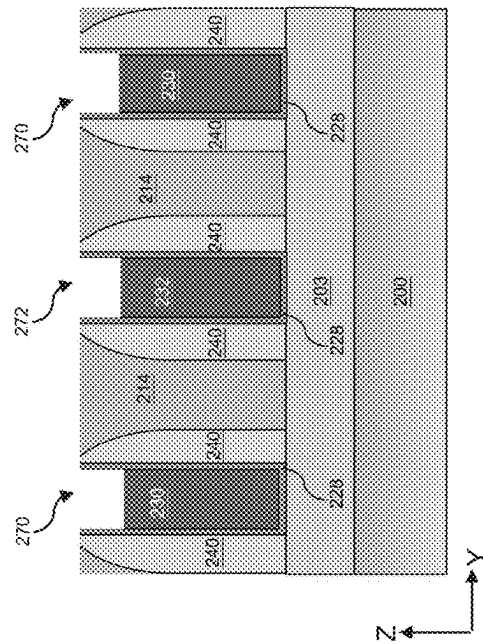


FIG. 16C

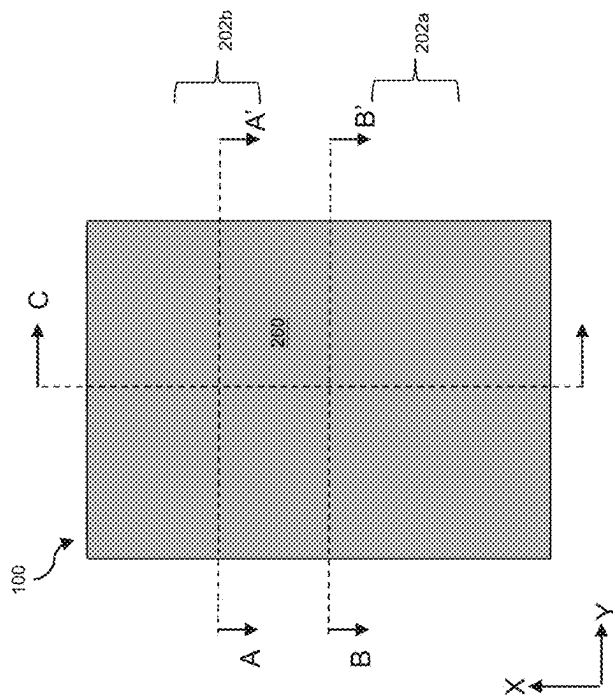


FIG. 17A

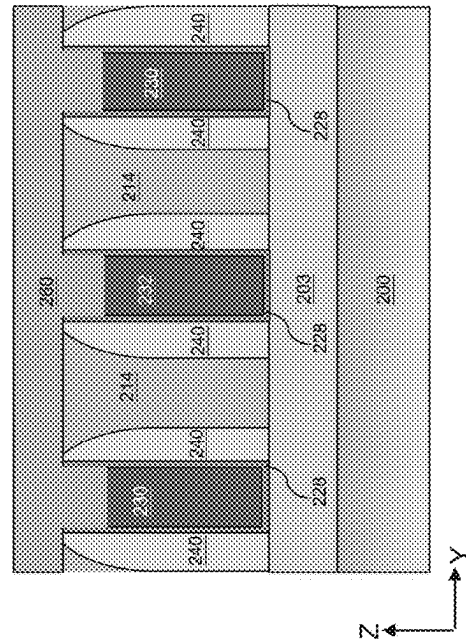


FIG. 17C

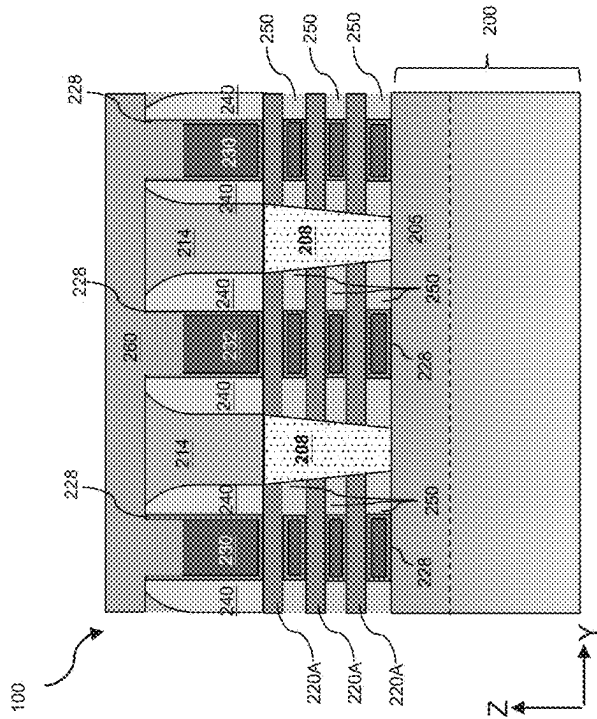


FIG. 17B

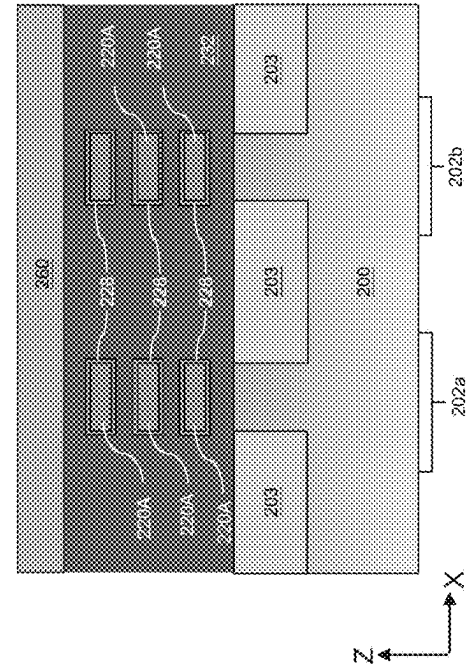


FIG. 17D

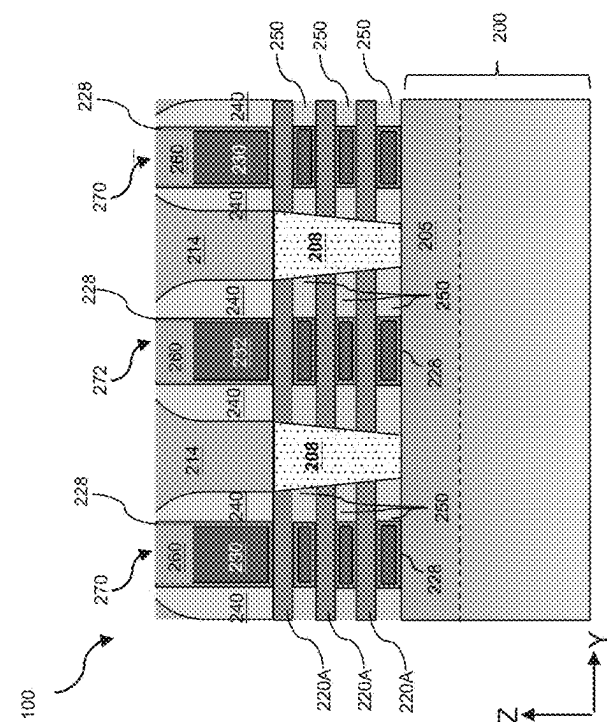


FIG. 18B

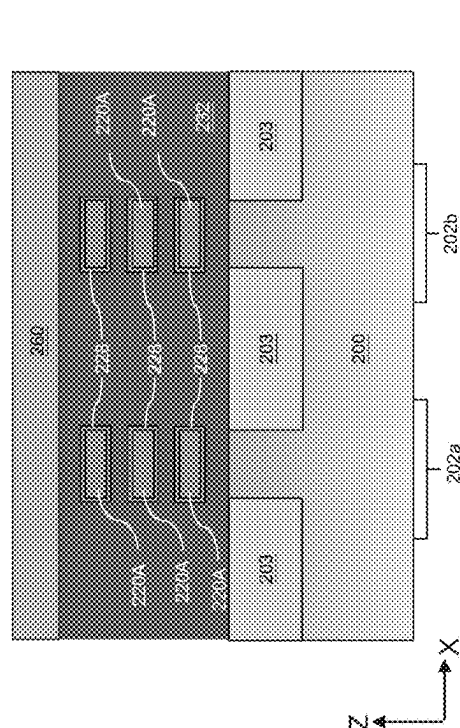


FIG. 18D

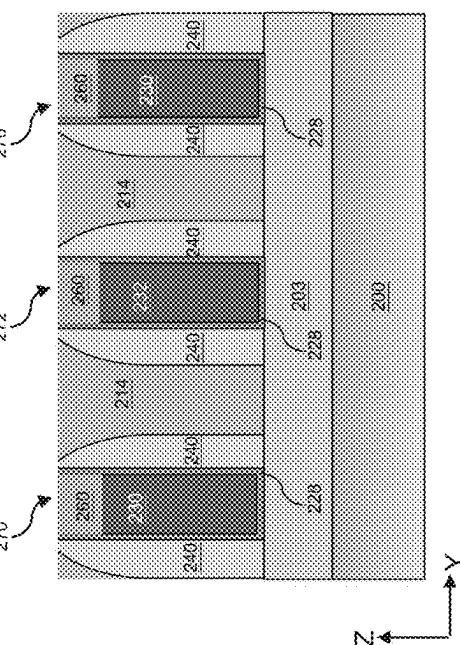
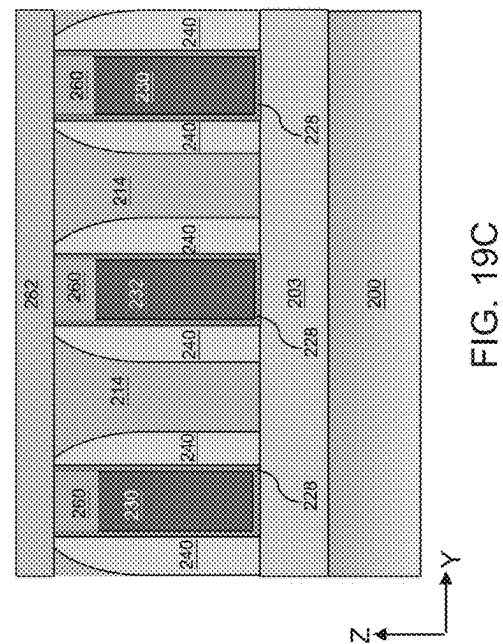
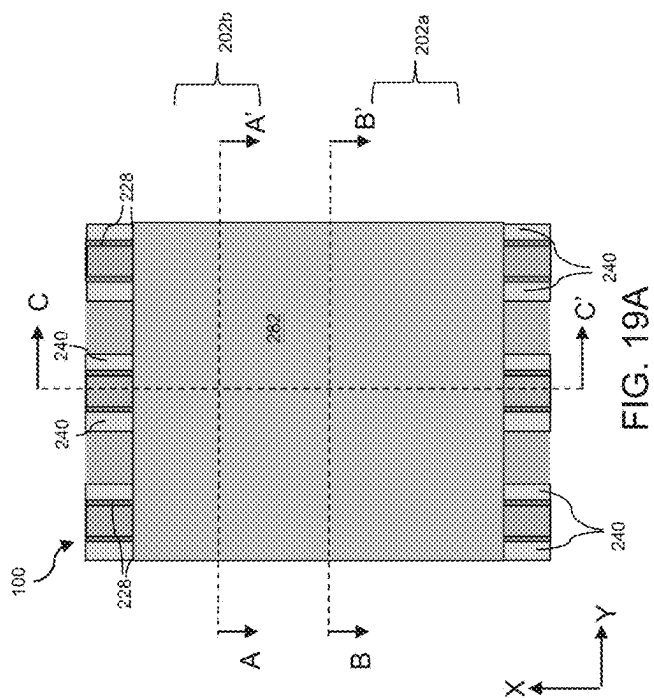
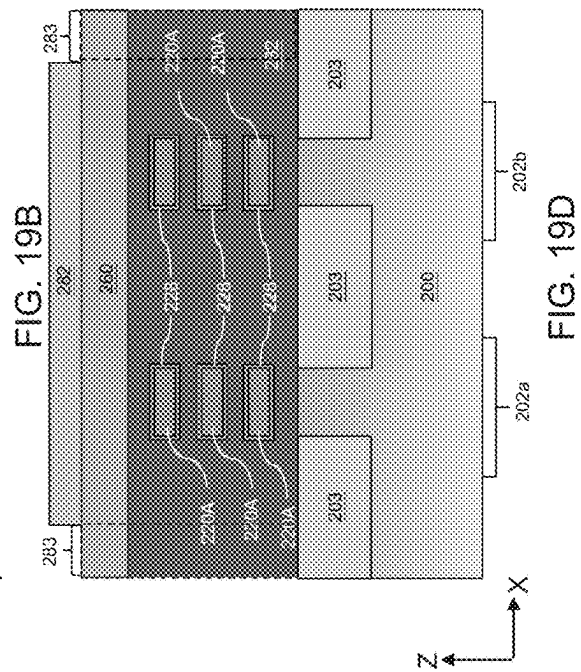
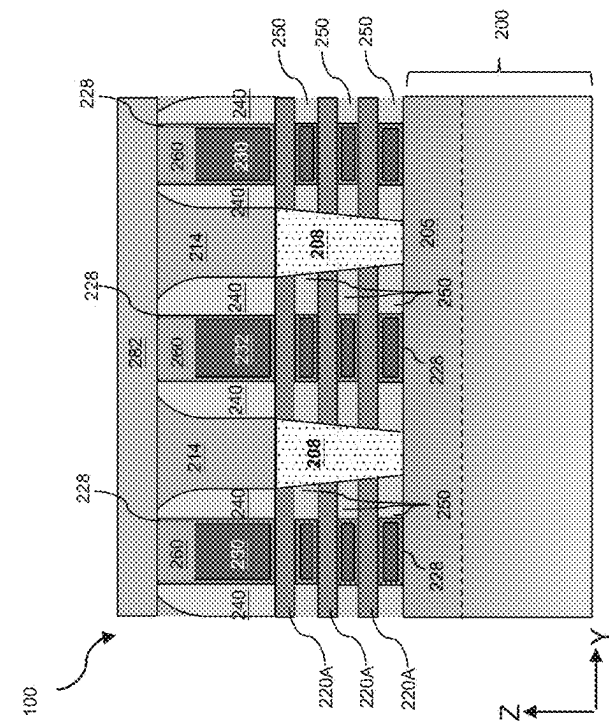
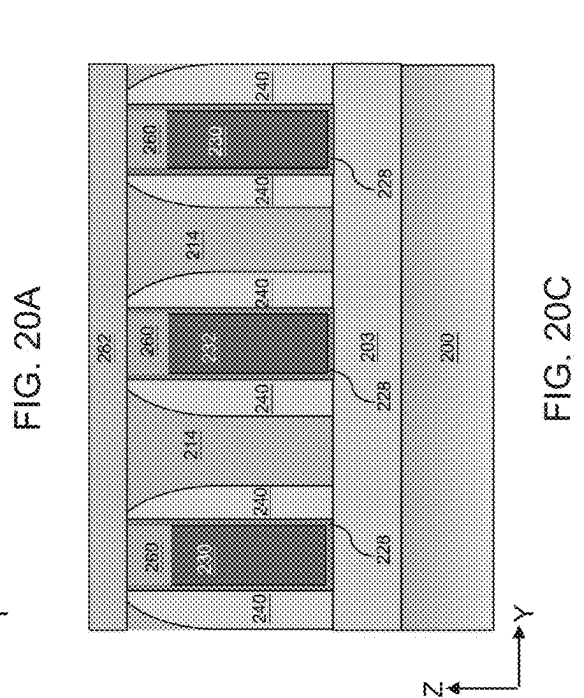
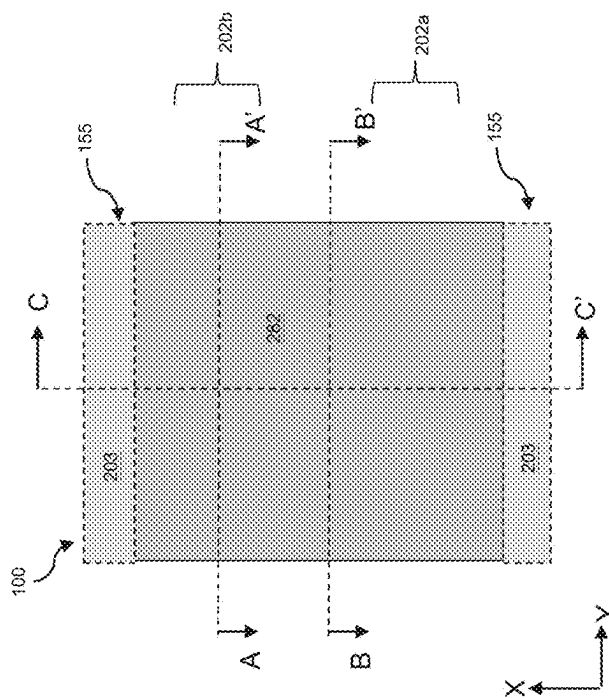
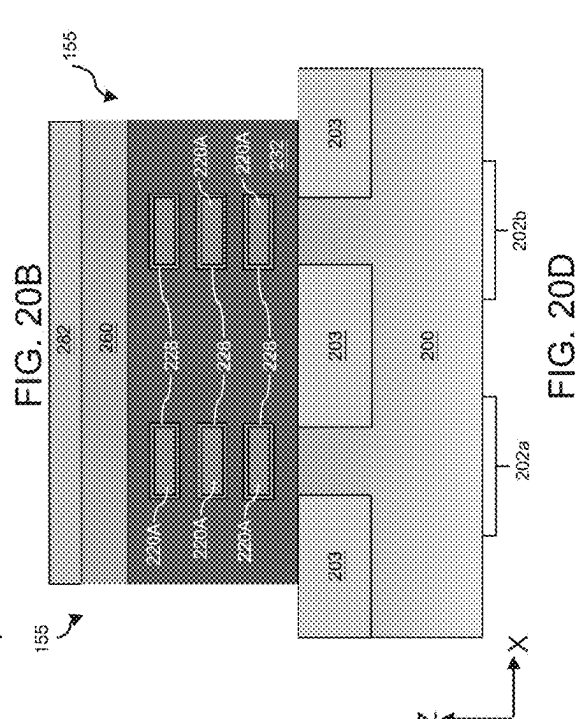
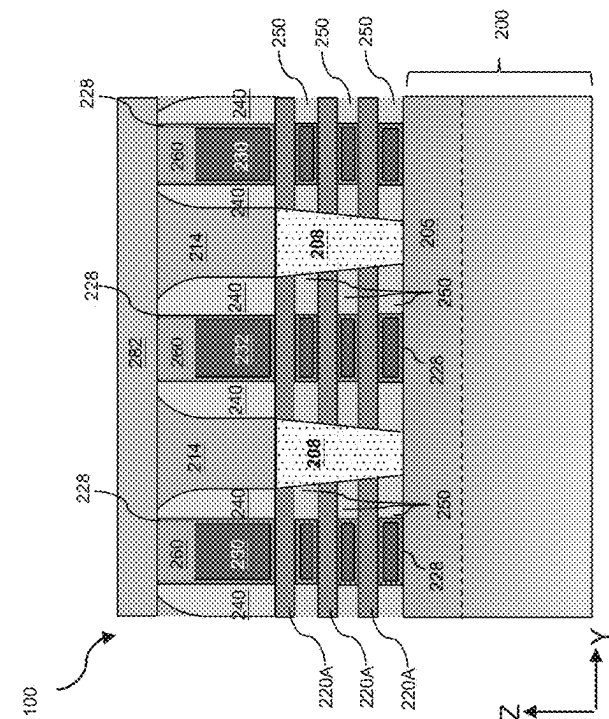


FIG. 18C





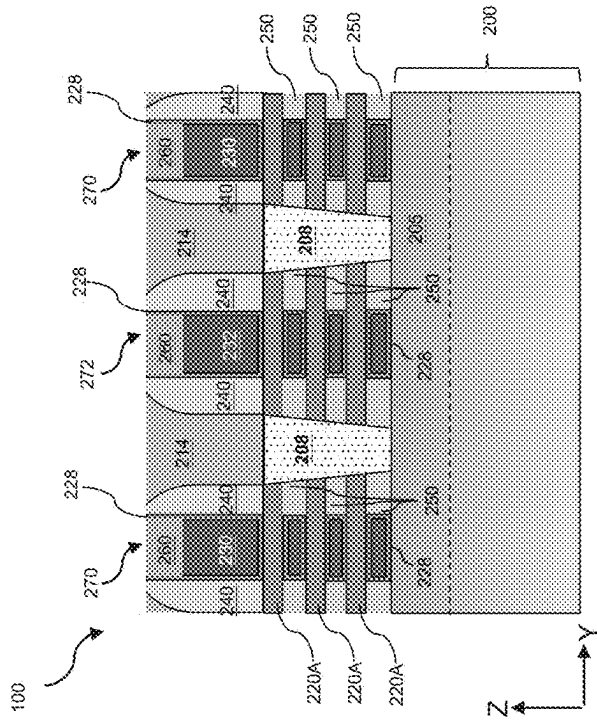


FIG. 21B

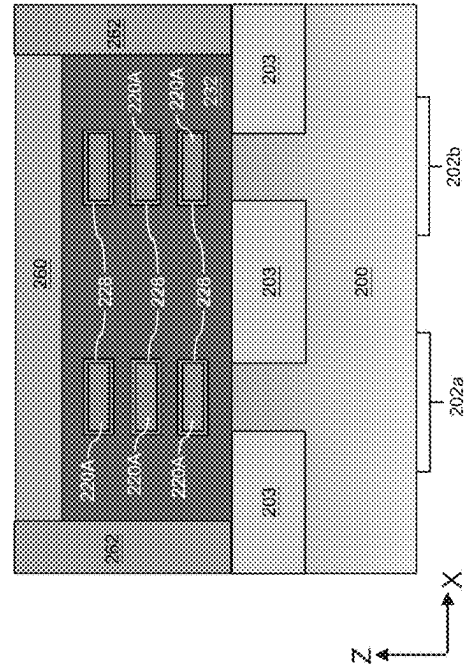


FIG. 21D

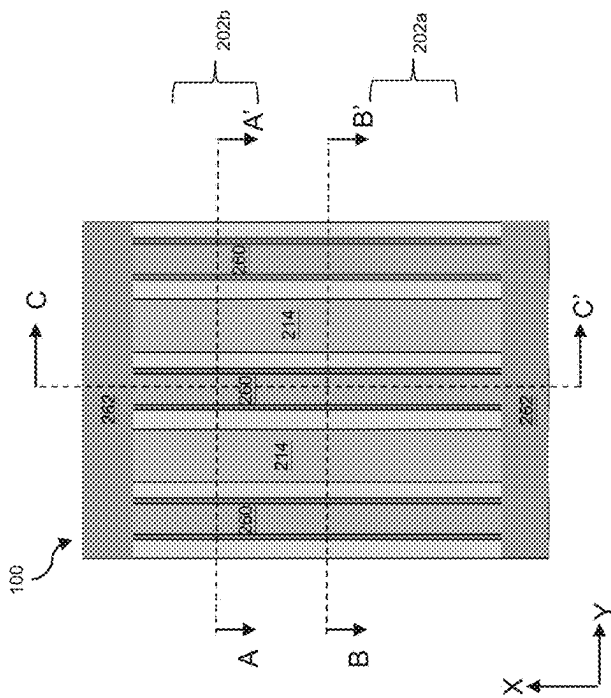


FIG. 21A

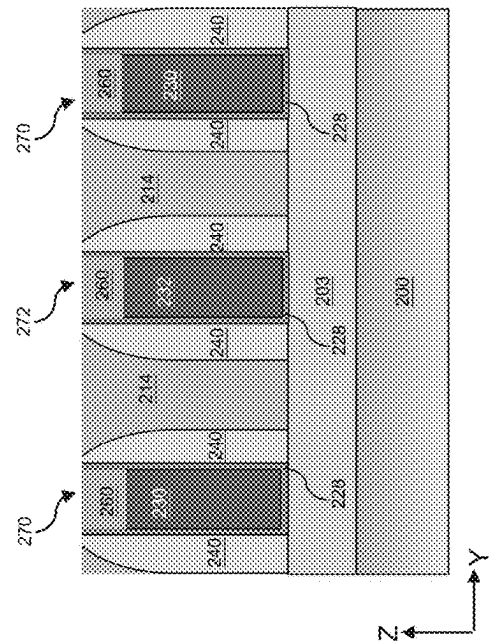
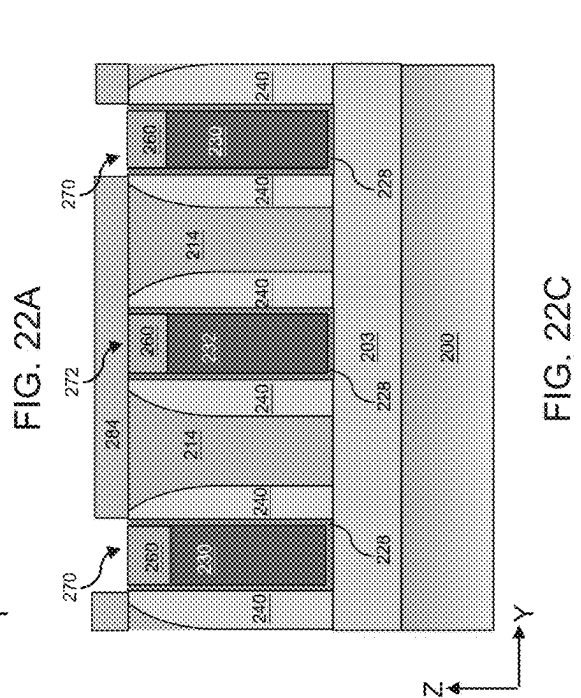
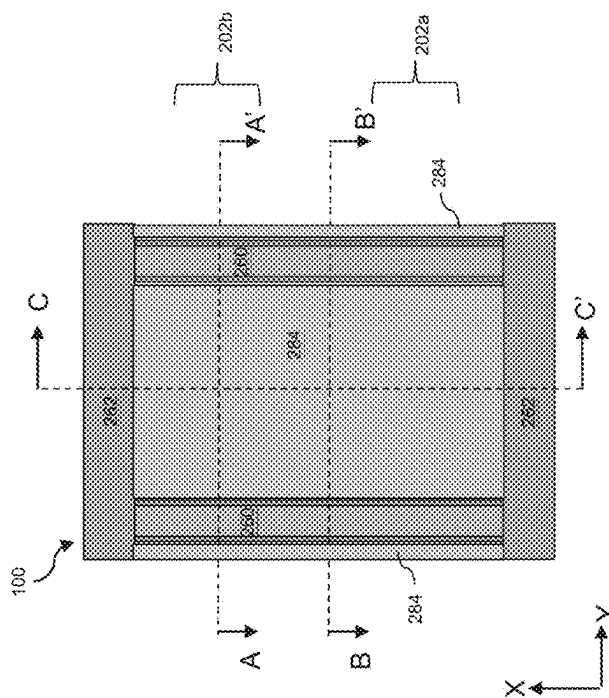
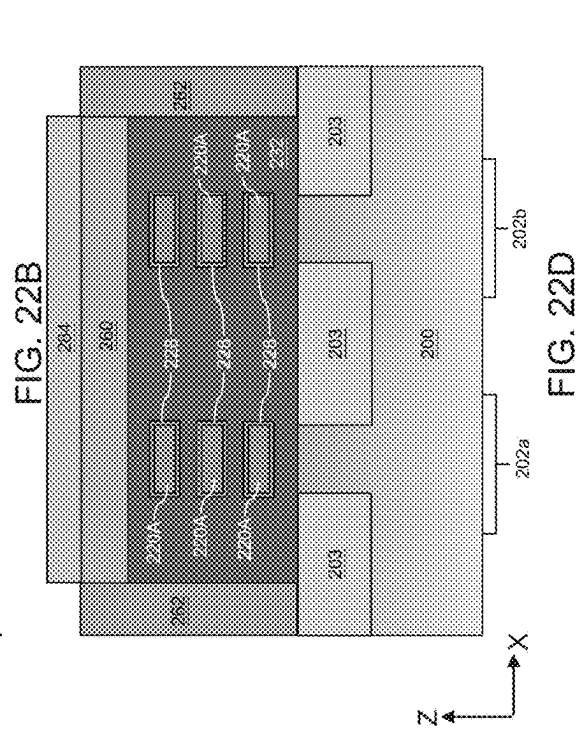
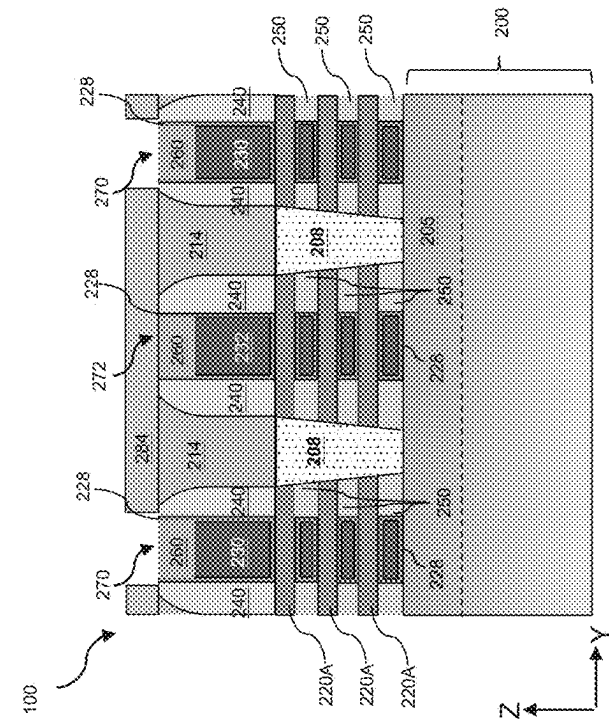


FIG. 21C



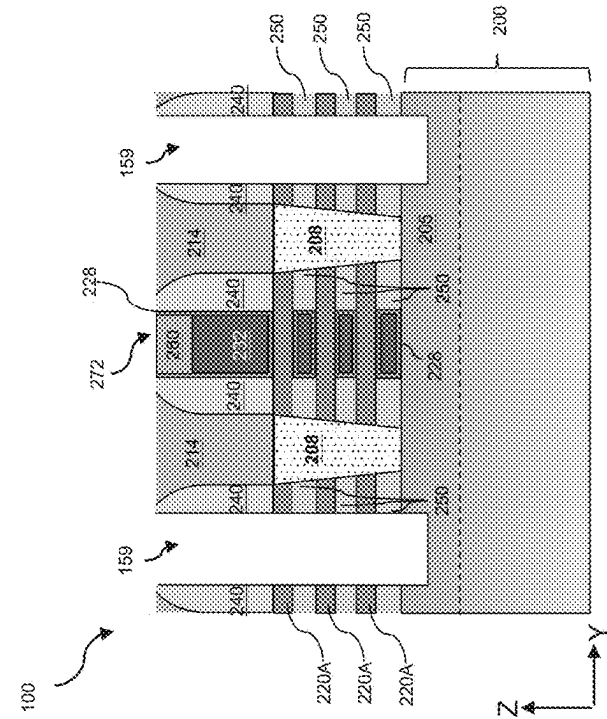


FIG. 23B

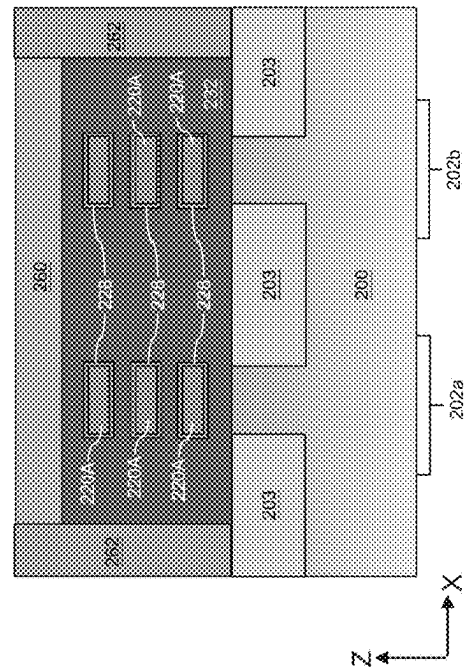


FIG. 23D

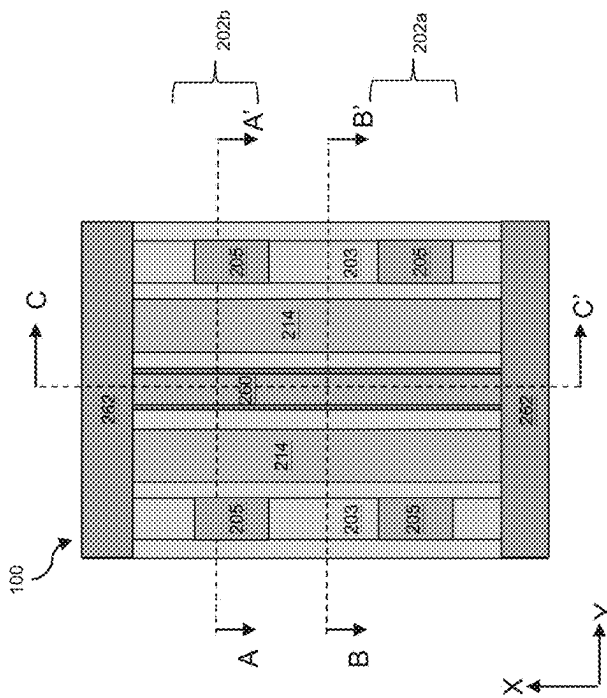


FIG. 23A

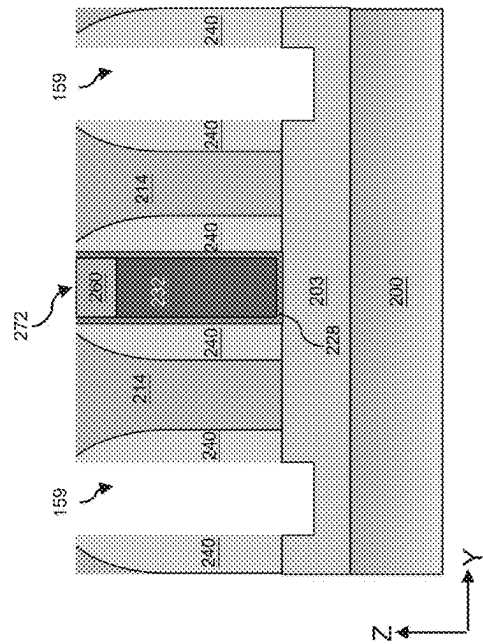


FIG. 23C

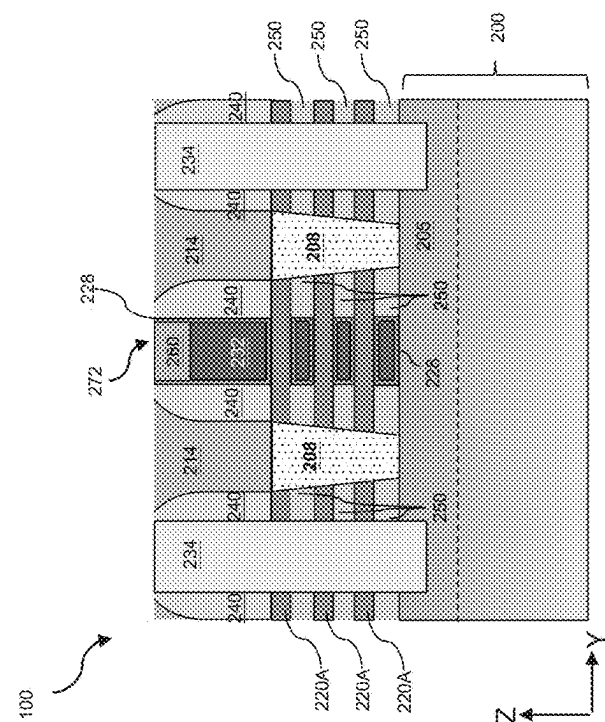


FIG. 24B

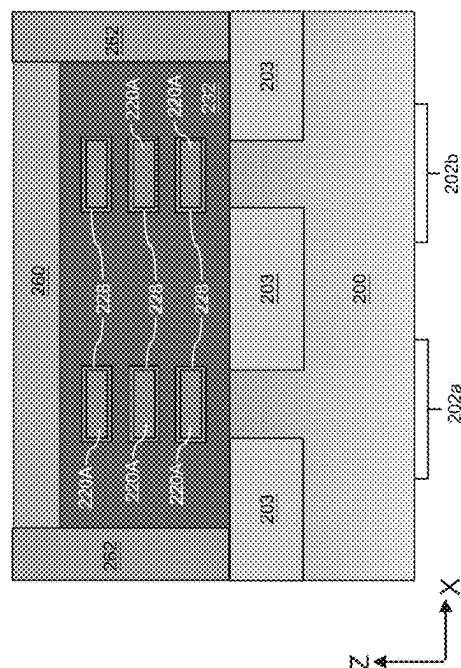


FIG. 24D

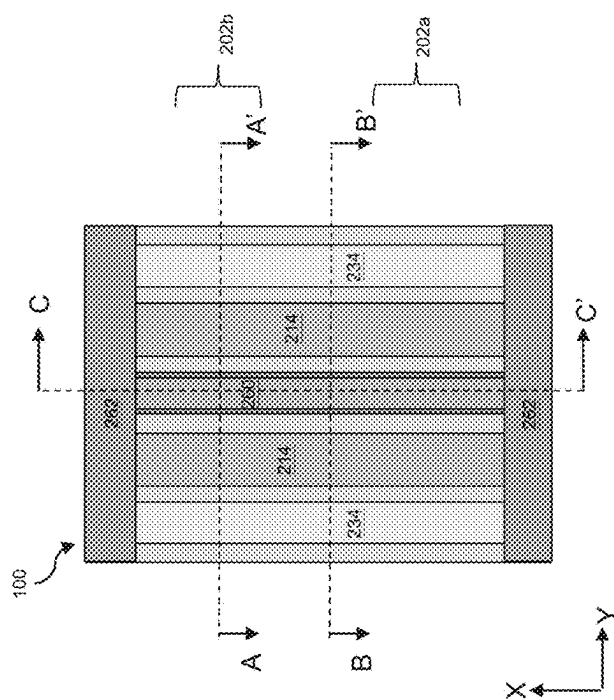


FIG. 24A

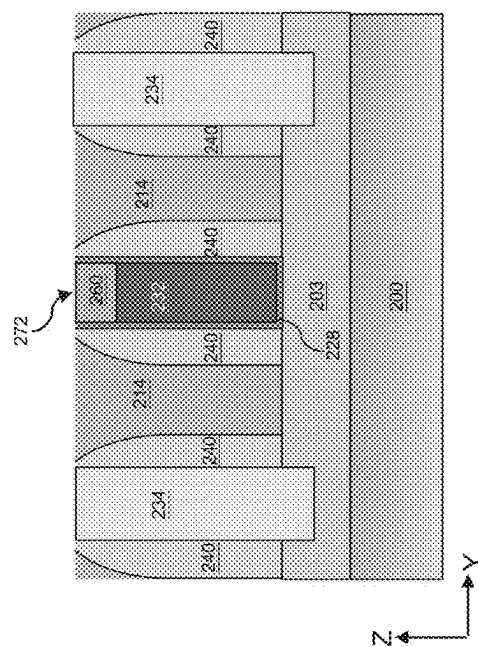


FIG. 24C

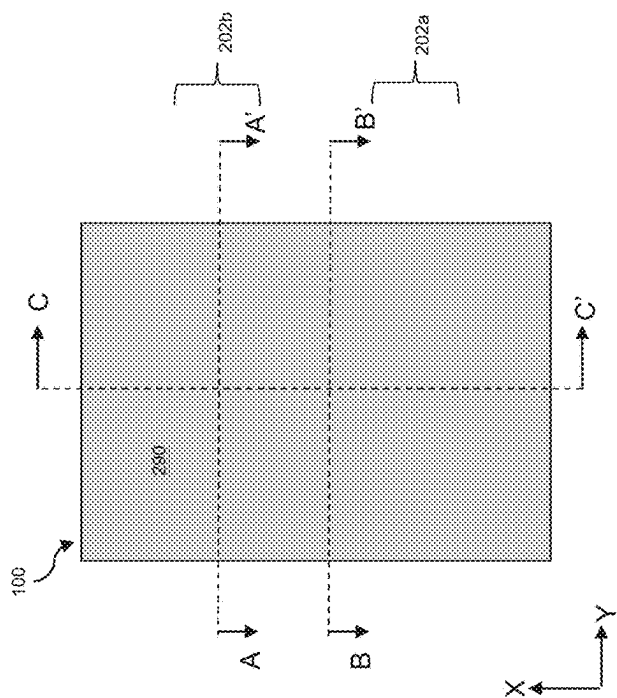


FIG. 25A

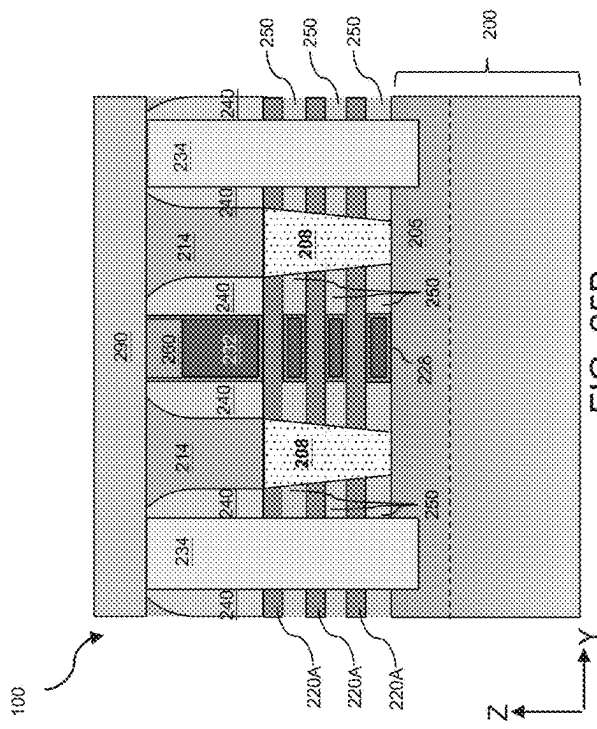


FIG. 25B

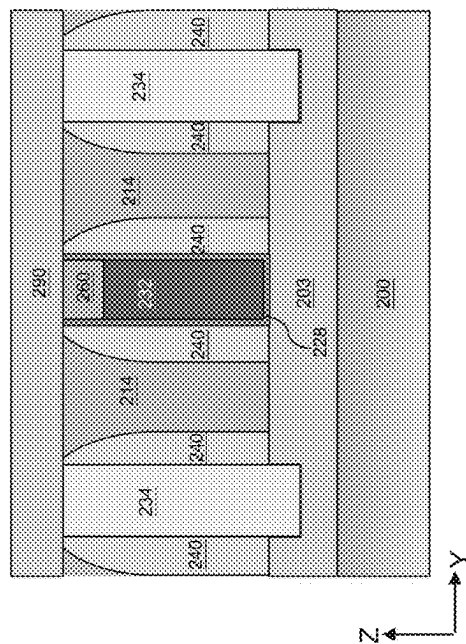


FIG. 25C

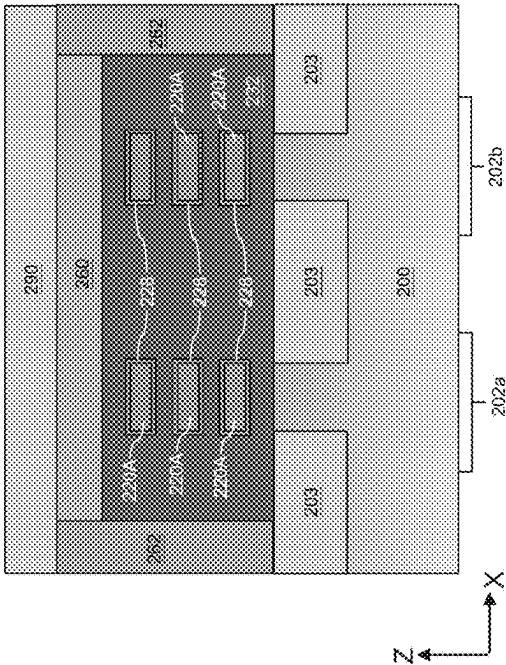


FIG. 25D

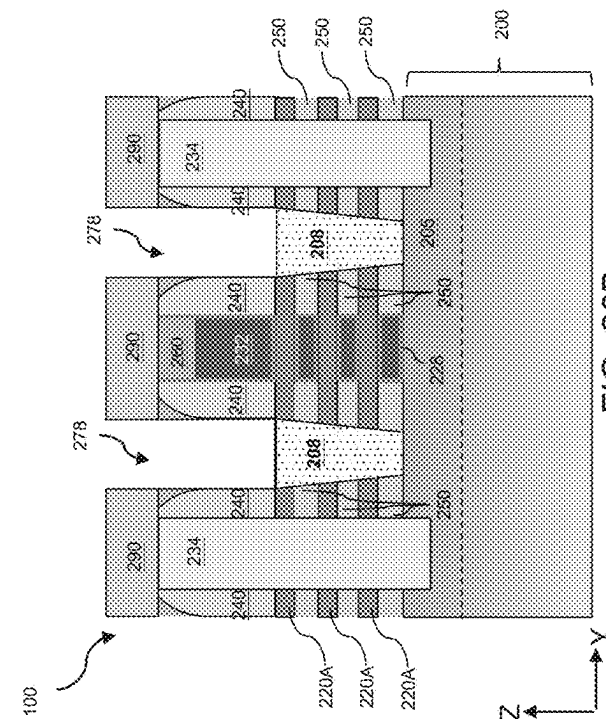


FIG. 26B

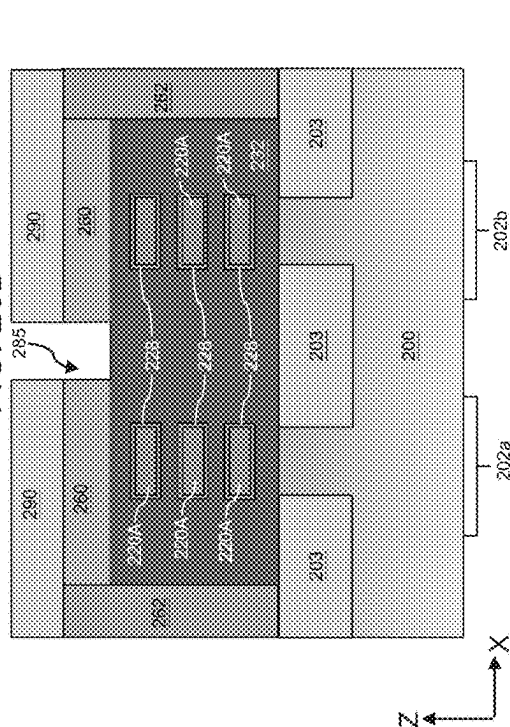


FIG. 26D

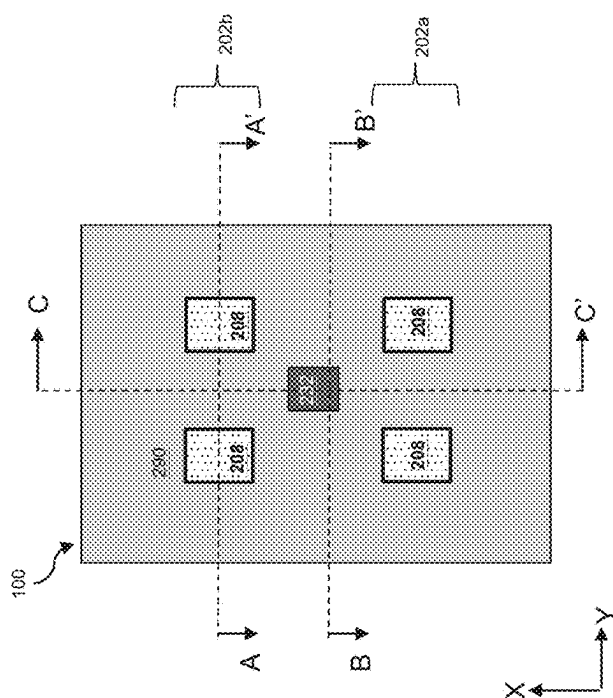


FIG. 26A

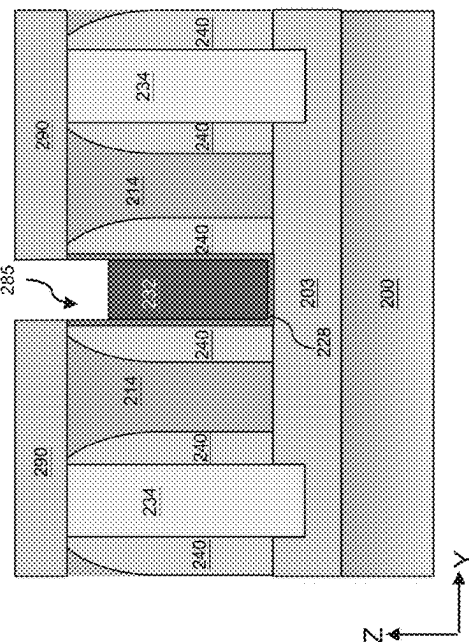


FIG. 26C

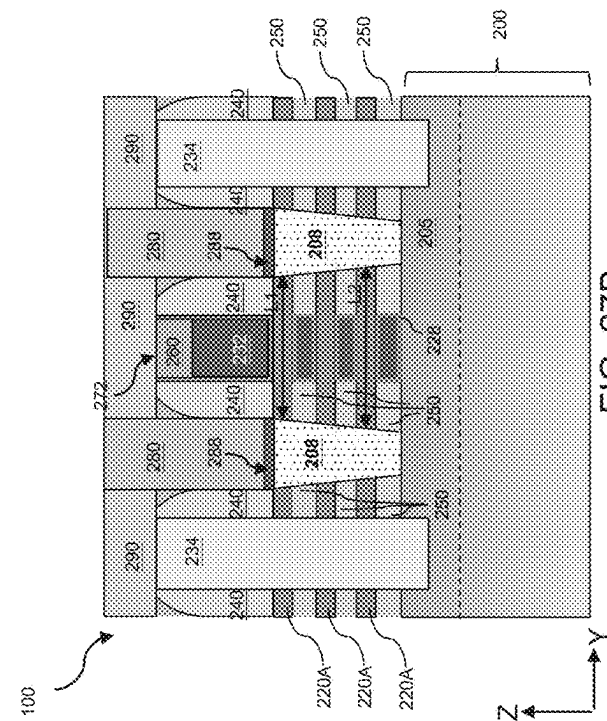


FIG. 27B

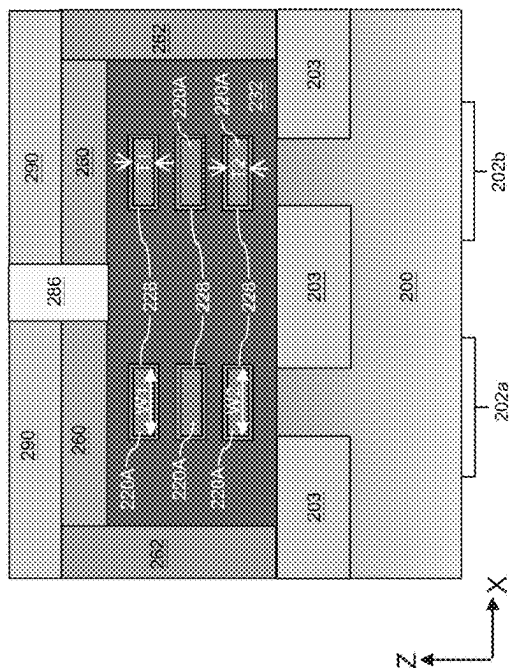


FIG. 27D

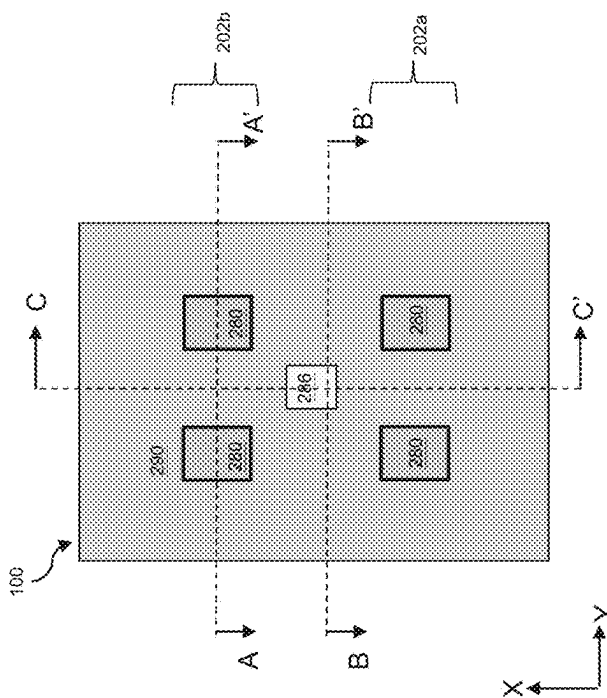


FIG. 27A

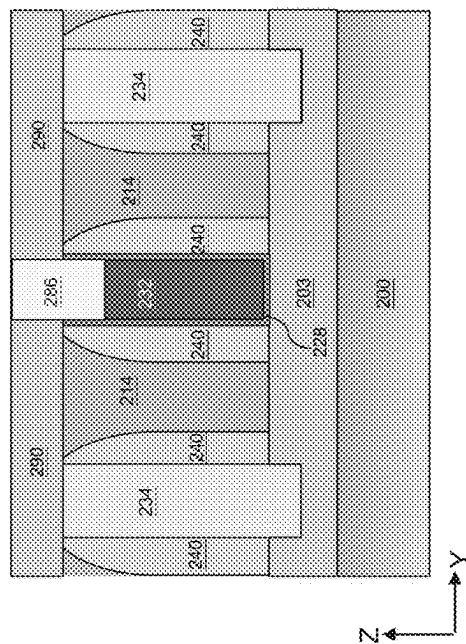


FIG. 27C

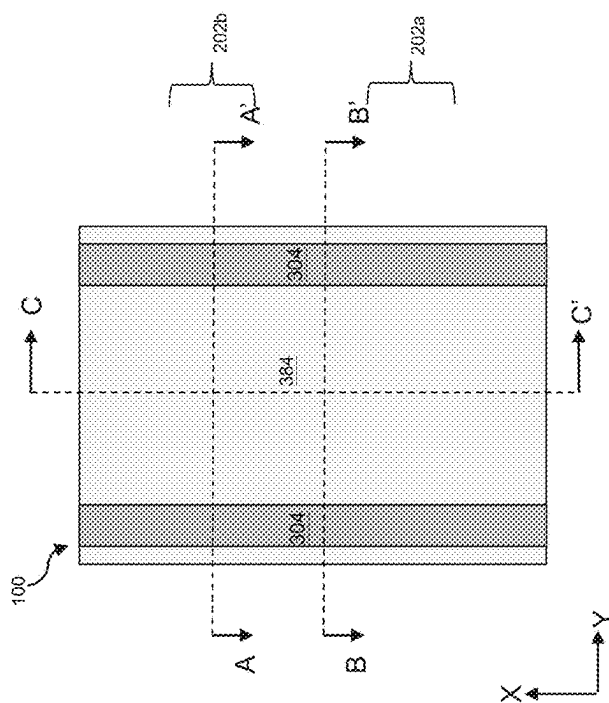


FIG. 28A

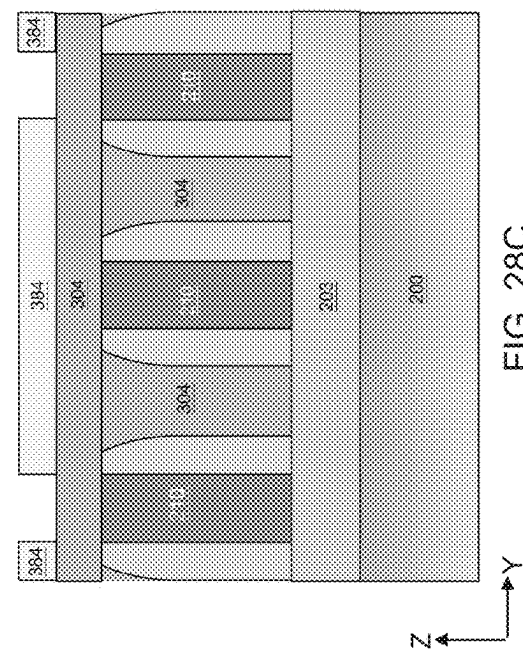


FIG. 28C

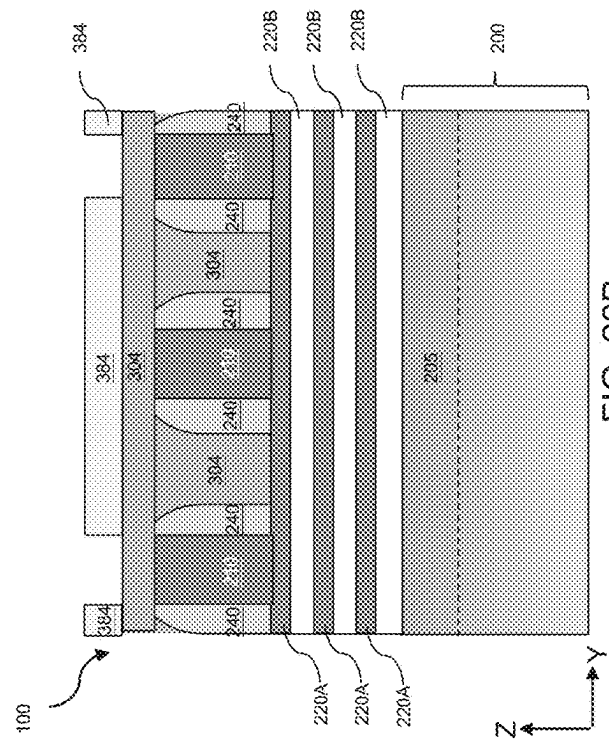


FIG. 28B

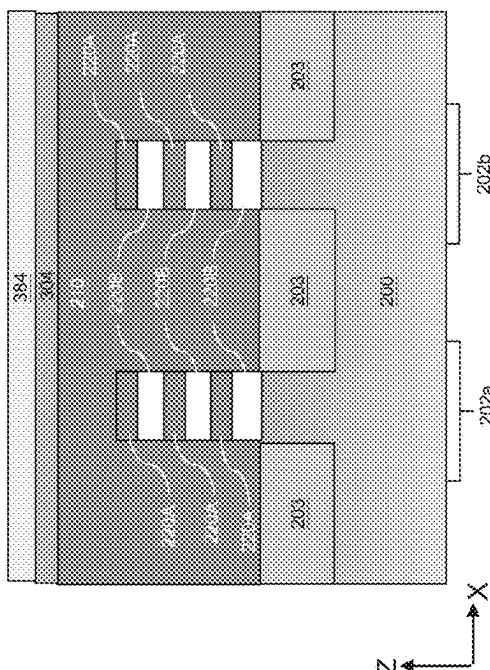
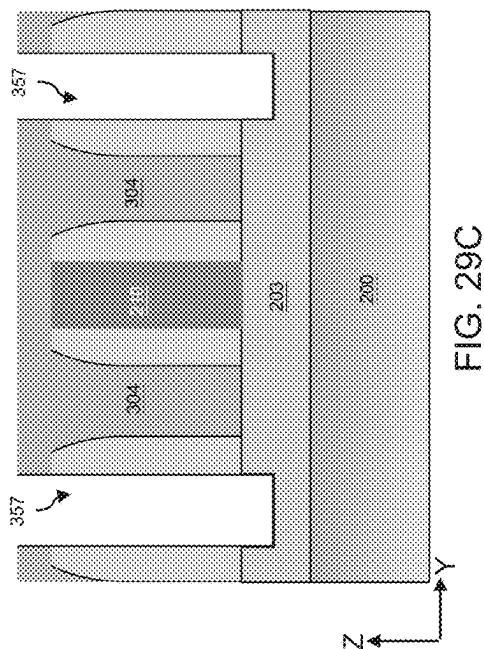
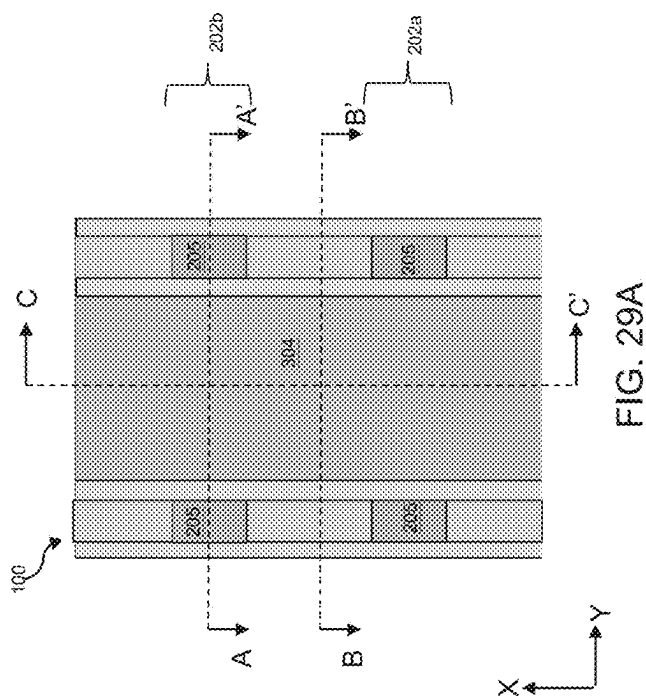
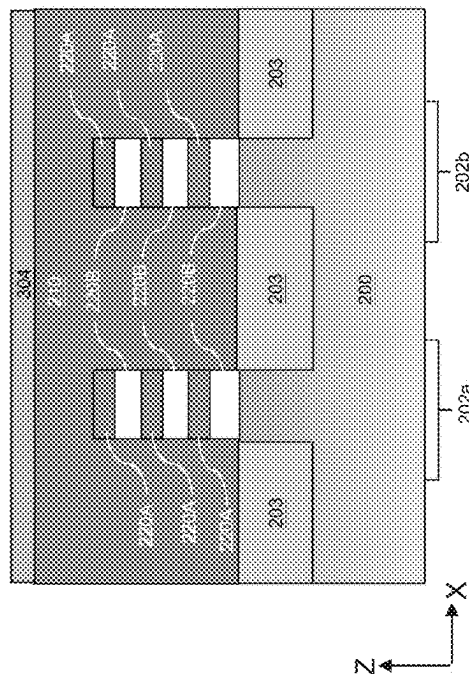
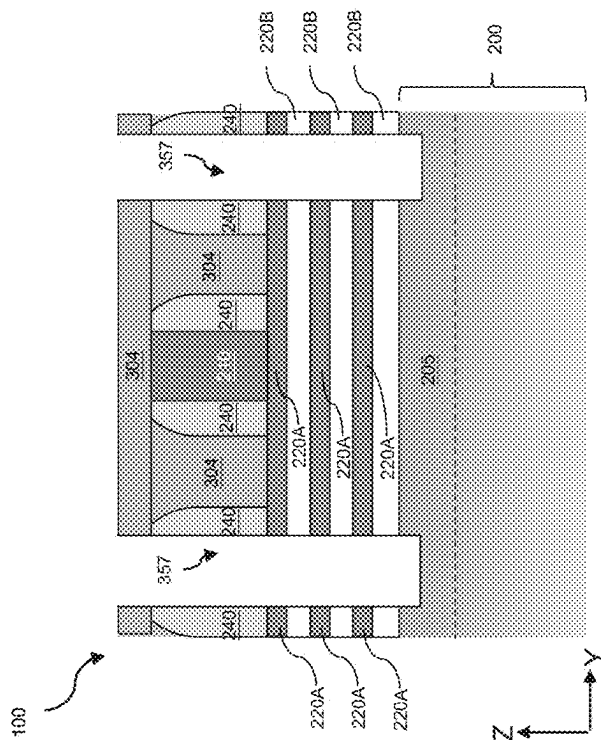


FIG. 28D



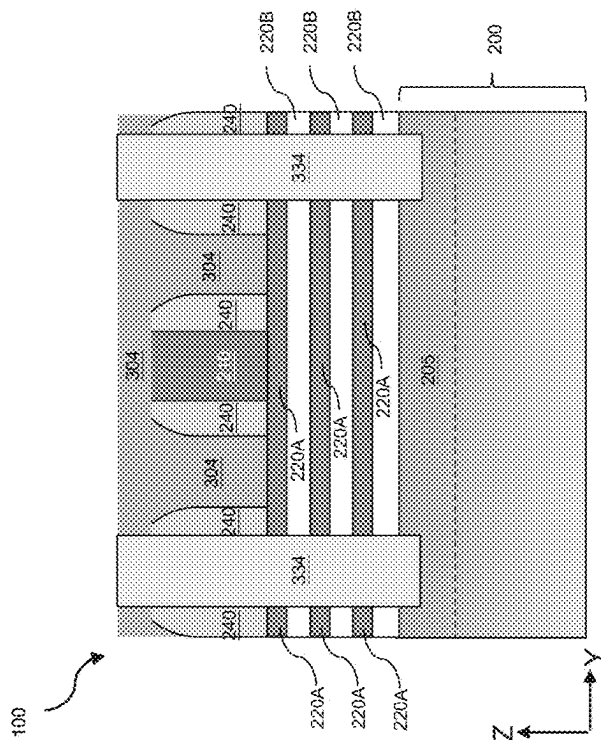


FIG. 30B

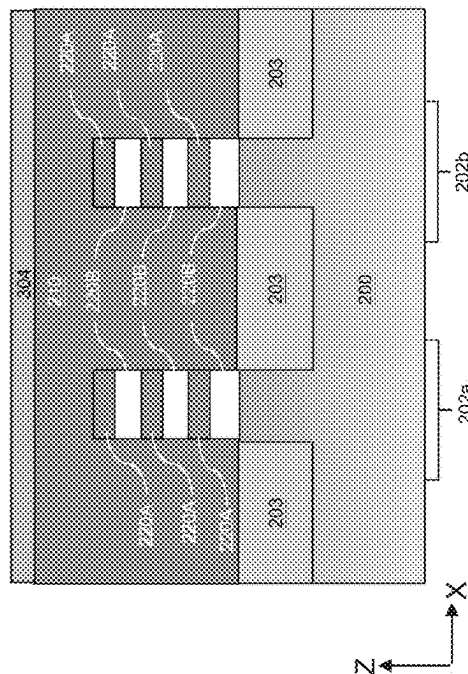


FIG. 30D

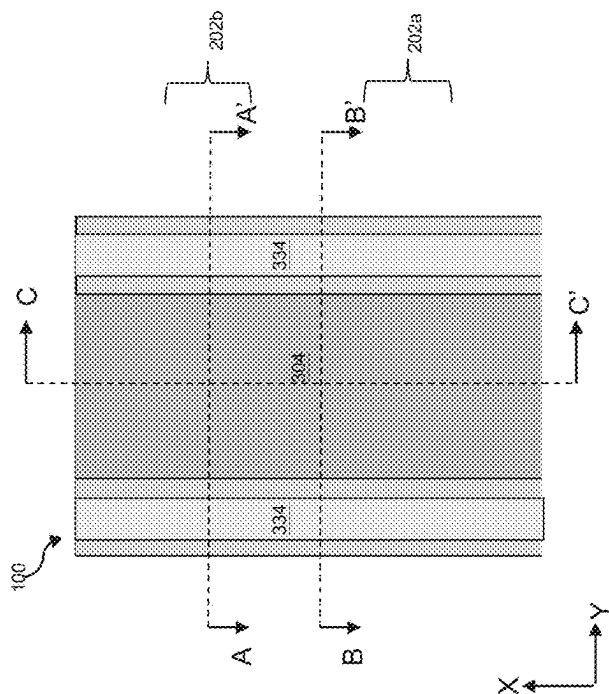


FIG. 30A

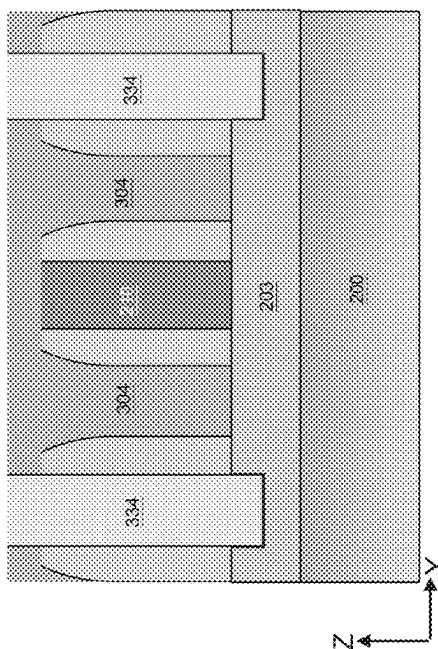


FIG. 30C

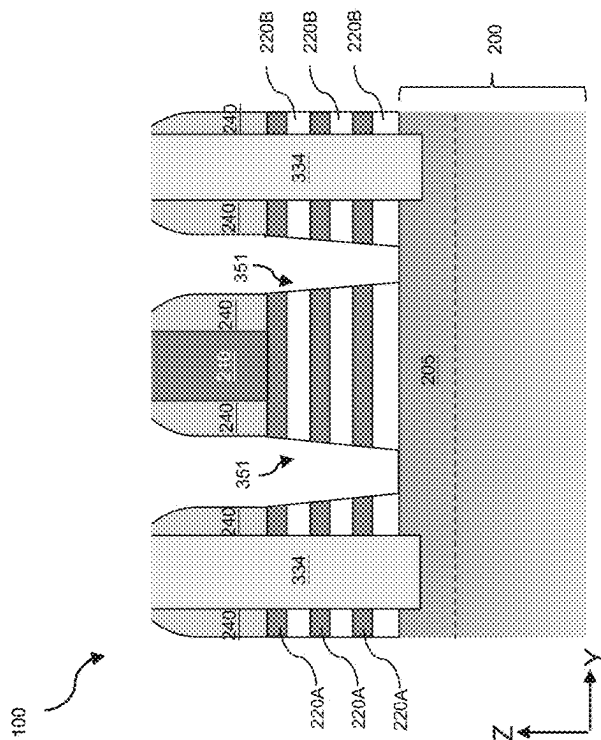


FIG. 31B

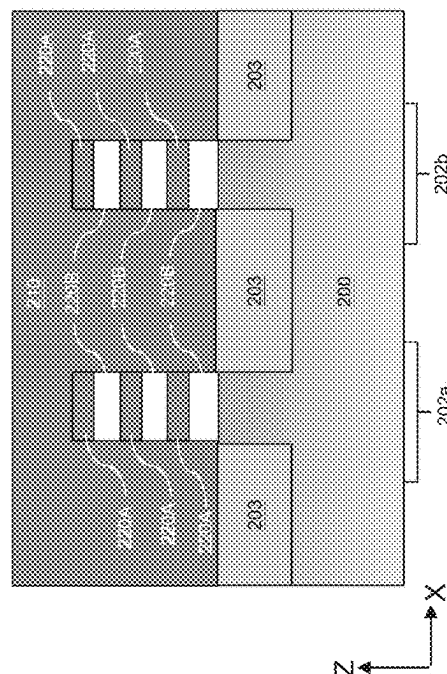


FIG. 31D

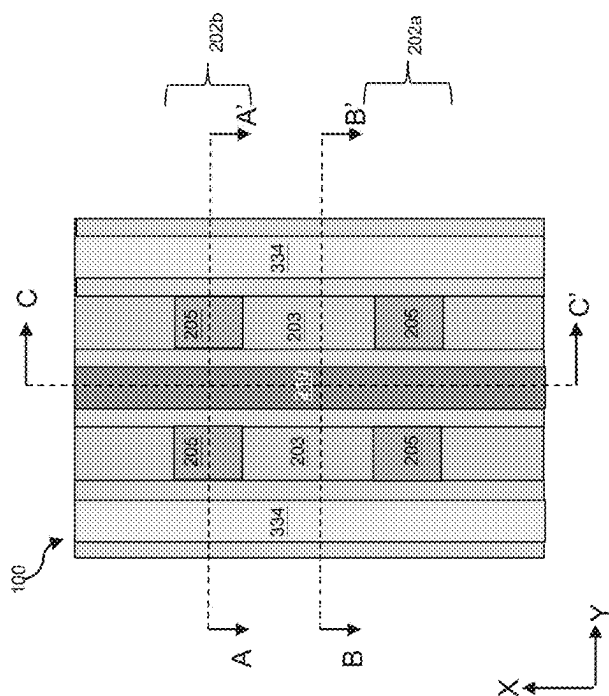


FIG. 31A

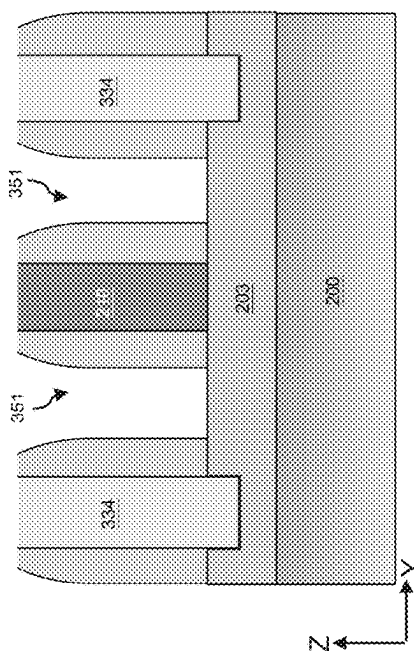


FIG. 31C

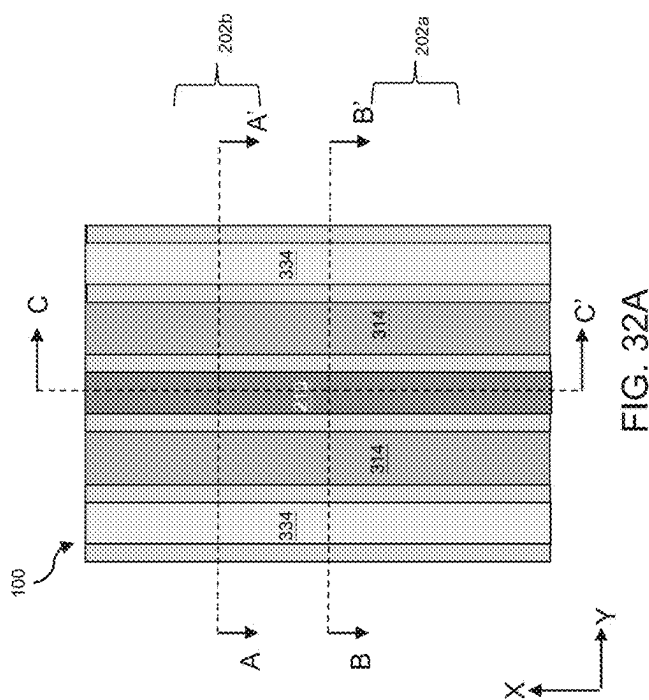


FIG. 32A

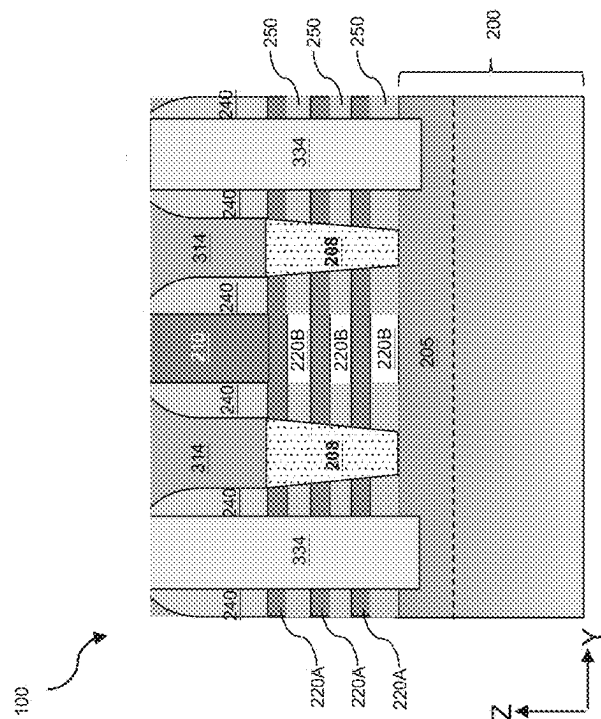


FIG. 32B

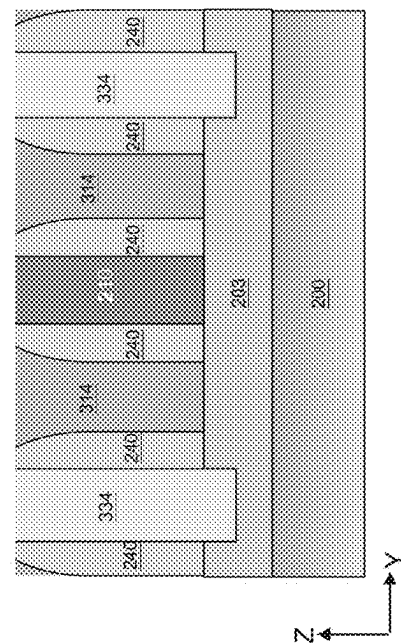


FIG. 32C

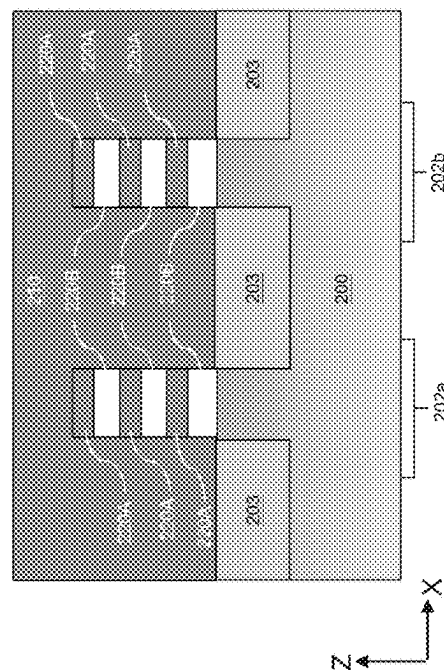


FIG. 32D

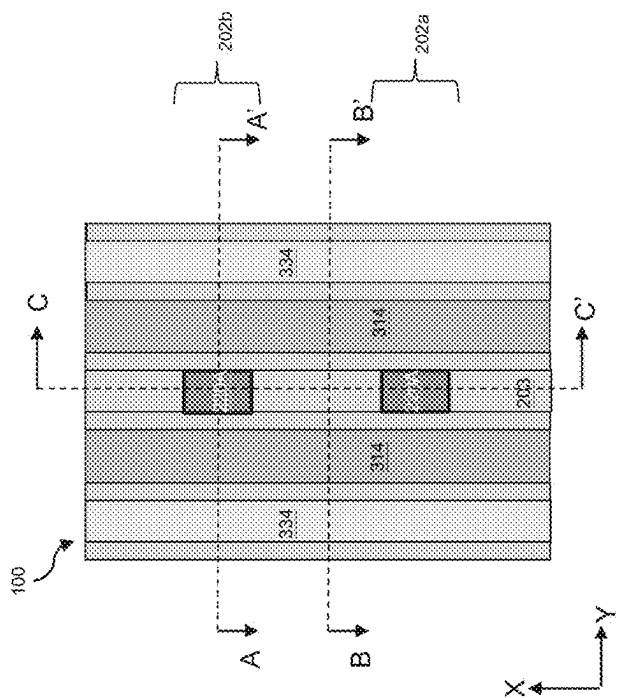


FIG. 33A

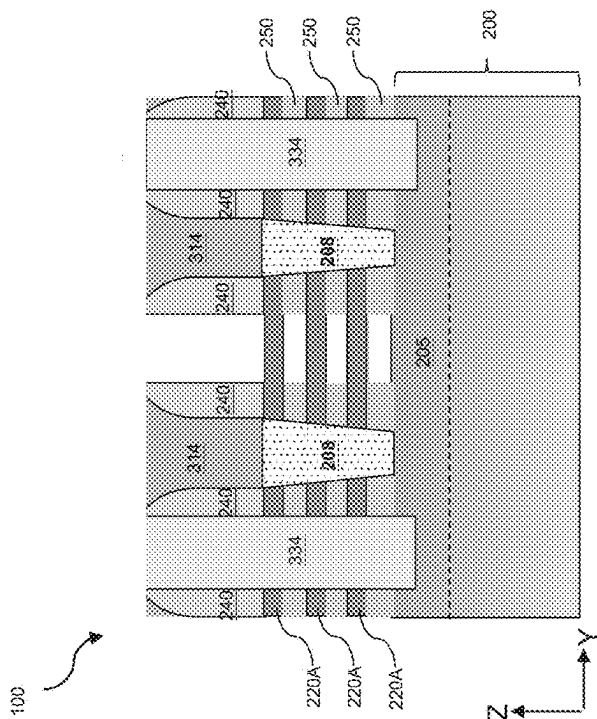


FIG. 33B

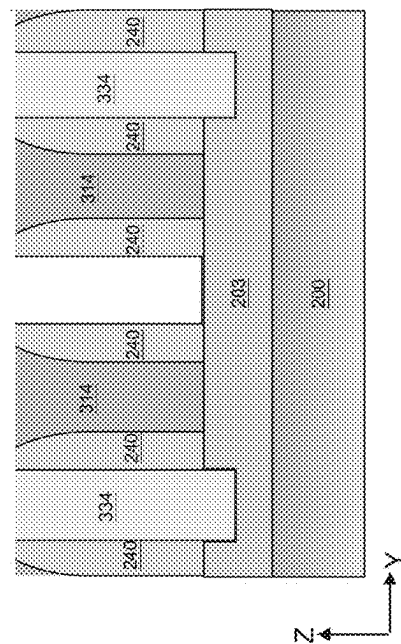


FIG. 33C

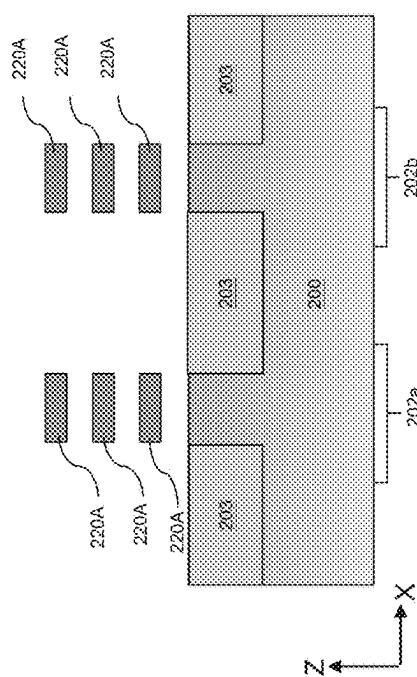
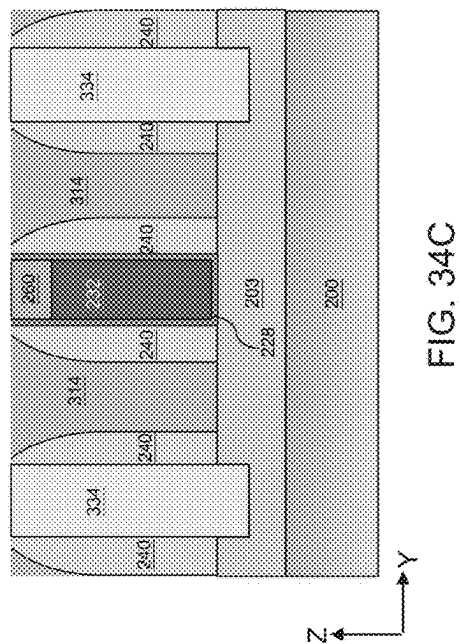
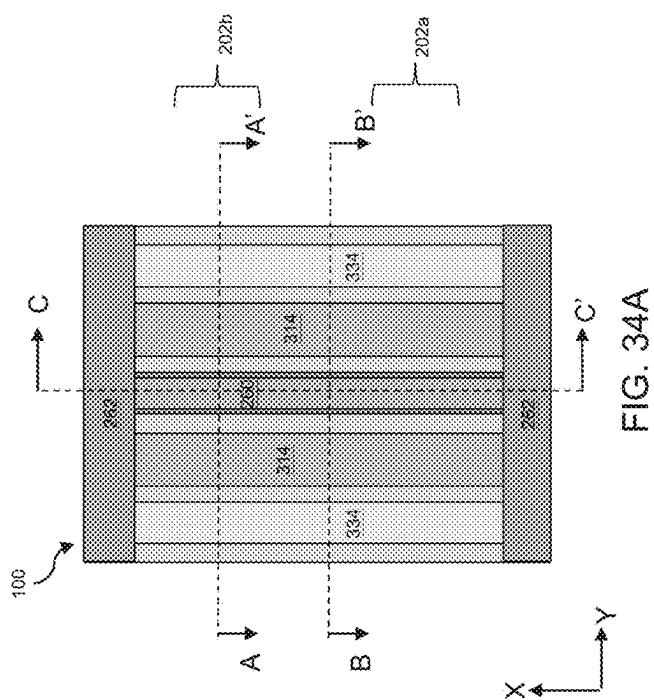
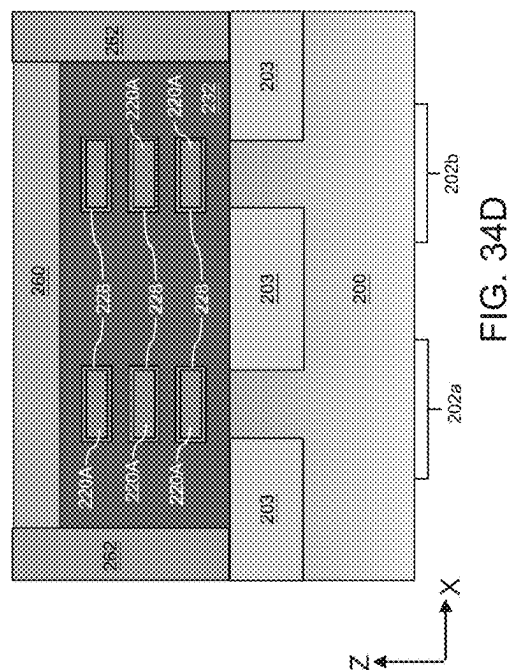
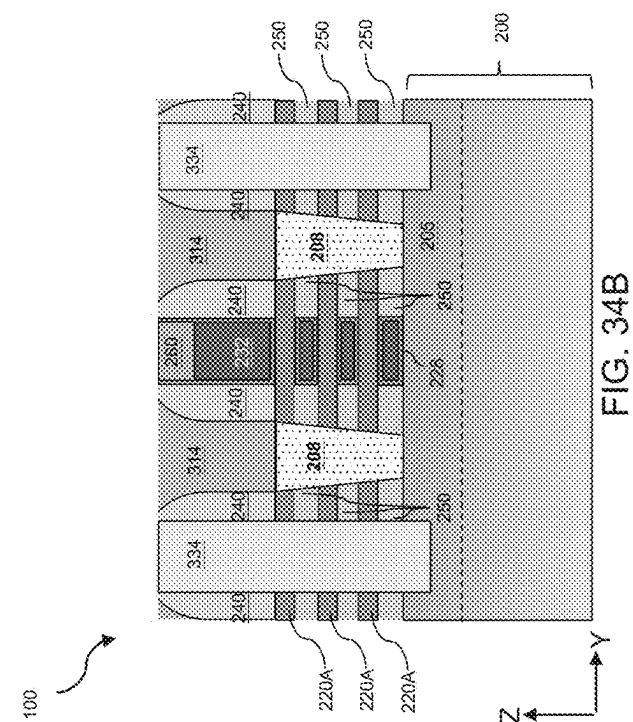


FIG. 33D



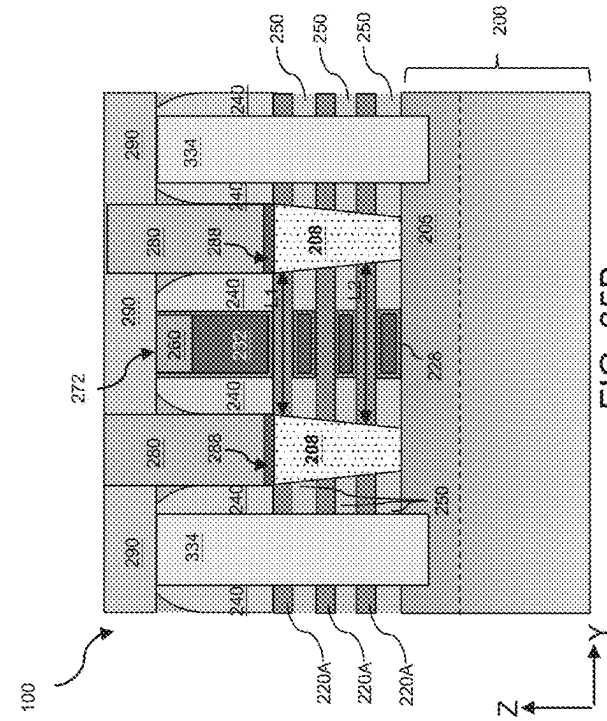


FIG. 35B

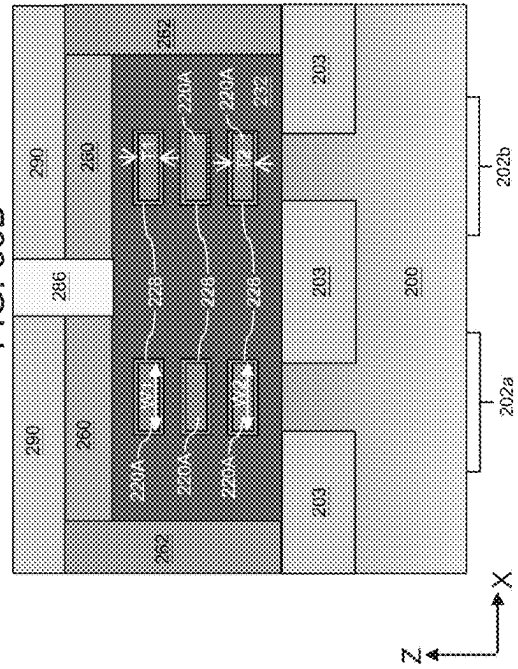


FIG. 35D

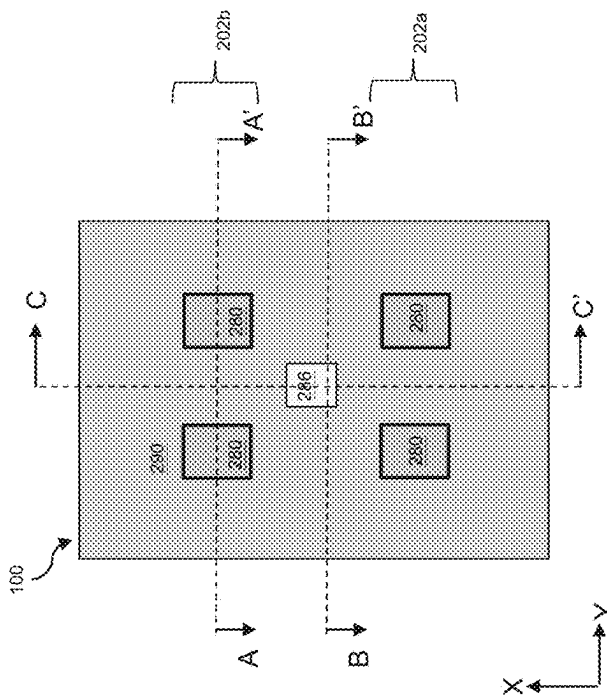


FIG. 35A

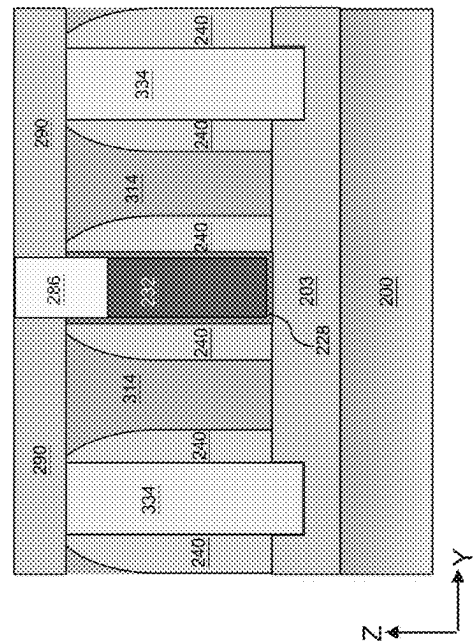


FIG. 35C

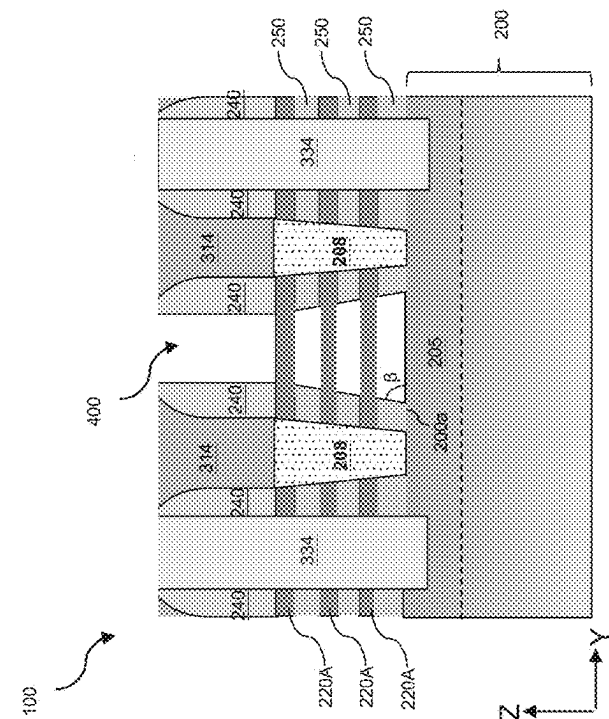


FIG. 36B

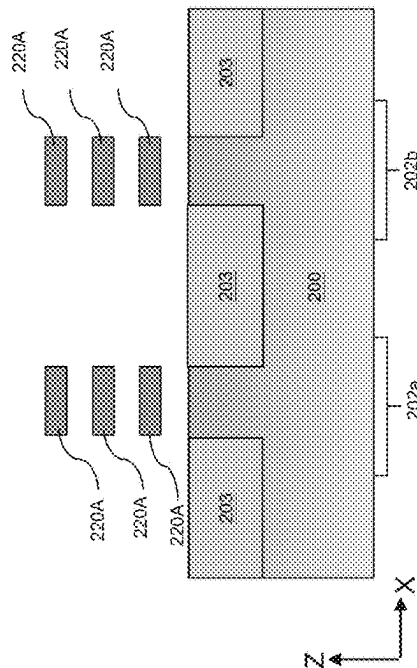


FIG. 36D

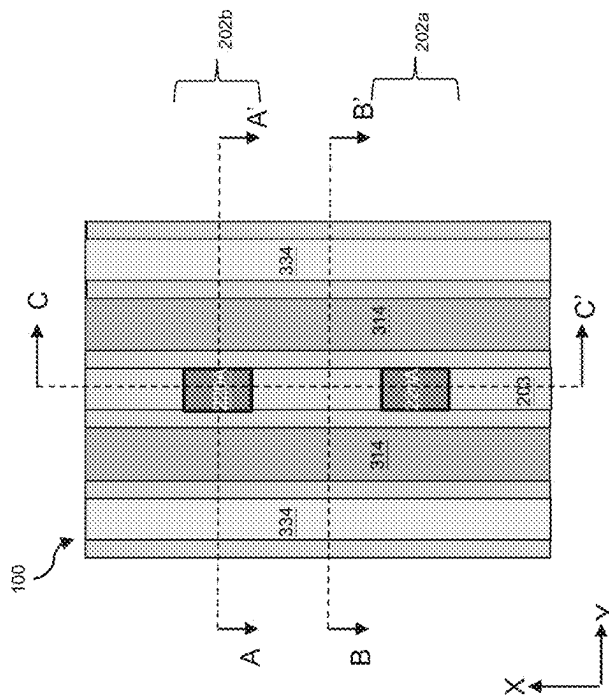


FIG. 36A

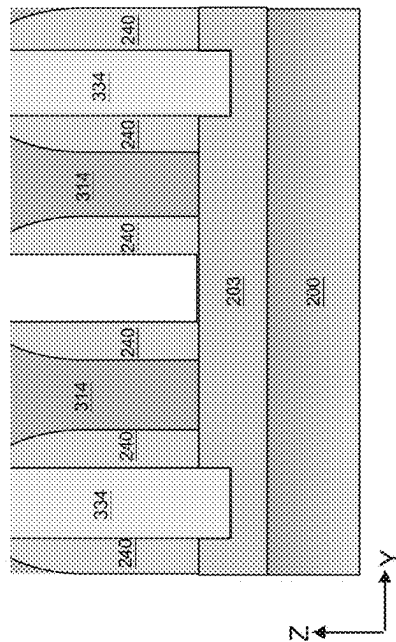
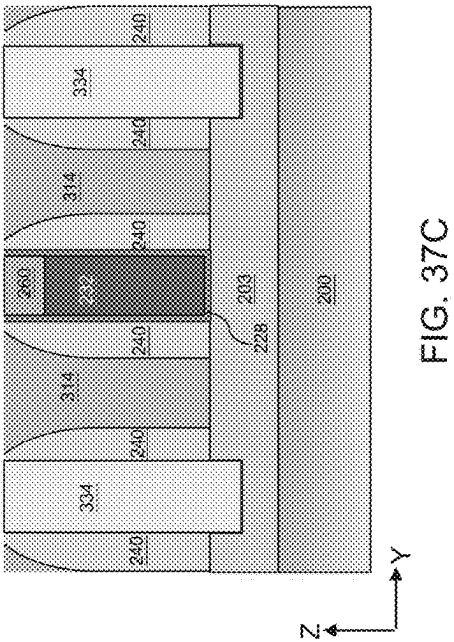
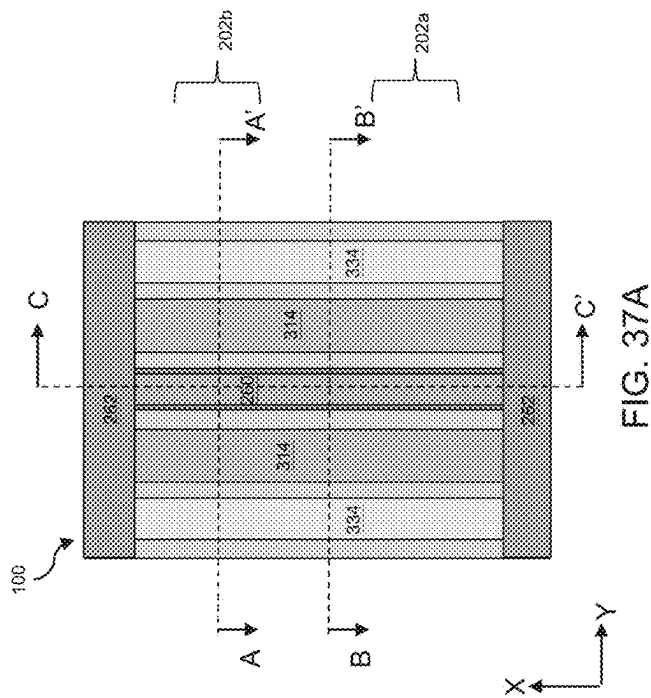
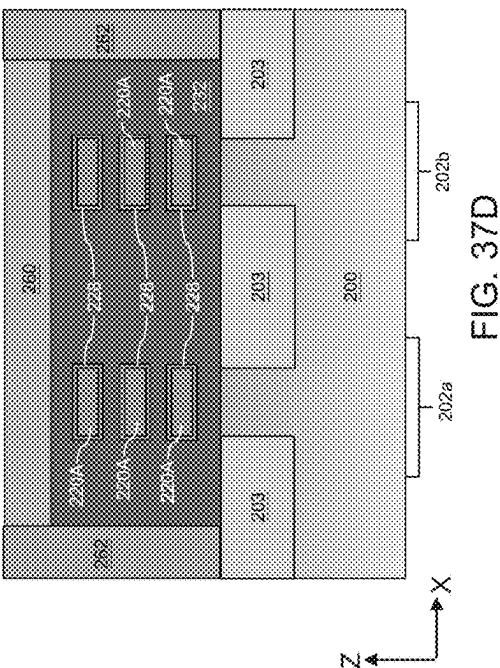
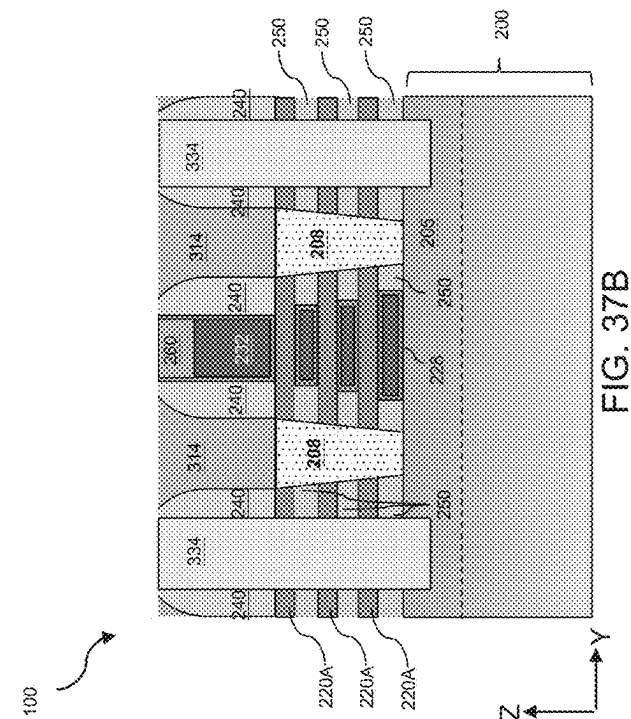


FIG. 36C



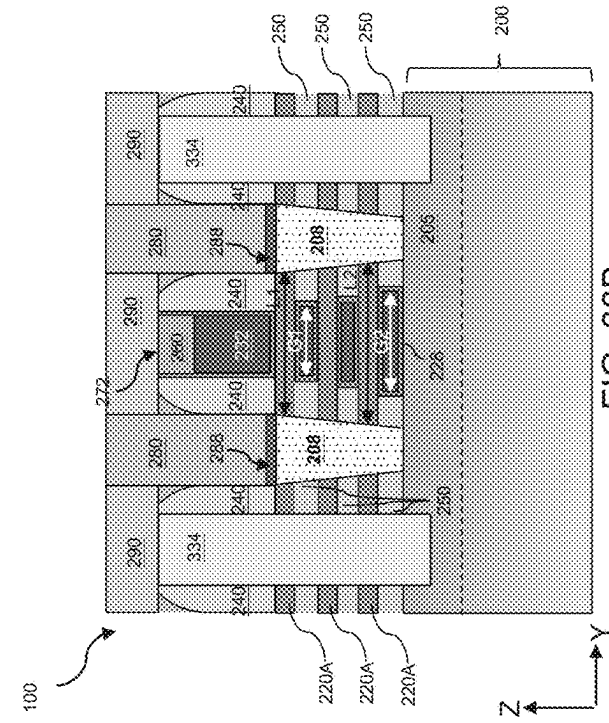


FIG. 38B

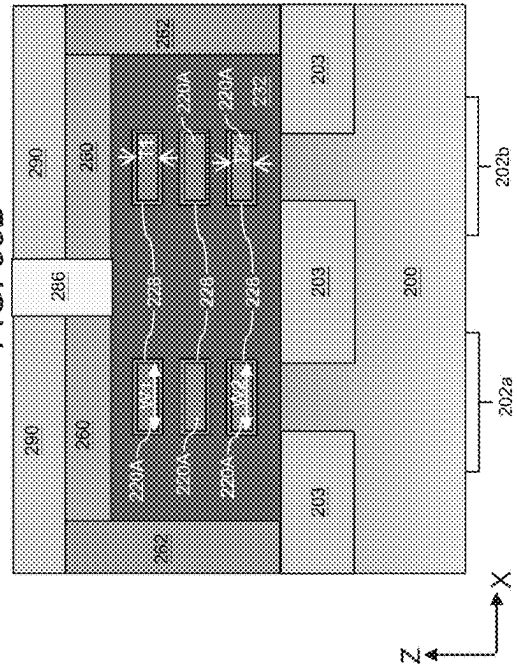


FIG. 38D

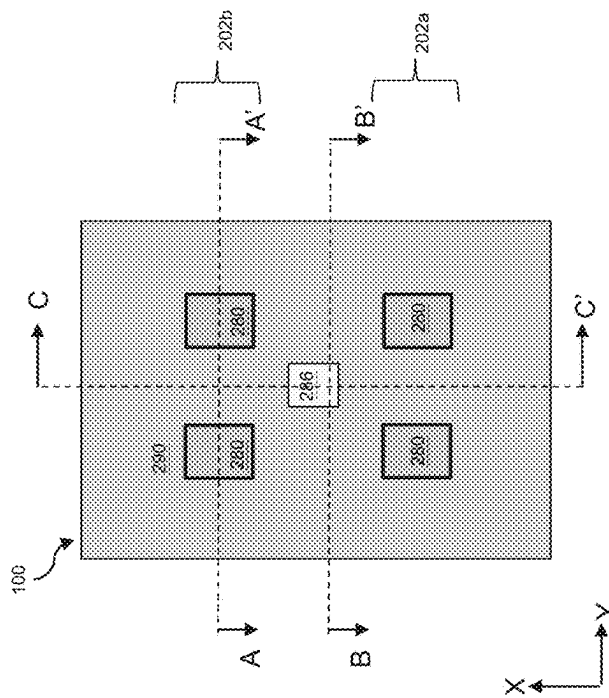


FIG. 38A

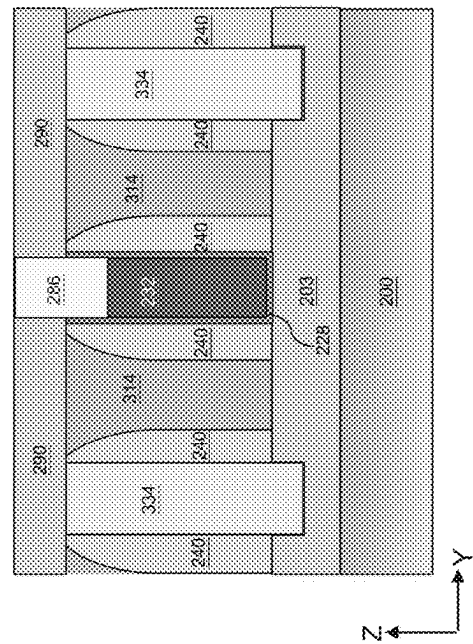


FIG. 38C

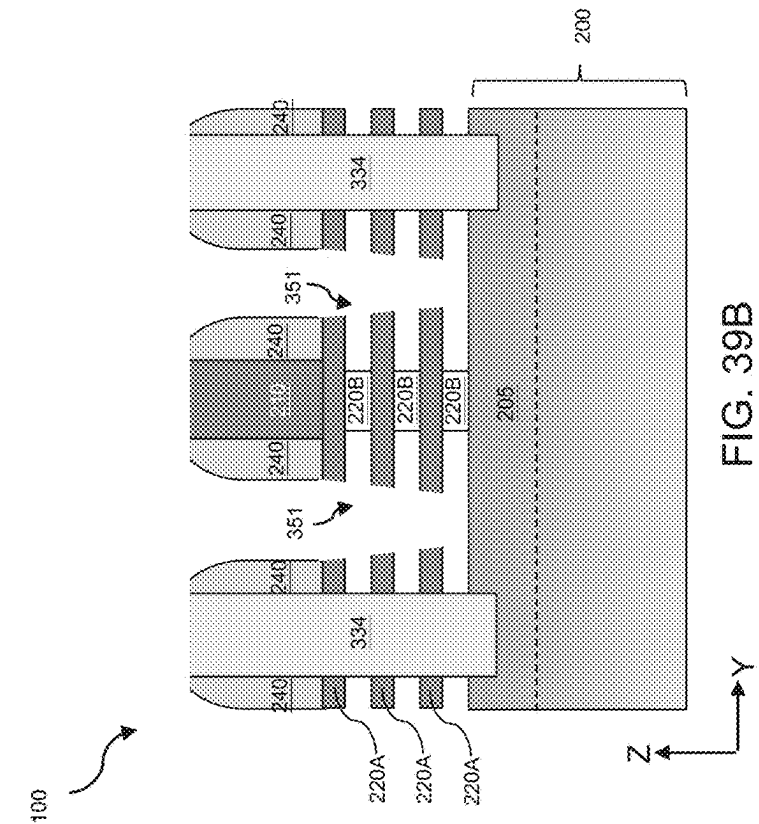


FIG. 39A

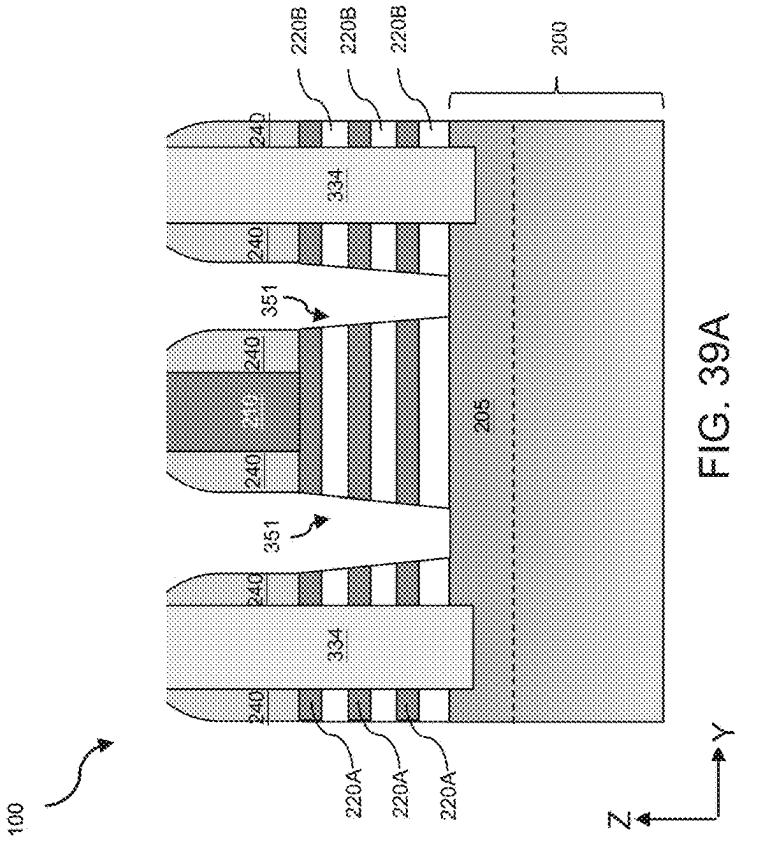
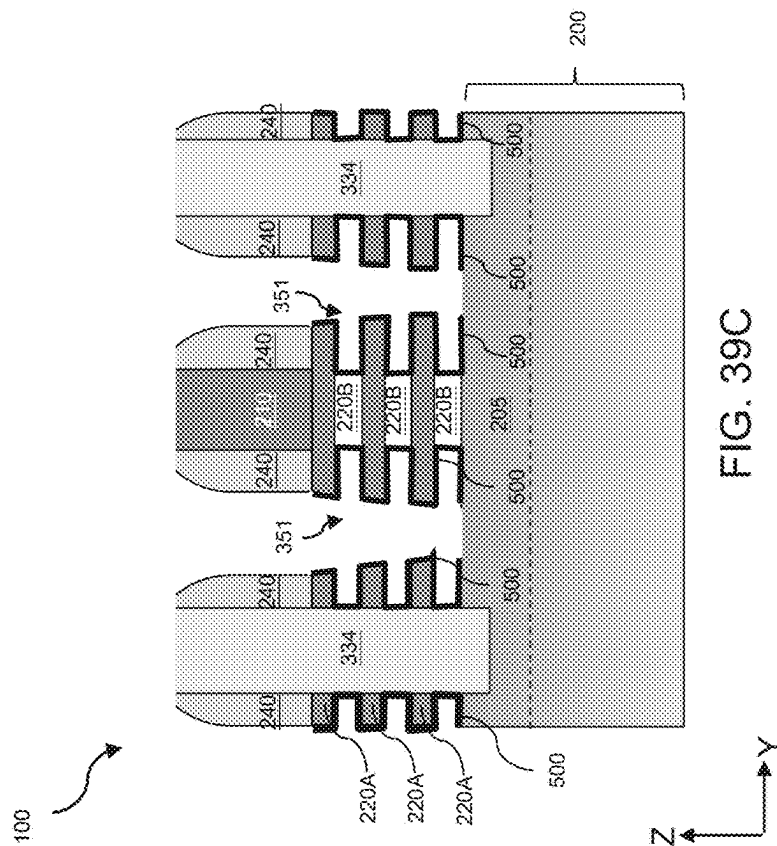
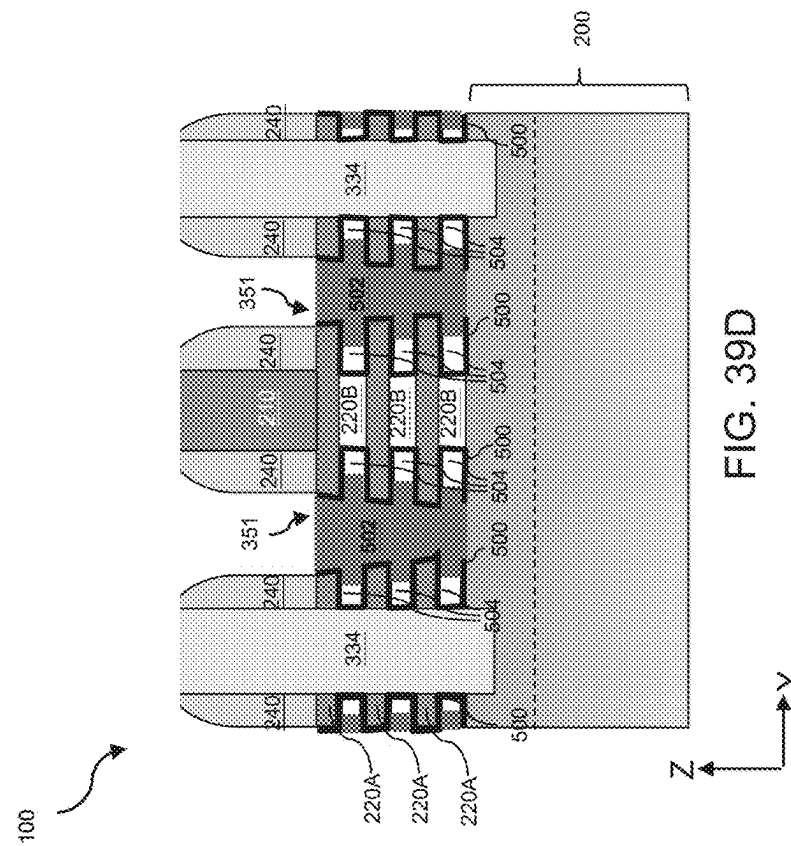
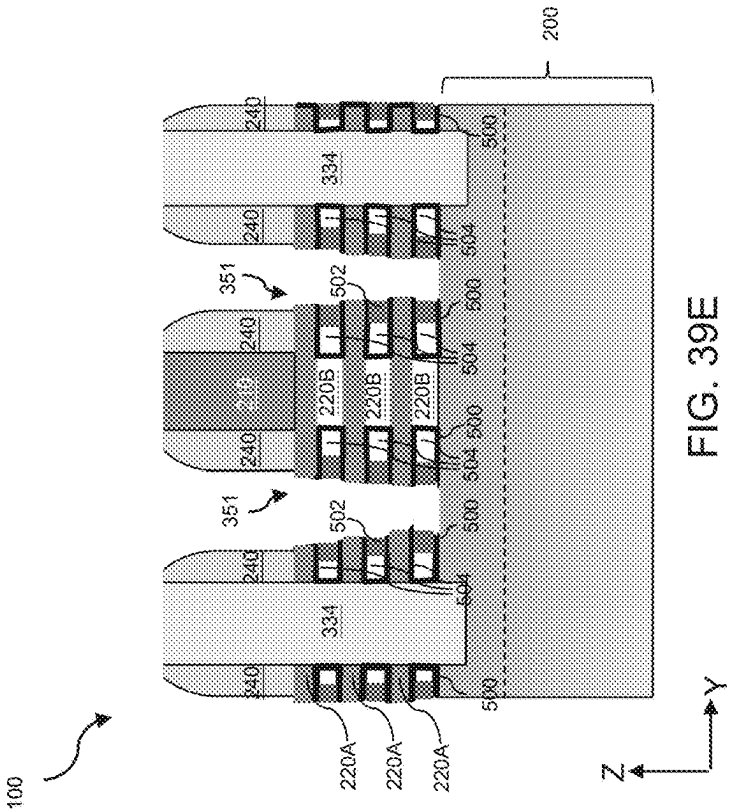
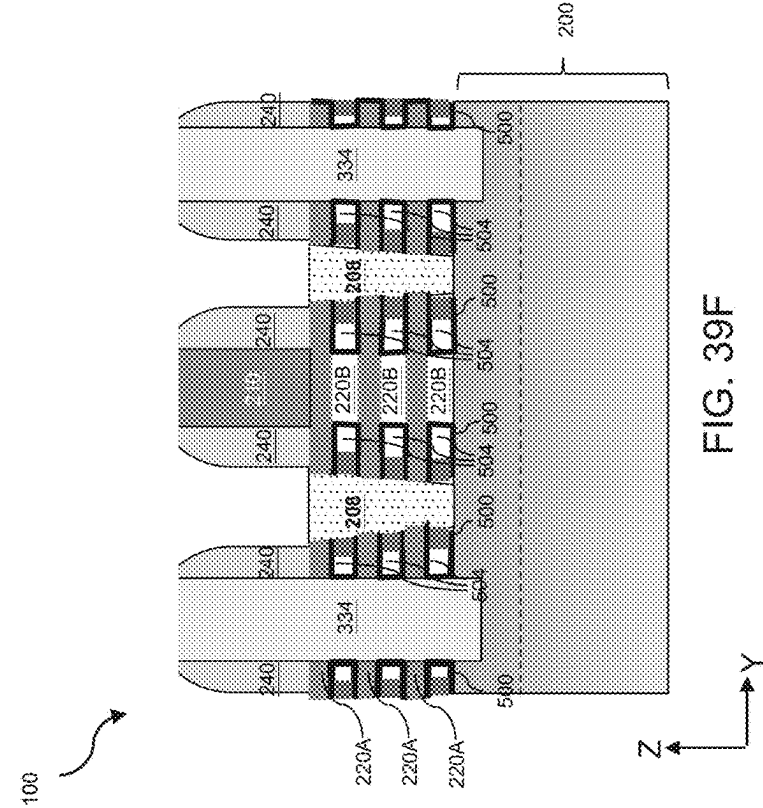
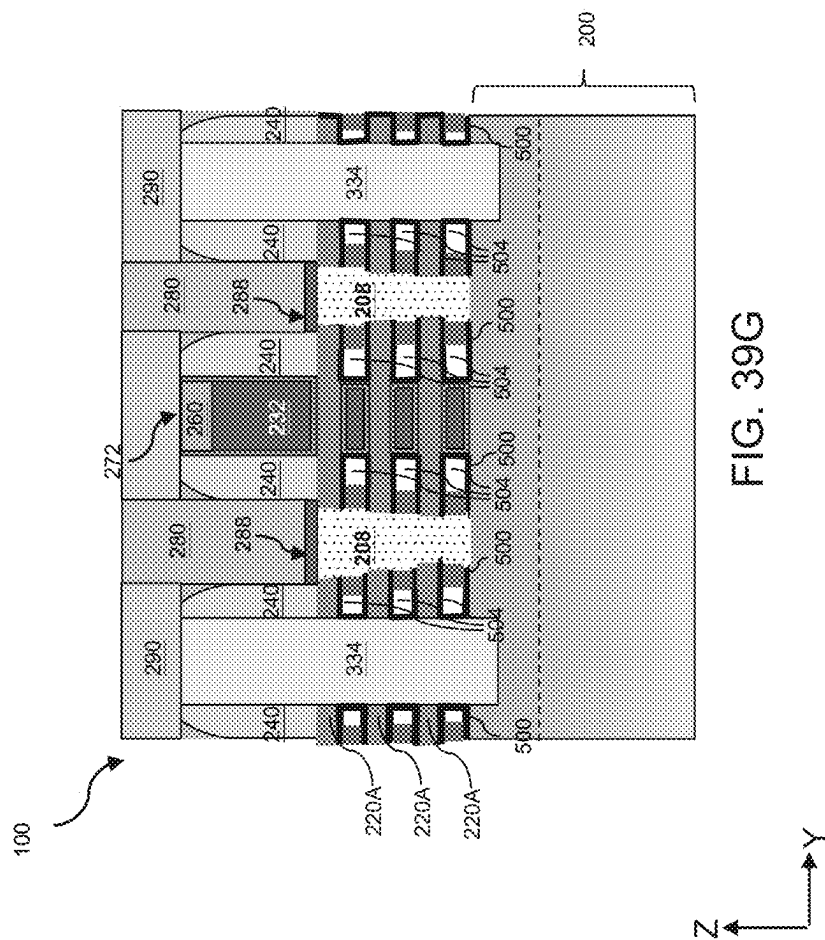


FIG. 39B







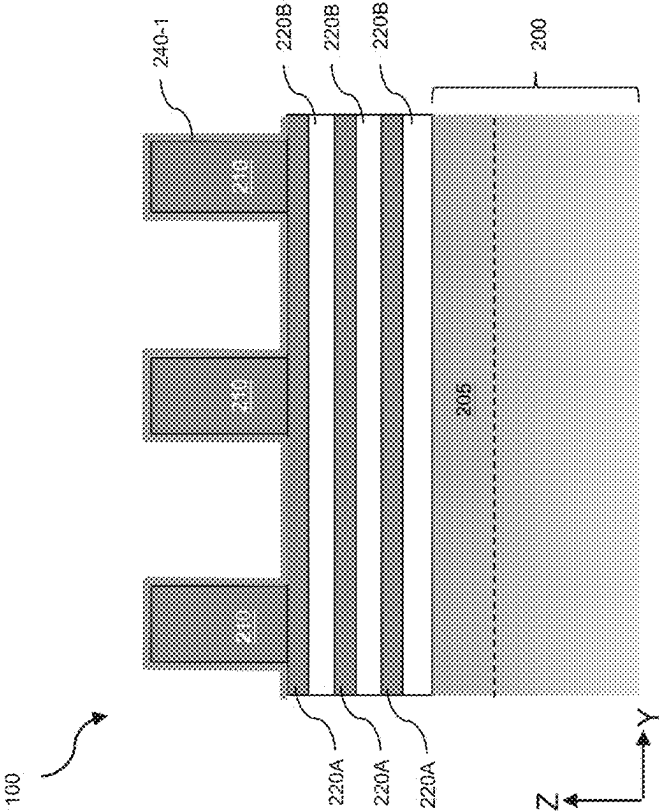


FIG. 40A

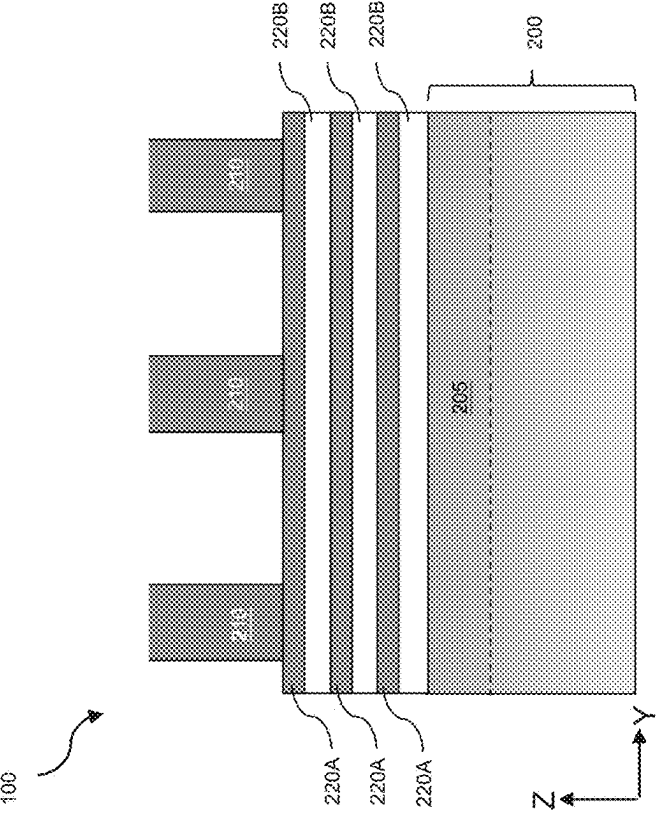
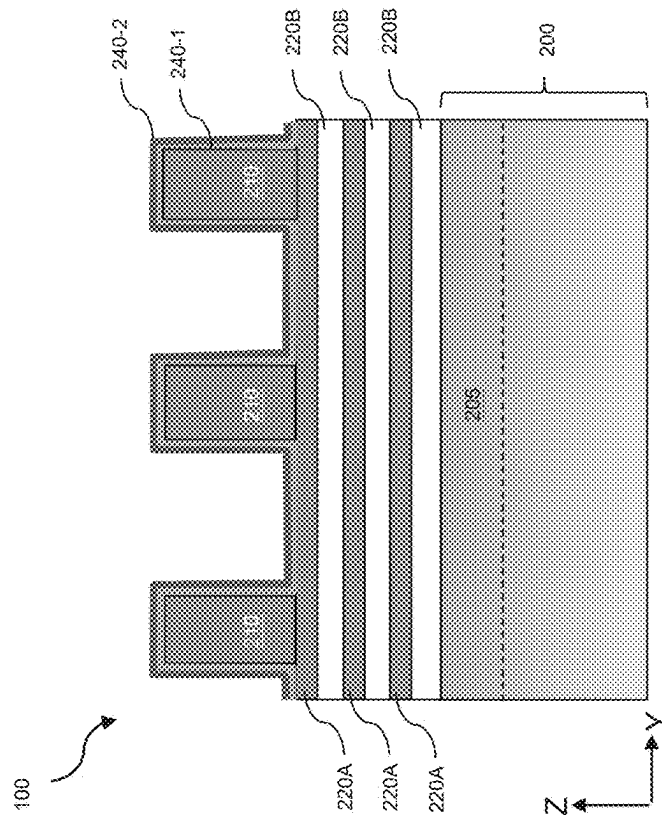
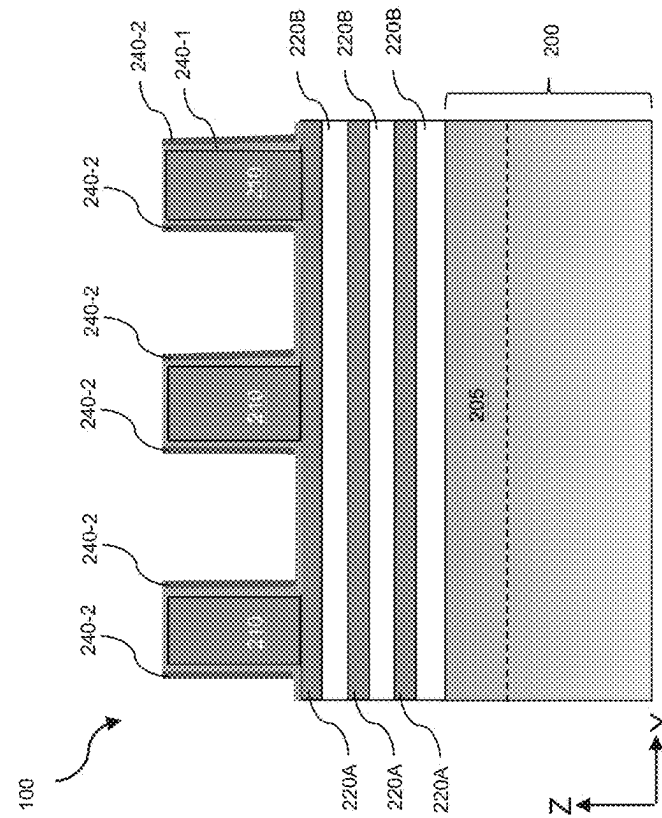


FIG. 40B



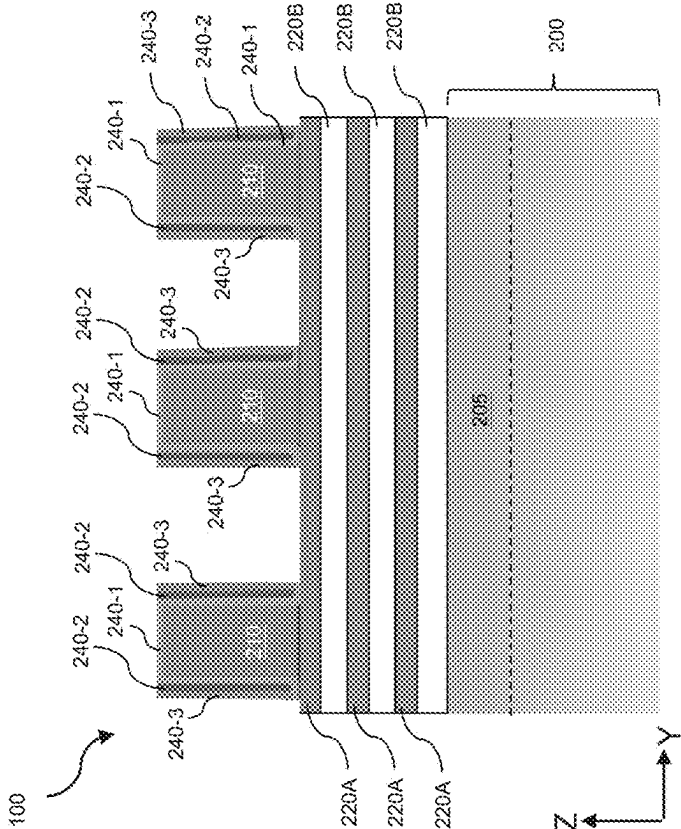


FIG. 40E

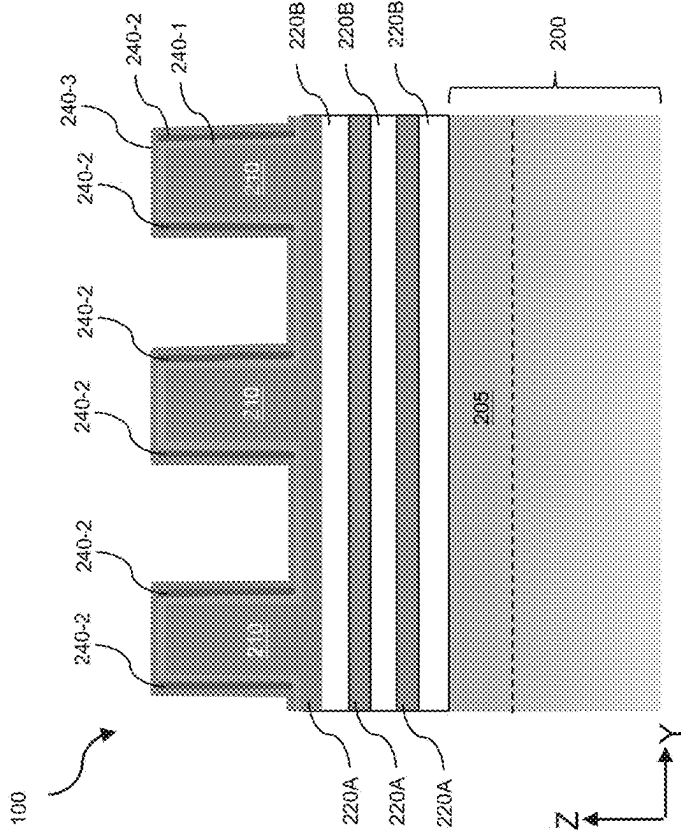


FIG. 40F

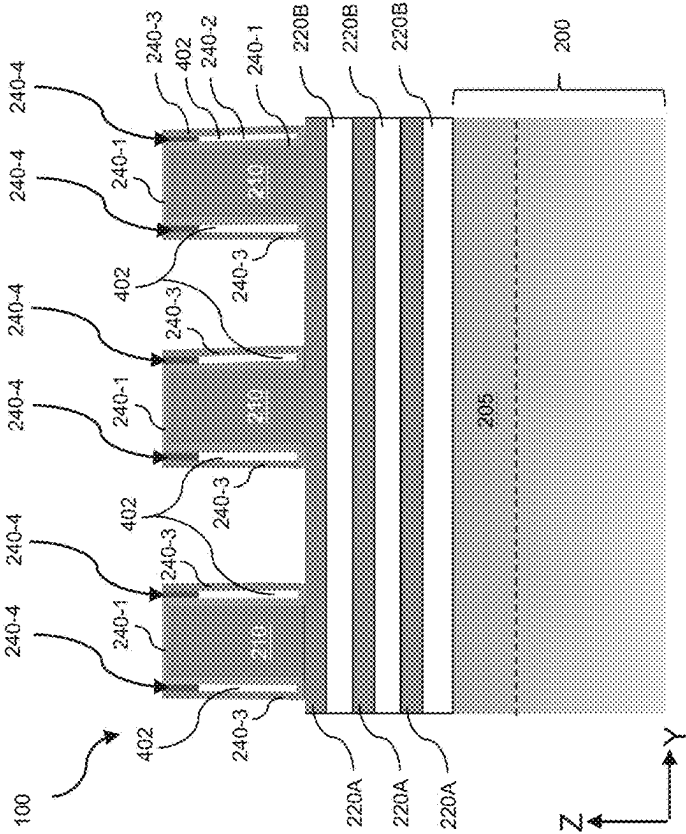


FIG. 40H

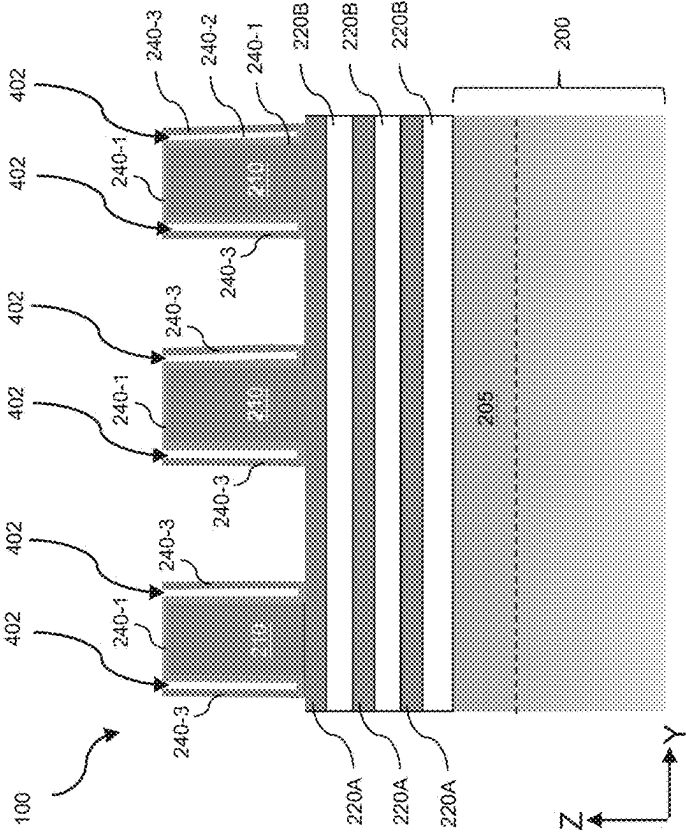


FIG. 40G

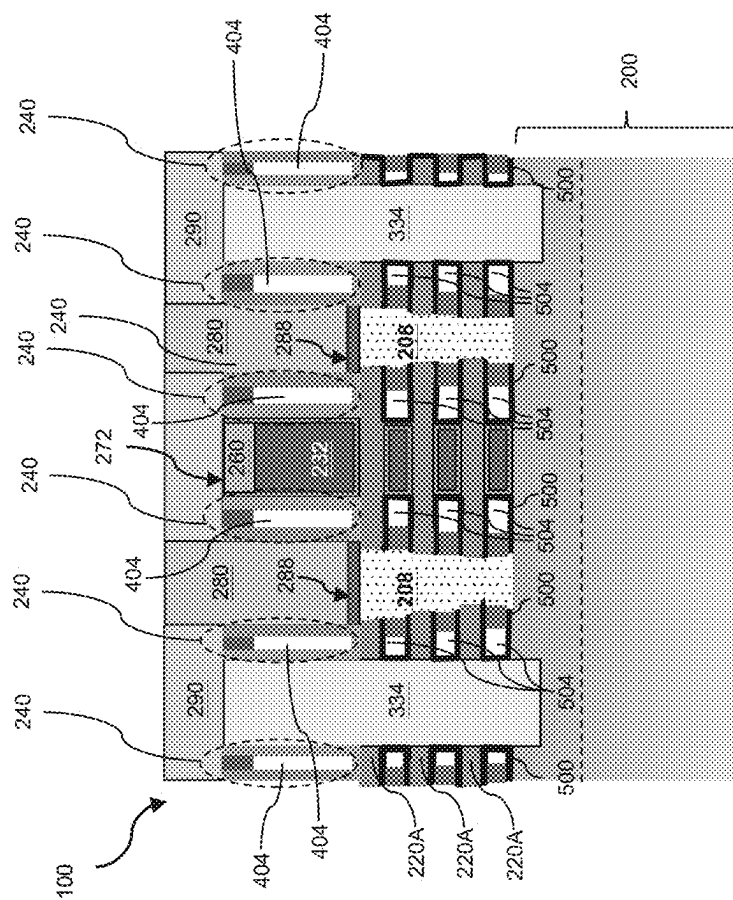


FIG. 40I

1

STRUCTURE AND METHOD FOR SEMICONDUCTOR DEVICES

This application is a Continuation of U.S. patent application Ser. No. 17/590,409, filed Feb. 1, 2022, which is a Divisional of U.S. patent application Ser. No. 16/585,636, filed Sep. 27, 2019, the entire disclosures of which are incorporated herein by reference.

BACKGROUND

The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of IC processing and manufacturing, and for these advancements to be realized, similar developments in IC processing and manufacturing are needed.

For example, multi-gate devices have been introduced in an effort to improve gate control by increasing gate-channel coupling, reduce OFF-state current, and reduce short-channel effects (SCEs). One such multi-gate device is a gate-all-around (GAA) transistor, whose gate structure extends around its channel region, thereby providing access to the channel region on all sides. Such GAA transistors are compatible with conventional complementary metal-oxide-semiconductor (CMOS) processes, allowing them to be aggressively scaled down while maintaining gate control and mitigating SCEs. However, conventional methods for GAA devices may experience challenges, including poor epitaxial growth in the source/drain region, small formation margin for gate dielectric and electrode in the narrow channel-channel spaces, and increased capacitance between adjacent conductive regions, such as the source/drain region and active gate structure, especially as device size is scaled down. Therefore, although conventional GAA devices have been generally adequate for their intended purposes, they are not satisfactory in every respect.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIGS. 1A, 1B, and 1C are flow charts of an example method for fabricating an embodiment of a GAA device according to some embodiments of the present disclosure;

FIGS. 2A to 38A are top views of embodiments of GAA devices of the present disclosure constructed at various fabrication stages according to some embodiments of the present disclosure;

FIGS. 2B to 38B are cross sectional views of embodiments of GAA devices of the present disclosure along the line A-A' in FIGS. 2A to 38A, respectively, according to some embodiments of the present disclosure;

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FIGS. 2C to 38C are cross sectional views of an embodiment of a GAA device of the present disclosure along the line B-B' in FIGS. 2A to 38C, respectively, according to some embodiments of the present disclosure;

FIGS. 2D to 38D are cross sectional views of an embodiment of a GAA device of the present disclosure along the line C-C' in FIGS. 2A to 38A, respectively, according to some embodiments of the present disclosure;

FIGS. 39A to 39G and 40A to 40I are cross-sectional views of example methods for fabricating various embodiments of a GAA device according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Still further, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term is intended to encompass numbers that are within a reasonable range including the number described, such as within +/-10% of the number described or other values as understood by person skilled in the art. For example, the term “about 5 nm” may encompass the dimension range from 4.5 nm to 5.5 nm.

Multi-gate devices (e.g. gate-all-around (GAA) devices) have been introduced in an effort to improve gate control by increasing gate-channel coupling, reduce OFF-state current, and reduce short-channel effects (SCEs). GAA devices can be aggressively scaled down while maintaining gate control and mitigating SCEs. However, conventional methods for GAA devices may experience challenges, including poor epitaxial growth in the source/drain region, small formation margin for gate dielectric and electrode in the narrow channel-channel spaces, and increased capacitance between adjacent conductive regions, such as a source/drain region and an adjacent active gate structure. These drawbacks are exacerbated as device size is scaled down.

The present disclosure is generally related to ICs and semiconductor devices and methods of forming the same. More particularly, the present disclosure is related to GAA

devices. A GAA device includes any device that has its gate structure, or portions thereof, formed around all-sides of a channel region (e.g. surrounding a portion of a channel region). In some instances, a GAA device may also be referred to as a quad-gate device where the channel region has four sides and the gate structure is formed on all four sides. The channel region of a GAA device may include one or more semiconductor layers, each of which may be in one of many different shapes, such as wire (or nanowire), sheet (or nanosheet), bar (or nano-bar), and/or other suitable shapes. In embodiments, the channel region of a GAA device may have multiple horizontal semiconductor layers (such as nanowires, nanosheets, or nano-bars) (hereinafter collectively referred to as “nanochannels”) vertically spaced, making the GAA device a stacked horizontal GAA device. The GAA devices presented herein may be a complementary metal-oxide-semiconductor (CMOS) GAA device, a p-type metal-oxide-semiconductor (pMOS) GAA device, or an n-type metal-oxide-semiconductor (nMOS) GAA device. Further, the GAA devices may have one or more channel regions associated with a single, contiguous gate structure, or multiple gate structures. One of ordinary skill may recognize other examples of semiconductor devices that may benefit from aspects of the present disclosure. For example, other types of metal-oxide semiconductor field effect transistors (MOSFETs), such as planar MOSFETs, FinFETs, other multi-gate FETs may benefit from the present disclosure. The GAA devices and methods of manufacture that are proposed in the present disclosure exhibit desirable properties, examples being: (1) a bottom-up epitaxial growth process that forms source/drain regions that are free from voids; (2) a large formation margin/window for gate dielectric and electrode in narrow channel-channel spaces; and (3) decreased capacitance between a source/drain region and an adjacent active gate structure.

In the illustrated embodiments, the IC device includes a GAA device **100**. The GAA device **100** may be fabricated during processing of the IC, or a portion thereof, that may include static random access memory (SRAM) and/or logic circuits, passive components such as resistors, capacitors, and inductors, and active components such as p-type field effect transistors (pFETs), n-type FETs (nFETs), FinFETs, MOSFETs, CMOS, bipolar transistors, high voltage transistors, high frequency transistors, other memory cells, and combinations thereof.

FIGS. 1A-1C are flow charts of an example method for fabricating an embodiment of a GAA device of the present disclosure according to some embodiments of the present disclosure. FIGS. 2A-27A are top views of an embodiment of a GAA device of the present disclosure constructed at various fabrication stages according to some embodiments of the present disclosure. FIGS. 2B-27B, 2C-27C, and 2D-27D are cross sectional views of an embodiment of a GAA device of the present disclosure along the lines A-A', B-B', and C-C' in FIGS. 2A-27A, respectively, according to some embodiments of the present disclosure.

Referring to block **810** of FIG. 1A and FIGS. 2A-2D, the GAA device **100** includes a substrate **200**. In some embodiments, the substrate **200** contains a semiconductor material, such as bulk silicon (Si). Alternatively or additionally, another elementary semiconductor, such as germanium (Ge) in a crystalline structure, may also be included in the substrate **200**. The substrate **200** may also include a compound semiconductor, such as silicon germanium (SiGe), silicon carbide (SiC), gallium arsenic (GaAs), gallium phosphide (GaP), indium phosphide (InP), indium arsenide (InAs), and/or indium antimonide (InSb), or combinations

thereof. The substrate **200** may also include a semiconductor-on-insulator substrate, such as Si-on-insulator (SOI), SiGe-on-insulator (SGOI), Ge-on-insulator (GOI) substrates. Portions of the substrate **200** may be doped, such as the doped portions **205**. The doped portions **205** may be doped with p-type dopants, such as boron (B) or boron fluoride (BF₃), or doped with n-type dopants, such as phosphorus (P) or arsenic (As). The doped portions **205** may also be doped with combinations of p-type and n-type dopants (e.g. to form a p-type well and an adjacent n-type well). The doped portions **205** may be formed directly on the substrate **200**, in a p-well structure, in an n-well structure, in a dual-well structure, or using a raised structure.

Referring to block **820** of FIG. 1A and FIGS. 2A-2D, a stack of semiconductor layers **220A** and **220B** are formed over the substrate **200** in an interleaving or alternating fashion, extending vertically (e.g. along the Z-direction) from the substrate **200**. For example, a semiconductor layer **220B** is disposed over the substrate **200**, a semiconductor layer **220A** is disposed over the semiconductor layer **220B**, another semiconductor layer **220B** is disposed over the semiconductor layer **220A**, so on and so forth. In the depicted embodiments, there are three layers of semiconductor layers **220A** and three layers of semiconductor layers **220B** alternating between each other. However, there may be any appropriate number of layers in the stack. For example, there may be 2 to 10 layers of semiconductor layers **220A**, alternating with 2 to 10 layers of semiconductor layers **220B** in the stack. The material compositions of the semiconductor layers **220A** and **220B** are configured such that they have an etching selectivity in a subsequent etching process. For example, in some embodiments, the semiconductor layers **220A** contain silicon germanium (SiGe), while the semiconductor layers **220B** contain silicon (Si). In some other embodiments, the semiconductor layers **220B** contain SiGe, while the semiconductor layers **220A** contain Si. In the depicted embodiment, each of the semiconductor layers **220A** has a substantially uniform thickness, depicted in FIG. 2B as the thickness **300**, while each of the semiconductor layers **220B** has a substantially uniform thickness, depicted in FIG. 2B as the thickness **310**.

Referring to block **820** of FIG. 1A and FIGS. 3A-3D, the stack of semiconductor layers **220A** and **220B** are patterned into a plurality of fin structures, for example, into fin structures (or fins) **130a** and **130b**. Each of the fins **130a** and **130b** includes a stack of the semiconductor layers **220A** and **220B** disposed in an alternating manner with respect to one another. The fins **130a** and **130b** each extends lengthwise (e.g. longitudinally) in a first direction (e.g. in the Y-direction) and are separated from each other (e.g. laterally) in a second direction (e.g. in the X-direction), as shown in FIGS. 3A and 3D. As illustrated in FIG. 3A, the fins may each have a lateral width along the X-direction, depicted in FIG. 3A as the width **350**. It is understood that the X-direction and the Y-direction are horizontal directions that are perpendicular to each other, and that the Z-direction is a vertical direction that is orthogonal (or normal) to a plane defined by the X-direction and the Y-direction. The substrate **200** may have its top surface aligned in parallel to the XY plane.

The fins **130a** and **130b** may be patterned by any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography

process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers, or mandrels, may then be used to pattern the fins. The patterning may utilize multiple etching processes which may include a dry etching and/or wet etching. The regions in which the fins are formed will be used to form active devices through subsequent processing and are thus referred to as active regions. For example, fin **130a** is formed in the active region **202a**, and the fin **130b** is formed in the active region **202b**. Both fins **130a** and **130b** protrude out of the doped portions **205**.

The structure **100** includes isolation features **203**, which may be shallow trench isolation (STI) features. In some examples, the formation of the isolation features **203** includes etching trenches into the substrate **200** between the active regions and filling the trenches with one or more dielectric materials such as silicon oxide, silicon nitride, silicon oxynitride, other suitable materials, or combinations thereof. Any appropriate methods, such as a chemical vapor deposition (CVD) process, an atomic layer deposition (ALD) process, a physical vapor deposition (PVD) process, a plasma-enhanced CVD (PECVD) process, a plasma-enhanced ALD (PEALD) process, and/or combinations thereof may be used for depositing the isolation features **203**. The isolation features **203** may have a multi-layer structure such as a thermal oxide liner layer over the substrate **200** and a filling layer (e.g., silicon nitride or silicon oxide) over the thermal oxide liner layer. Alternatively, the isolation features **203** may be formed using any other isolation formation techniques. As illustrated in FIG. 3D, the fins **130a** and **130b** are located above the top surface **203a** of the isolation features **203** (e.g. protrude out of the isolation features **203**) and are also located above the top surface **200a** of the substrate **200**.

Referring to block **830** of FIG. 1A and FIGS. 4A-4D, dummy gate structures **210** are formed over a portion of each of the fins **130a** and **130b**, and over the isolation features **203**, in between the fins **130a** and **130b**. The dummy gate structures **210** may be configured to extend lengthwise (e.g. longitudinally) in parallel to each other, for example, each along the X-direction, as shown in FIG. 4A. In some embodiments, as illustrated in FIG. 4D, each of the dummy gate structures wraps around the top surface and side surfaces of each of the fins **130a**, **130b**. The dummy gate structures **210** may include polysilicon. In some embodiments, the dummy gate structures **210** also include one or more mask layers, which are used to pattern the dummy gate electrode layers. The dummy gate structures **210** may undergo a gate replacement process through subsequent processing to form metal gates, such as a high-k metal gate, as discussed in greater detail below. Some of the dummy gate structures **210** may also undergo a second gate replacement process to form a dielectric based gate that electrically isolates the GAA device **100** from neighboring devices, as also discussed in greater detail below. The dummy gate structures **210** may be formed by a procedure including deposition, lithography patterning, and etching processes. The deposition processes may include CVD, ALD, PVD, other suitable methods, and/or combinations thereof.

Referring to block **840** of FIG. 1A and FIGS. 5A-5D, gate spacers **240** are formed on the sidewalls of the dummy gate structures **210**. The gate spacers **240** may include silicon nitride (Si_3N_4), silicon oxide (SiO_2), silicon carbide (SiC), silicon oxycarbide (SiOC), silicon oxynitride (SiON), sili-

con oxycarbon nitride (SiOCN), carbon doped oxide, nitrogen doped oxide, porous oxide, or combinations thereof. The gate spacers **240** may include a single layer or a multi-layer structure. In some embodiments, each of the gate spacers **240** may have a thickness **241** (e.g. measured in the Y-direction) in a range from about 3 nm to about 10 nm. A thickness within the stated range of values may be needed for device performance, especially for advanced technology nodes. In some embodiments, the gate spacers **240** may be formed by depositing a spacer layer (containing the dielectric material) over the dummy gate structures **210**, followed by an anisotropic etching process to remove portions of the spacer layer from the top surfaces of the dummy gate structures **210**. After the etching process, portions of the spacer layer on the sidewall surfaces of the dummy gate structures **210** substantially remain and become the gate spacers **240**. In some embodiments, the anisotropic etching process is a dry (e.g. plasma) etching process. Additionally or alternatively, the formation of the gate spacers **240** may also involve chemical oxidation, thermal oxidation, ALD, CVD, and/or other suitable methods. In the active regions, the gate spacers **240** are formed over the top layer of the semiconductor layers **220A**. Accordingly, the gate spacers **240** may also be interchangeably referred to as the top spacers **240**. In some examples, one or more material layers (not shown) may also be formed between the dummy gate structures **210** and the corresponding top spacers **240**. The one or more material layers may include an interfacial layer and/or a high-k dielectric layer, as examples.

Referring to block **850** of FIG. 1A and FIGS. 6A-6D, portions of the fins **130a** and **130b** exposed by the dummy gate structures **210** and the gate spacers **240** are at least partially recessed (or etched away) to form tapered trenches **151** for subsequent epitaxial source and drain growth. As described in greater detail below, the tapering of the trenches **151** is a deliberate feature of the proposed process, an effect being an efficient epitaxial growth process that prevents voids from being induced in the subsequently-formed source/drain regions. In effect, the tapered trenches **151** result in a bottom-up epitaxial growth process that conformally fills the tapered trenches **151**. The formation of the tapered trenches **151** exposes sidewalls of the stack of semiconductor layers **220A** and **220B**. In the depicted embodiments, an acute angle α subtended by a sidewall **151w** of the tapered trenches **151** and the top surface **200a** of the substrate may be in a range from about 80 degrees to about 88 degrees (e.g. about 85 degrees). In the examples shown in FIGS. 6A-6D, the bottom **151a** of the tapered trenches **151** is substantially aligned (e.g. substantially coplanar) with the top surface **200a** of the substrate **200**. Alternatively, in some other embodiments (not shown), the recess process removes only some, but not all, of the semiconductor layers **220A** and **220B**. In other words, the bottom **151a** of the tapered trenches **151** is located above the top surface **200a** of the substrate **200** (e.g. in the Z-direction). In yet some other embodiments (not shown), the recess process may remove not only the exposed fins **130a** and **130b**, but also remove a portion of the underlying doped region **205** of the substrate **200**. In other words, in such embodiments, the bottom **151a** of the tapered trenches **151** may be located below the top surface of the substrate **200** (e.g. in the Z-direction).

In the depicted embodiments (e.g. as seen in FIG. 6B), the remaining stack of semiconductor layers **220A** and **220B** includes two regions—a first region that is vertically beneath the dummy gate structures **210** (referred to as the “center portions”) and a second region that is vertically beneath the

top spacers **240** (referred to as the “side portions”). Accordingly, the portion of the semiconductor layers **220A** vertically beneath the dummy gate structures **210** are referred to as the center portions **220A-center**; while the portions of the semiconductor layers **220A** vertically beneath the top spacers **240** and that extend laterally towards the tapered trenches **151** are referred to as the side portions **220A-side**. Similarly, the portion of the semiconductor layers **220B** vertically beneath the dummy gate structures **210** are referred to as the center portions **220B-center**; while the portions of the semiconductor layers **220B** vertically beneath the top spacers **240** and that extend laterally towards the tapered trenches **151** are referred to as the side portions **220B-side**.

The process used to form the tapered trenches **151** may include multiple lithography and etching steps, and may use any suitable methods, such as dry etching and/or wet etching. As an example, one or more of the multiple lithography and etching steps used to form the tapered trenches **151** may include a first etch process having a first etch chemistry and a second etch process having a second etch chemistry that is different from the first etch chemistry. The first etch process may be a main-etch process that initially forms an opening in the stack of semiconductor layers **220A** and **220B**, while the second etch process may be an over-etch process that shapes the initially-formed opening to produce the tapered profile observed in the trenches **151**. The first etch chemistry may include hydrogen bromide (HBr) combined with argon (Ar), helium (He), oxygen (O₂), or a combination thereof. The second etch chemistry may include hydrogen bromide (HBr) combined with nitrogen, methane (CH₄), or a combination thereof. The second etch process (e.g. the over-etch process) may be performed at a high bias power (e.g. a bias power in a range from about 150 Watts to about 600 Watts).

Referring to block **860** of FIG. **1A** and FIGS. **7A-7D**, portions of the semiconductor layers **220B** are removed through the exposed sidewall surfaces in the tapered trenches **151** via a selective etching process. The selective etching process may be any suitable processes, such as a wet etching or a dry etching process. The extent to which the semiconductor layers **220B** are recessed (or the size of the portion removed) is determined by the processing conditions such as the duration the semiconductor layers **220B** is exposed to an etching chemical. In the depicted embodiments, the duration is controlled such that the side portions **220B-side** are removed in their entirety, while the center portions **220B-center** remain substantially unchanged. In other words, the remaining portions of the semiconductor layers **220B** each has a sidewall that is substantially aligned with a sidewall of the dummy gate structures **210** (e.g. the sidewall in the XZ plane, defined by the X-direction and the Z-direction). As illustrated in FIG. **7B**, the selective etching process creates openings **161**, which extend the trenches **151** into areas beneath the semiconductor layers **220A** and top spacers **240**. The openings **161** are referred to as “first gaps” in block **860** of FIG. **1A**.

Meanwhile, the semiconductor layers **220A** are only slightly affected during the selective etching process. For example, prior to the selective etching process, the side portions **220A-side** each has a thickness **300**, and side portions **220B-side** each has a thickness **310** (see FIG. **2B**). After the selective etching process, the side portions **220A-side** have a thickness **305**, and the openings **161** have a height **315** (or interchangeably referred to as thickness **315**). Thickness **305** is only slightly smaller than thickness **300**, and thickness **315** is only slightly larger than thickness **310**. For example, thickness **305** may be about 1% to 10%

smaller than thickness **300**; and thickness **315** may be about 1% to 10% larger than thickness **310**. The etch selectivity between the semiconductor layers **220A** and **220B** is made possible by the different material compositions between these layers. For example, the semiconductor layers **220B** may be etched away at a substantially faster rate (e.g. more than about 5 times to about 10 times faster) than the semiconductor layers **220A**.

As discussed above, the selective etching process may be a wet etching process. In an embodiment, the semiconductor layers **220A** includes Si and the semiconductor layers **220B** includes SiGe. In such an embodiment, a Standard Clean 1 (SC-1) solution may be used to selectively etch away the SiGe semiconductor layers **220B**. For example, the SiGe semiconductor layers **220B** may be etched away at a substantially faster rate than the Si semiconductor layers **220A**. As a result, desired portions of the semiconductor layers **220B** (e.g. the side portions **220B-side**) are removed, while the semiconductor layers **220A** remain substantially unchanged. The SC-1 solution includes ammonia hydroxide (NH₄OH), hydrogen peroxide (H₂O₂), and water (H₂O). The etching duration is adjusted such that the size of the removed portions of SiGe layers are controlled. The optimal condition may be reached by additionally adjusting the etching temperature, dopant concentration, as well as other experimental parameters.

In another embodiment, the semiconductor layers **220A** include SiGe and the semiconductor layers **220B** includes Si. In such an embodiment, a cryogenic deep reactive ion etching (DRIE) process may be used to selectively etch away the Si semiconductor layer **220B**. For example, the DRIE process may implement a sulfur hexafluoride-oxygen (SF₆—O₂) plasma. The optimal condition may be reached by adjusting the etching temperature, the power of the Inductively Coupled Plasma (ICP) power source and/or Radio Frequency (RF) power source, the ratio between the SF₆ concentration and the O₂ concentration, the dopant (such as boron) concentrations, as well as other experimental parameters. For example, the etching rate of a Si semiconductor layer **220B** using a SF₆—O₂ plasma (with approximately 6% O₂) may exceed about 8 μm/min at a temperature of about -80° C.; while the SiGe semiconductor layers **220A** are not substantially affected during the process.

Referring to block **870** of FIG. **1A** and FIGS. **8A-8D**, a dielectric material **248** is deposited into both the trenches **151** and the openings **161**. The dielectric material **248** may be selected from SiO₂, SiON, SiOC, SiOCN, or combinations thereof. In some embodiments, the proper selection of the dielectric material **248** may be based on its dielectric constant. In an embodiment, this dielectric material **248** may have a dielectric constant lower than that of the top spacers **240**. In some other embodiments, this dielectric material **248** may have a dielectric constant higher than that of the top spacers **240**. This aspect of the dielectric material **248** will be further discussed later. The deposition of the dielectric material **248** may be any suitable methods, such as CVD, PVD, PECVD, MOCVD, ALD, PEALD, or combinations thereof. A chemical-mechanical polishing (CMP) process may be performed to planarize the top surfaces of the device **100**, and to expose the top surfaces of the dummy gate structures **210**. In the operation depicted in FIGS. **8A-8D**, the dielectric material **248** completely fills both the trenches **151** and the openings **161**.

Referring to block **880** of FIG. **1A** and FIGS. **9A-9D**, the dielectric material **248** is etched back such that the top surface **200a** of the substrate **200** is exposed. In the depicted embodiment, the etching-back is a self-aligned anisotropic

dry-etching process, such that the top spacers **240** are used as the masking element. Alternatively, a different masking element (e.g. a photoresist) may be used. The etching-back process may be similar to the process described above in reference to FIGS. **6A-6D** where formation of the tapered trenches **151** was described. The etching-back process removes the dielectric materials **248** within the tapered trenches **151** but does not substantially affect the dielectric materials **248** within the openings **161**. As a result, the dielectric material **248** filling the openings **161** become inner spacers **250**. In other words, the inner spacers **250** are formed between vertically adjacent (e.g. along in the Z-direction) side portions **220A**-side of the semiconductor layers **220A** (see FIG. **9B**). In the present embodiment, the inner spacers **250** are only present in the active regions. As illustrated in FIG. **9C**, no inner spacers **250** are present over the isolation features **203**. Rather, only top spacers **240** are present over the isolation features **203**. As illustrated in FIG. **9B**, the sidewall surfaces of the inner spacers **250**, the top spacers **240**, and side surfaces of the semiconductor layers **220A** form continuous sidewall surfaces **171**. In other words, the continuous sidewall surfaces **171** include both exposed side surfaces of semiconductor materials from the semiconductor layers **220A** and exposed side surfaces of dielectric material from the top spacers **240** and the inner spacers **250**. Furthermore, due to the tapered profile of the sidewalls of continuous sidewall surfaces **171**, a distance between horizontally adjacent portions of the semiconductor layer **220A** (e.g. along the Y-direction) decreases from a mouth of the trench **151** to the bottom **151a** of the trench **151**. For example, in FIG. **9B**, the distance **D3** between horizontally adjacent portions of the semiconductor layer **220A** at the mouth of the trench **151** is greater than the distance **D2** between horizontally adjacent portions of the semiconductor layer **220A** at a middle region of the trench **151**. Similarly, the distance **D2** is greater than the distance **D1** between horizontally adjacent portions of the semiconductor layer **220A** near the bottom **151a** of the trench **151**.

Referring to block **890** of FIG. **1B** and FIGS. **10A-10D**, the method **800** continues to forming epitaxial source/drain features **208** in the trenches **151**. In some embodiments, one source/drain feature is a source electrode, and the other source/drain feature is a drain electrode. The semiconductor layers **220A** that extend from one source/drain feature **208** to the other source/drain feature **208** may form channels of the GAA device **100**. Multiple processes including etching and growth processes may be employed to grow the epitaxial source/drain features **208**. In the depicted embodiment, the epitaxial source/drain features **208** have top surfaces that are substantially aligned with the top surface of the topmost semiconductor layer **220A**. However, in other embodiments, the epitaxial source/drain features **208** may alternatively have top surfaces that extend higher than the top surface of the topmost semiconductor layer **220A** (e.g. in the Z-direction). In the depicted embodiment, the epitaxial source/drain features **208** occupy a lower portion of the trenches **151** (e.g. the portion defined by the inner spacers **250** and the semiconductor layers **220A**), leaving an upper portion of the trenches **151** (e.g. the portion defined by the top spacers **240**) open. In some embodiments, the epitaxial source/drain features **208** may merge together, for example, along the X-direction, to provide a larger lateral width than an individual epitaxial feature. In the depicted embodiments, as shown in FIG. **10A**, the epitaxial source/drain features **208** are not merged.

The epitaxial source/drain features **208** may include any suitable semiconductor materials. For example, the epitaxial

source/drain features **208** in an n-type GAA device may include Si, SiC, SiP, SiAs, SiPC, or combinations thereof; while the epitaxial source/drain features **208** in a p-type GAA device may include Si, SiGe, Ge, SiGeC, or combinations thereof. The source/drain features **208** may be doped in-situ or ex-situ. For example, the epitaxially grown Si source/drain features may be doped with carbon to form silicon:carbon (Si:C) source/drain features, phosphorous to form silicon:phosphor (Si:P) source/drain features, or both carbon and phosphorous to form silicon carbon phosphor (SiCP) source/drain features; and the epitaxially grown SiGe source/drain features may be doped with boron. One or more annealing processes may be performed to activate the dopants in the epitaxial source/drain features **208**. The annealing processes may include rapid thermal annealing (RTA) and/or laser annealing processes.

The epitaxial source/drain features **208** directly interface with the continuous sidewall surfaces **171**. During the epitaxial growth, semiconductor materials grow from the exposed top surface **200a** of the substrate **200** (e.g. the exposed top surface of doped region **205**) as well as the exposed side surfaces of the semiconductor layers **220A**. It is noted that semiconductor materials do not grow from the surfaces of the inner spacers **250** and the top spacers **240** during the epitaxial growth process. Since the distance between horizontally adjacent portions of the semiconductor layer **220A** decreases from the mouth of the trench **151** to the bottom **151a** of the trenches **151**, the epitaxial growth process fills up the bottom of the trench **151** prior to the top of the trenches **151**. Consequently, the tapered profile of the trenches **151** causes the epitaxial growth process to be a bottom-up conformal epitaxial growth process that fills the tapered trenches **151**, thereby preventing voids from being formed in the epitaxial source/drain features **208**.

Referring to block **900** of FIG. **1B** and FIGS. **11A-11D**, an interlayer dielectric (ILD) layer **214** is formed over the epitaxial source/drain features **208** in the remaining spaces of the trenches **151**, as well as vertically over the isolation features **203**. The ILD layer **214** may also be formed in between the adjacent gates **210** along the Y-direction, and in between the source/drain features **208** along the X-direction. The ILD layer **214** may include a dielectric material, such as a high-k material, a low-k material, or an extreme low-k material. For example, the ILD layer **214** may include SiO₂, SiOC, SiON, or combinations thereof. The ILD layer **214** may include a single layer or multiple layers, and may be formed by a suitable technique, such as CVD, ALD, and/or spin-on techniques. After forming the ILD layer **214**, a CMP process may be performed to remove excessive portions of the ILD layer **214**, thereby planarizing the top surface of the ILD layer **214**. Among other functions, the ILD layer **214** provides electrical isolation between the various components of the GAA device **100**.

Referring to block **910** of FIG. **1B** and FIGS. **12A-12D**, the dummy gate structures **210** are selectively removed through any suitable lithography and etching processes. In some embodiments, the lithography process may include forming a photoresist layer (resist), exposing the resist to a pattern, performing a post-exposure bake process, and developing the resist to form a masking element, which exposes a region including the dummy gate structures **210**. Then, the dummy gate structures **210** are selectively etched through the masking element. In some other embodiments, the top spacers **240** may be used as the masking element or a part thereof. For example, the dummy gate structures **210** may include polysilicon, the top spacers **240** and the inner spacers **250** may include dielectric materials, and the semi-

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conductor layers **220A-center** includes a semiconductor material. Therefore, an etch selectivity may be achieved by selecting appropriate etching chemicals, such that the dummy gate structures **210** may be removed without substantially affecting the features of the GAA device **100**. The removal of the dummy gate structures **210** creates gate trenches **153**. The gate trenches **153** expose the top surfaces and the side surfaces of the stack of semiconductor layers **220A**, **220B**, as depicted in FIG. 12D. In other words, the center portions **220A-center** and **220B-center** are exposed at least on two side surfaces in the gate trenches **153**. Additionally, the gate trenches **153** also expose the top surfaces of the isolation features **203**.

Referring to block **920** of FIG. 1B and FIGS. 13A-13D, any remaining center portions **220B-center** are also selectively removed through the gate trenches **153**, for example using wet or dry etching process. The etching chemical is selected such that the center portions **220B-center** has a sufficiently different etching rate as compared to the center portions **220A-center** and the inner spacers **250**. As a result, the center portions **220A-center** and the inner spacers **250** remain substantially unchanged. This selective etching process may include one or more etching steps.

As illustrated in FIGS. 13A-13D, in the present embodiment, the removal of the semiconductor layers **220B** forms suspended semiconductor layers **220A-center** and openings **157** in between the vertically adjacent layers (e.g. in the Z-direction), thereby exposing the top and bottom surfaces of the center portions **220A-center**. Each of the center portions **220A-center** are now exposed circumferentially in the X-Z plane. In addition, the portion of the doped regions **205** beneath the center portions **220A-center** are also exposed in the openings **157**. In some other embodiments however, the removal process only removes some but not all of the center portions **220B-center**.

In the examples depicted in FIGS. 12A-12D and FIGS. 13A-13D, the gate trench **153** and the opening **157** vertically adjacent to the gate trench **153** (e.g. in the Z-direction) collectively form an opening having a vertical profile. In other words, the opening collectively formed by the gate trench **153** and its corresponding opening **157** have vertical sidewalls. In some embodiments, such openings having the vertical sidewalls may be formed by a plurality of etch processes. For example, the etch chemistry of the etch process used to remove the dummy gate structures **210** and thereby form the gate trenches **153** (e.g. in FIGS. 12A-12D) may include hydrogen bromide (HBr) combined with chlorine (Cl₂), tetrafluoromethane (CF₄), oxygen, or a combination thereof. Furthermore, the etch process used to selectively remove the semiconductor layers **220B** and thereby form the openings **157** (e.g. in FIGS. 13A-13D) may have an initial etch chemistry including hydrogen bromide (HBr) combined with chlorine (Cl₂), oxygen, or a combination thereof. This initial etch chemistry is followed by a subsequent etch chemistry including hydrogen bromide (HBr) combined with tetrafluoromethane (CF₄), oxygen, or a combination thereof that induces the vertical profile of the opening collectively formed by the gate trench **153** and its corresponding opening **157**. As described in further detail below, in other embodiments, however, the opening collectively formed by a gate trench **153** and its corresponding opening **157** may have a tapered profile. Such a tapered profile may be achieved by omitting the above-described subsequent etch chemistry that includes hydrogen bromide (HBr) combined with tetrafluoromethane (CF₄), oxygen, or

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a combination thereof. In such examples, a gate structure that is subsequently formed in the tapered opening has a tapered profile as well.

Referring to blocks **930** and **940** of FIG. 1B, FIGS. 14A-14D, and FIGS. 15A-15D, a gate structure is formed. The gate structure includes a gate dielectric layer and a gate electrode disposed over the gate dielectric layer. For example, the gate structure may include a polysilicon gate electrode over a SiON gate dielectric layer. As another example, the gate structure may include a metal gate electrode over a high-k dielectric layer. In some instances, a refractory metal layer may interpose between the metal gate electrode (such as an aluminum gate electrode) and the high-k dielectric layer. As yet another example, the gate structure may include silicide. In the depicted embodiment, the gate structures each includes a gate dielectric layer **228** and a gate electrode that includes one or more metal layers **230**, **232**. The gate dielectric layers **228** are formed between the metal layers **230**, **232** and the channels formed by the semiconductor layers **220A** (e.g. the center portions **220A-center**).

In some embodiments, the gate dielectric layers **228** are formed conformally on the device **100** (see FIGS. 14A-14D). The gate dielectric layers **228** at least partially fill the gate trenches **153**. In some embodiments, dielectric interfacial layers may be formed over the center portions **220A-center** of the semiconductor layers **220A** prior to forming the gate dielectric layers **228**. Such dielectric interfacial layers improve the adhesion between the center portions **220A-center** of the semiconductor layers **220A** and the gate dielectric layers **228**. In the examples depicted in this disclosure, such dielectric interfacial layers are omitted. Instead, in the embodiments shown, the gate dielectric layers **228** is formed around the exposed surfaces of each of the semiconductor layers **220A**, such that they wrap around the center portions **220A-center** of each of the semiconductor layers **220A** in 360 degrees. Additionally, the gate dielectric layers **228** also directly contact vertical sidewalls of the inner spacers **250** and vertical sidewalls of the top spacers **240**. The gate dielectric layers **228** may include a dielectric material having a dielectric constant greater than a dielectric constant of SiO₂, which is approximately 3.9. For example, the gate dielectric layers **228** may include hafnium oxide (HfO₂), which has a dielectric constant in a range from about 18 to about 40. As various other examples, the gate dielectric layers **228** may include ZrO₂, Y₂O₃, La₂O₃, Gd₂O₃, TiO₂, Ta₂O₅, HfErO, HfLaO, HfYO, HfGdO, HfAlO, HfZrO, HfTiO, HfTaO, SrTiO, or combinations thereof. The formation of the gate dielectric layers **228** may be by any suitable processes, such as CVD, PVD, ALD, or combinations thereof.

Referring to block **940** of FIG. 1B and FIGS. 15A-15D, metal layers **230**, **232** are formed over the gate dielectric layers **228** to fill the remaining spaces of the gate trenches **153**. The metal layers **230**, **232** may include any suitable materials, such as titanium nitride (TiN), tantalum nitride (Ta₂N), titanium aluminide (TiAl), titanium aluminum nitride (TiAlN), tantalum aluminide (TaAl), tantalum aluminum nitride (TaAlN), tantalum aluminum carbide (TaAlC), tantalum carbonitride (TaCN), aluminum (Al), tungsten (W), copper (Cu), cobalt (Co), nickel (Ni), platinum (Pt), or combinations thereof. In some embodiments, a CMP is performed to expose a top surface of the ILD **214**. The dielectric layers **228** and the metal layers **230** collectively form the gate structures **270**, while the dielectric layers **228** and the metal layers **232** collectively form gate structure

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272. Each of the gate structures 270, 272 engages multiple layers within the center portions 220A-center (e.g. multiple nanochannels).

In some embodiments, a gate top hard mask layer 260 may optionally be formed over the gate structures 270, 272. For example, referring to FIGS. 16A-16D, the metal layers 230, 232 may optionally be recessed, such that a top surface of the metal layers 230, 232 extends below a top surface of the ILD 214. Subsequently, as illustrated in FIGS. 17A-17D, a gate top hard mask layer 260 is formed over the GAA device 100 such that it covers the gate structures 270, 272 (specifically, the metal layers 230, 232), the ILD layers 214, and fills the space created by the recess process. A CMP may be conducted to planarize the top surface of the gate top hard mask layer 260. In some embodiments, as illustrated in FIGS. 18A-18D, the CMP exposes the top surfaces of the ILD layers 214, the top surfaces of the top spacers 240, and the top surfaces of the gate dielectric layers 228. The gate top hard mask layers 260 may include a dielectric material, such as SiO₂, SiOC, SiON, SiOCN, nitride-based dielectric, metal oxide dielectric, HfO₂, Ta₂O₅, TiO₂, ZrO₂, Al₂O₃, Y₂O₃, or combinations thereof. The gate top hard mask layer 260 protects the gate structure 272 in the subsequent etching processes to form the source/drain contact features, and also insulates the gate structure 272. However, in some other embodiments (not shown), recessing of the metal layers 230, 232 and/or the formation of the gate top hard mask layers 260 is omitted.

Referring to block 950 of FIG. 1C and FIGS. 19A-19D, a mask layer 282 (e.g. a photoresist layer) is formed over the top surface of the device 100. The mask layer 282 may cover the main body (or the center portion) of the device 100 but not the two end portions 283 (along the X-direction) of the device 100. Referring to block 960 of FIGS. 1C and 20A-20D, an end-cut process is subsequently conducted. The end-cut process forms end-cut trenches 155, which split the gate structures 270, 272 along the X direction into individual gates. The individual gates may extend over an n-type region only (e.g. for an NMOS gate), over a p-type region only (e.g. for a PMOS gate), or over both an n-type region and a p-type region (e.g. for a CMOS gate). The end-cut process may include any suitable lithography and etching processes such that the end portions 283 are etched down to expose the isolation structure 203.

Referring to block 970 of FIG. 1C and FIGS. 21A-21D, a dielectric material is deposited into the end-cut trenches 155 to form the gate end dielectric features 262, which extends from a top surface of the isolation features 203 and fully covers an end of the gates, such as the gate structures 270, 272. The gate end dielectric features 262 may include a nitride-based dielectric material (e.g., Si₃N₄), a metal oxide, SiO₂, or combinations thereof. As described in further detail below, a subsequent step that replaces gate features 270 with dielectric based gates removes the top spacers 240 and the inner spacers 250 without substantially affecting the gate end dielectric features 262. Therefore, there needs to be sufficient etching selectivity between the gate end dielectric features 262 and the spacer layers (i.e., top spacers 240 and inner spacers 250). For example, the etching rate for the top spacers 240 and the inner spacers 250 in the etching chemical may be substantially faster than the etching rate for the gate end dielectric features 262 in the same solution, e.g. more than about 5 to 50 times faster. This difference in etching rate is a result of the different characteristics of the materials in these different layers, which may also be manifested in their different dielectric constants. In many embodiments, the gate end dielectric material may have a

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dielectric constant higher than both that of the top spacers 240 and that of the inner spacers 250. For example, the gate end dielectric features 262 may include a dielectric material with a dielectric constant larger than about 6.9 to about 7. For example, the gate end dielectric features 262 may include nitride. The nitride may have a dielectric constant larger than about 7.8 to about 8.0. On the other hand, the top spacers 240 and/or the inner spacers 250 may include oxide-based dielectric materials. For example, the top spacers 240 and/or the inner spacers 250 may include oxides with a dielectric constant in the range from about 3.9 to about 5.0. For another example, the top spacers 240 and/or the inner spacers 250 may include doped oxides, such as nitrogen-doped oxides and/or carbon-doped oxides. The nitrogen-doped oxide may have a dielectric constant between about 4 and about 5. The carbon-doped oxide may have a dielectric constant between about 3 and about 4. In some embodiments, the gate end dielectric features 262 may include a single layer. In some other embodiments, the gate end dielectric features 262 may include multiple layers, such as a nitride layer and an oxide layer.

Referring to block 980 of FIG. 1C and FIGS. 22A-22D, a mask layer 284 (e.g. a photoresist layer) is formed over the GAA device 100. In an embodiment, the mask layer 284 covers one or more gate structures 272 but does not cover one or more of the other gate structures 270. Subsequently, referring to block 990 of FIG. 1C and FIGS. 23A-23D, the exposed gate structures 270 are removed via any suitable processes to form gate trenches 159. As a result, the doped regions 205 as well as the isolation features 203 beneath the gate structures 270 are exposed in the gate trenches 159. The etching process may be a wet etching or a dry etching process, using the mask layer 284 as the masking elements. In the depicted embodiment, the etching process not only removes the exposed gate structures 270, but also removes the gate dielectric layer 228, portions of the top spacers 240, inner spacers 250, and semiconductor layers 220A, and partially recesses the doped region 205 of the substrate 200. However, in other embodiments, the removal of the gate dielectric layer 228 and/or the recess of the doped region 205 may be omitted. Alternatively or additionally, the sidewalls of the top spacers 240 may be used as masking elements.

As illustrated in block 1000 of FIG. 1C and FIGS. 24A-24D, the gate trenches 159 are filled with one or more dielectric materials to form the dielectric based gates 234. The dielectric materials may include SiO₂, SiOC, SiON, SiOCN, carbon-doped oxide, nitrogen-doped oxide, carbon-doped and nitrogen-doped oxide, dielectric metal oxides such as HfO₂, Ta₂O₅, TiO₂, ZrO₂, Al₂O₃, Y₂O₃, lanthanum-(La-) doped oxide, oxide doped with multiple metals, or combinations thereof. The dielectric based gates 234 may include a single layer or multiple layers. The formation processes may use any suitable processes, such as ALD, CVD, PVD, PEALD, PECVD, or combinations thereof. A CMP process may be performed to remove excessive dielectric materials and provide a top surface that is substantially coplanar with the ILD layer 214, the top spacers 240, and the gate end dielectric features 262.

Referring to block 1010 of FIG. 1C and FIGS. 25A-25D, a gate top dielectric layer 290 is formed over the GAA device 100. The gate top dielectric layer 290 may be formed by any suitable processes, such as CVD, PECVD, flowable CVD (FCVD), or combinations thereof. The gate top dielectric layer 290 covers top surfaces of the dielectric based gates 234, the ILD 214, the top spacers 240, the gate structure 272, and the gate top hard mask layer 260, if present. The gate top dielectric layer 290 may include a

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dielectric material, such as SiO₂, SiOC, SiON, SiOCN, nitride-based dielectric, metal oxide dielectric, HfO₂, Ta₂O₅, TiO₂, ZrO₂, Al₂O₃, Y₂O₃, or combinations thereof. The gate top dielectric layer **290** may have a thickness between about 3 nm and about 30 nm. A thickness within the stated range of values may be needed for device performance (e.g. to meet transistor switching speed requirements), especially for advanced technology nodes. In some embodiments, the gate top dielectric layer **290** protect the gate structure **272** in the subsequent etching processes to form the source/drain contact features, and also insulate the gate structure **272**.

Referring to block **1020** of FIG. **1C** and FIGS. **26A-26D**, a portion of the gate top dielectric layer **290** and ILD **214** are removed to form contact holes **278** over the epitaxial source/drain features **208**. Any appropriate methods may be used to form the contact holes **278**, such as multiple lithography and etching steps. In an embodiment, a self-aligned contact formation process may be utilized. For example, the ILD **214** may include a dielectric material that has an etching rate substantially faster than that of the top spacers **240** and that of the gate top hard mask layer **260**. Therefore, the top spacers **240** and the gate top hard mask layer **260** are not substantially affected when the ILD **214** is etched away to form the contact holes **278**. As the top spacers **240** and the gate top hard mask layer **260** protect the gate structure **272** from the etching chemical, the integrity of the gate structure **272** are preserved. The contact holes **278** expose the top surfaces of the epitaxial source/drain features **208** for subsequent contact layer formation. Additionally, a portion of the gate top dielectric layer **290** and the gate top hard mask layer **260** (if present) are also removed to form via holes **285** over the metal layers **232** of the gate structure **272**. The via holes **285** expose the metal layers **232** for subsequent via feature formation. Any appropriate methods may be used to form the via holes **285** and may include multiple lithography and etching steps.

Referring to block **1030** of FIG. **1C** and FIGS. **27A-27D**, contact features **280** are formed within the contact holes **278**. Accordingly, the contact features **280** are embedded within the gate top dielectric layer **290** and ILD **214**, and electrically connect the epitaxial source/drain features **208** to external conductive features (not shown). Additionally, via features **286** are also formed in the via holes **285**. Accordingly, the via features **286** are embedded within the gate top dielectric layer **290** (and within the gate top hard mask layer **260**, if present) and electrically connect the gate structure **272** to external conductive features (not shown). The contact features **280** and the via features **286** may each include Ti, TiN, TaN, Co, Ru, Pt, W, Al, Cu, or combinations thereof, respectively. Any suitable methods may be used to form the contact features **280** and the via features **286**. In some embodiments, additional features are formed in between the source/drain features **208** and the contacts **280**, such as self-aligned silicide features **288**. A CMP process may be performed to planarize the top surface of the GAA device **100**.

As discussed above, the dielectric constants for the top spacers **240** and the inner spacers **250** may be different. Whether the top spacer or the inner spacer should use a material with a lower dielectric constant may be a design choice. For example, the design choice may be made based on a comparison between the relative importance of the capacitance values of different device regions. For example, a designer may assign the material with the lower dielectric constant to the top spacer **240** rather than the inner spacer **250**. On the other hand, if it is more important to have a higher capacitance in the source/drain-metal gate region, the

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designer may assign the material with the lower dielectric constant to the inner spacer **250** rather than the top spacer **240**.

More specifically, the top spacer **240** may be considered to be the dielectric medium of a capacitor between a pair of vertically aligned conductive plates, i.e., the sidewall of the contact **280** and the sidewall of the gate structure **272**. Similarly, the inner spacer **250** may be considered to be the dielectric medium of another capacitor between another pair of vertically aligned conductive plates, i.e. the sidewall of the source/drain feature **208** and the sidewall of the gate structure **272**. The capacitance is proportional to the dielectric constant of the dielectric medium, according to the following equation:

$$C = \epsilon \frac{A}{d} = k\epsilon_0 \frac{A}{d}$$

wherein C is the capacitance of the capacitor, ϵ is the permittivity of the dielectric medium, ϵ_0 is the permittivity of vacuum, A is the area of the capacitor, d is the separation distance of the capacitor, and k is the dielectric constant of the dielectric medium. Therefore, a smaller dielectric constant leads to a smaller capacitance. If, according to the design needs, it is more important to have a higher capacitance in the contact-to-metal gate region than in the source/drain-to-metal gate region, the designer may assign the material with the lower k to the top spacer **240** rather than the inner spacer **250**. On the other hand, if it is more important to have a higher capacitance in the source/drain-metal gate region, the designer may assign the material with the lower k to the inner spacer **250** rather than the top spacer **240**. Referring to block **1040** of FIG. **1C**, additional layers and/or features may also be formed above and/or within the gate top dielectric layer **290** to complete the fabrication of the GAA device **100**.

Referring to FIGS. **27B** and **27D**, several structural features may be observed. First, as seen in FIG. **27B**, due to the tapered sidewalls of the epitaxial source/drain features **208**, the semiconductor layers **220A** that form the channels (e.g. nanochannels) of the GAA device **100** have different lengths along the Y-direction, depending on the position of the semiconductor layer **220A** in the Z-direction. For example, the semiconductor layer **220A** in closest proximity to the substrate **200** in the Z-direction has a length L2 along the Y-direction, while the semiconductor layer **220A** that is farthest from the substrate **200** in the Z-direction has a length L1 along the Y-direction, where the length L2 is greater than the length L1. In some embodiments, the length L2 is larger than the length L1 by at least 0.5 nm (e.g. by at least 1 nm). Lengths within the stated range of values may be needed for device performance (e.g. to meet transistor switching speed requirements), especially for advanced technology nodes.

Second, as seen in FIG. **27D**, the semiconductor layers **220A** that form the channels (e.g. nanochannels) of the GAA device **100** have different widths along the X-direction, depending on the position of the semiconductor layer **220A** in the Z-direction. For example, the semiconductor layer **220A** in closest proximity to the substrate **200** in the Z-direction has a width W2 along the X-direction, while the semiconductor layer **220A** that is farthest from the substrate **200** in the Z-direction has a width W1 along the X-direction, where the width W2 is greater than the width W1. In some embodiments, a difference between the width W2 and the width W1 is in a range from about 0.5 nm to about 5 nm.

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Each of the widths W1 and W2 may be in a range from about 6 nm to about 50 nm. Widths within the stated range of values may be needed for device performance (e.g. to meet transistor switching speed requirements), especially for advanced technology nodes.

Third, the semiconductor layers 220A that form the channels (e.g. nanochannels) of the GAA device 100 may have a respective thickness T measured in the Z-direction. For example, in the embodiment depicted in FIG. 27D, the semiconductor layer 220A in closest proximity to the substrate 200 in the Z-direction has a thickness T2 along the Z-direction, while the semiconductor layer 220A that is farthest from the substrate 200 in the Z-direction has a thickness T1 along the Z-direction. The thicknesses T1 and T2 may be substantially equal, with each of the thicknesses T1 and T2 being in a range from about 3 nm to about 10 nm. A thickness within the stated range of values may be needed for device performance (e.g. to meet transistor switching speed requirements), especially for advanced technology nodes.

Fourth, as seen in FIG. 27B, the inner spacers 250 wrap around the side portions 220A-side of the semiconductor layers 220A that form the channels (e.g. nanochannels) of the GAA device 100 in 360 degrees. Furthermore, the gate structure 272 (including the gate dielectric layers 228 and the metal layers 232) wrap around the center portions 220A-center of the semiconductor layers 220A that form the channels of the GAA device 100.

Fifth, as seen in FIG. 27C, the non-active regions of the GAA device 100 are devoid of the inner spacers 250. Instead, the top spacers 240 extend to the isolation features 203, and the sidewalls of the gate structure 272 (including the gate dielectric layers 228 and the metal layers 232) physically contact the top spacers 240.

The above process flow describes one embodiment of the present invention. In this embodiment, the dielectric based gates 234 are formed after the formation of gate structures 270, 272. However, other embodiments are also contemplated without departing from the scope of the present disclosure. For example, rather than forming the dielectric based gates 234 by removing the gate structures 270 and subsequently filling the gate trenches, the dielectric based gates 234 may alternatively be formed prior to the formation of the gate structures 270, 272. Additional details for this alternative embodiment may be found in related patents, such as U.S. Pat. No. 9,613,953, entitled "Semiconductor device, semiconductor device layout, and method of manufacturing semiconductor device" by Jhon Jhy Liaw, U.S. Pat. No. 9,805,985, entitled "Method of manufacturing semiconductor device and semiconductor device" by Jhon Jhy Liaw, and U.S. Pat. No. 9,793,273, entitled "Fin-based semiconductor device including a metal gate diffusion break structure with a conformal dielectric layer" by Jhon Jhy Liaw. These patents are herein incorporated in their entities.

In one such implementation, after the top spacers 240 are formed (e.g. as illustrated in FIGS. 5A-5D), an ILD 304 may be formed over the GAA device 100, as seen in FIGS. 28A-28D. A mask layer 384 may be formed over the ILD 304 to cover the entire area except the region in which the dielectric based gates are to be formed. Subsequently, as seen in FIGS. 29A-29D, an etching process may be used to remove the exposed portions of the ILD 304, as well as the dummy gate structures 210 beneath the exposed portions of the ILD 304. The etching process may also remove a portion of the doped regions 205 under the dummy gate structures 210. This etching process forms dielectric based gate trenches 357, which are similar to those trenches 159

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illustrated in FIGS. 23A-23D. The mask layer 384 may then be removed (see FIGS. 29A-29D). Once the dielectric based gate trenches 357 are formed, a dielectric material, similar to the materials described above for dielectric based gates 234, are used to fill in the trenches 357 to form the dielectric based gates 334 (see FIGS. 30A-30D). The method then proceeds to conduct a CMP and to etch a portion of the stack to form source/drain trenches 351, similar to those trenches 151 illustrated in the FIGS. 6A-6D (see FIGS. 31A-31D). Subsequent processes, such as those depicted in FIGS. 32A-35A, 32B-35B, 32C-35C, and 32D-35D, may proceed in ways similar to those illustrated in FIGS. 7A-27A, 7B-27B, 7C-27C, and 7D-27D. The final structure (see FIGS. 35A-35D) may be similar to that of FIGS. 27A-27D.

As mentioned above, in some embodiments, the gate structure 272 may have a tapered profile. In one such implementation, after the epitaxial source/drain features 208 are formed (e.g. as illustrated in FIGS. 32A-32D), the dummy gate structure 210 and the semiconductor layers 220B are selectively removed through suitable lithography and etching processes (see FIGS. 36A-36D). For example, the etch chemistry of the etch process used to remove the dummy gate structure 210 and thereby form a top portion of the trench 400 (e.g. in FIG. 36B) may include hydrogen bromide (HBr) combined with chlorine (Cl₂), tetrafluoromethane (CF₄), oxygen, or a combination thereof. The etch process then proceeds to selectively remove the semiconductor layers 220B to thereby extend the trench 400 to expose a portion of the top surface of the substrate 200 by using an etch chemistry including hydrogen bromide (HBr) combined with chlorine (Cl₂), oxygen, or a combination thereof. Use of such a combination of etch chemistries results in the tapered trench 400, where an acute angle β subtended by a sidewall of the trench 400 and the top surface 200a of the substrate 200 is in a range from about 80 degrees to about 88 degrees. Subsequent processes, such as those shown in FIGS. 37A-38A, 37B-38B, 37C-38C, and 37D-38D, may proceed in ways similar to those illustrated in FIGS. 34A-35A, 34B-35B, 34C-35C, and 34D-35D.

Referring to FIG. 38B, several structural features may be observed. Due to the tapered sidewalls of the trench 400, the resultant gate structure 272 (including gate dielectric layer 228 and metal layer 232) also has tapered sidewalls. In some embodiments, the gate lengths of the gate structure 272 along the Y-direction is different, depending on the position of the gate structure 272 in the Z-direction. For example, the gate length of the portion of the gate structure 272 in closest proximity to the substrate 200 in the Z-direction has a length G2 along the Y-direction, while the gate length of the portion of the gate structure 272 that is farthest from the substrate 200 in the Z-direction has a length G1 along the Y-direction, where the length G2 is greater than the length G1. In some embodiments, the length G2 is larger than the length G1 by at least 0.5 nm (e.g. by at least 1 nm). In general, the gate structure 272 seen in FIG. 38B controls the channels in the semiconductor layers 220A as well as a parasitic planar channel in the substrate 200. In a typical GAA device, the parasitic planar channel in the substrate 200 can be a limiting factor in the performance of the GAA device. However, with the tapered gate structure 272 seen in FIG. 38B, the longer gate length G2 of the portion of the gate structure 272 in closest proximity to the substrate 200 reduces the off-state current of the parasitic planar channel in the substrate 200 and junction leakage, thereby improving the performance of the GAA device 100.

In some implementations, the inner spacers 250 may have air gaps formed therein. FIGS. 39A to 39G show an embodi-

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ment method of forming airgaps in the inner spacers **250** in an effort to decrease the capacitance between the sidewall of the source/drain feature **208** and the sidewall of the gate structure **272**. FIG. **39A** is similar to the structure shown in FIG. **31B** and may be formed using the processes described above in reference to FIG. **31B**. Subsequently, as described above in reference to FIGS. **7A-7D**, portions of the semiconductor layers **220B** are removed through the exposed sidewall surfaces in the trenches **151** via a selective etching process to yield the structure illustrated in FIG. **39B**. A result of the selective etching process is the extension of the trenches **351** into areas beneath the semiconductor layers **220A** and top spacers **240**.

Referring to FIG. **39C**, a conformal dielectric layer **500** is formed on exposed portions of the semiconductor layers **220A** and **220B**. In some embodiments, the conformal dielectric layer **500** may be formed by an ALD process and may include a material similar to those discussed above in reference to the inner spacer **250**. Referring to FIG. **39D**, a deposition process (e.g. a low-pressure CVD, abbreviated as LPCVD) is performed to non-conformally deposit a further dielectric layer **502** in the source/drain trenches **351**. The further dielectric layer **502** may include a material similar to those discussed above in reference to the inner spacer **250**. However, due to the extension of the trenches **351** into areas beneath the semiconductor layers **220A** and top spacers **240** and the presence of the conformal dielectric layer **500** therein, the further dielectric layer **502** does not fully fill the areas beneath the semiconductor layers **220A**, thereby forming airgaps **504**. Referring to FIG. **39E**, an etch back process is performed on the further dielectric layer **502** to expose end regions of the semiconductor layers **220A** and the top surface of the substrate **200**, while leaving behind a vestigial portion of the further dielectric layer **502** in the areas beneath the semiconductor layers **220A**. The etch back process may include a wet etch process or a dry etch process that selectively etches the further dielectric layer **502** without substantially perturbing or consuming the top spacers **240**. Referring to FIG. **39F**, the epitaxial source/drain features **208** are then formed using similar processes described above in reference to FIGS. **10A-10D**. Subsequent processes may proceed in ways similar to those illustrated in FIGS. **33B-35B**. The final structure (see FIG. **39G**) includes airgaps **504** in the inner spacers that can reduce the capacitance between the sidewall of the source/drain feature **208** and the sidewall of the active gate structure **272**.

In some implementations, the top spacers **240** may have air gaps formed therein. FIGS. **40A** to **40I** show an embodiment method of forming airgaps in the top spacers **240** to decrease the capacitance between the sidewall of the contact **280** and the sidewall of the gate structure **272**. FIG. **40A** is similar to the structure shown in FIG. **4B** and may be formed using the processes described above (e.g. LPCVD and/or ALD processes) in reference to FIG. **4B**. FIG. **40B** shows a first gate spacer **240-1** being conformally formed on exposed surfaces of the dummy gate structures **210** and the top-most semiconductor layer **220A**. The first gate spacer **240-1** may include similar materials and may be formed using similar methods as described above in reference to gate spacer **240**.

Referring to FIG. **40C**, a sacrificial layer **240-2** is formed (e.g. using LPCVD and/or ALD processes) conformally over the first gate spacer **240-1**. The sacrificial layer **240-2** may include polysilicon or a dielectric material that is different from the material of the first gate spacer **240-1**. Referring to FIG. **40D**, the sacrificial layer **240-2** is etched back such that vestigial portions thereof remain at sidewalls (e.g. vertical sidewalls) of the first gate spacer **240-1**. Suitable etch back

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process may include a wet etch process or a dry etch process that selectivity etches the sacrificial layer **240-2** without substantially perturbing or consuming the first gate spacer **240-1**.

Referring to FIG. **40E**, a second gate spacer **240-3** is conformally formed on exposed surfaces of the sacrificial layer **240-2** and the first gate spacer **240-1**. The second gate spacer **240-3** may include similar materials and may be formed using similar methods as described above in reference to first gate spacer **240-1**. In FIG. **40F**, the second gate spacer **240-3** is etched back such that vestigial portions thereof remain at sidewalls (e.g. vertical sidewalls) of the second gate spacer **240-3**. The etch back process also exposes top surfaces of the top-most semiconductor layer **220A**, as seen in FIG. **40F**. The etch back process used to remove portions of the second gate spacer **240-3** may include a wet etch process or a dry etch process. As seen in FIG. **40F**, as a result of the process flow shown in FIGS. **40A** to **40F**, the dummy gate structures **210** have the first gate spacer **240-1** disposed on a top surface and sidewalls thereof. The sacrificial layer **240-2** is disposed on sidewalls of the first gate spacer **240-1**, and the second gate spacer **240-3** is disposed on sidewalls of the sacrificial layer **240-2**.

Referring to FIG. **40G**, the sacrificial layer **240-2** is removed using a selective etch process that selectively removes the material of sacrificial layer **240-2** without substantially perturbing or consuming the material of the first gate spacer **240-1** and the material of the second gate spacer **240-3**. As discussed above, the sacrificial layer **240-2** may include polysilicon (e.g. a material similar to the dummy gate structures **210**). However, since the first gate spacer **240-1** is disposed on a top surface and sidewalls of the dummy gate structures **210**, the first gate spacer **240-1** functions as a protective layer that prevents removal of the dummy gate structures **210** during the process of FIG. **40G**. The result of FIG. **40G** is the formation of a space **402** between adjacent ones of the first gate spacer **240-1** and the second gate spacer **240-3**.

Referring to FIG. **40H**, deposition and etch back processes are performed to form dielectric material **240-4** in top regions of the spaces **402**. With regards to the deposition process, since the space **402** is narrow (e.g. less than or equal to about 1 nm), the dielectric material **240-4** does not completely fill the spaces **402**, but instead fills up top regions thereof. Bottom regions of the spaces **402** (e.g. spaces proximate to the substrate **200**) remain unfilled. Consequently, the deposition process of FIG. **40H** produces air gaps **404**, with remaining portions of the first gate spacer **240-1**, the second gate spacer **240-3**, and the dielectric material **240-4** collectively forming the top spacers **240** having air gaps **404** therein. Subsequent processes may proceed in ways similar to those illustrated in FIGS. **6A-38A**, **6B-38B**, **6C-38C**, and **6D-38D**. The final structure (see FIG. **40I**) includes airgaps **404** in the top spacers that can reduce the capacitance between the sidewall of the contact **280** and the sidewall of the gate structure **272**.

Though not intended to be limiting, embodiments of the present disclosure offer benefits for semiconductor processing and semiconductor devices. For example, the disclosed method allows larger process margins for forming gate dielectric layers and metal layers within the limited spacing between semiconductor channel layers of a GAA device than other technologies, thereby eliminating or reducing voids and/or other defects in those layers. For a specific example, tapered trenches are formed to induce a bottom-up epitaxial growth process that conformally fills the tapered trenches. Additionally, the present method allows the epi-

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taxial source/drain features to be grown on side surfaces that include larger area of semiconductor materials, rather than dielectric materials. This improves the qualities of the epitaxial source/drain features, and eventually improves the performance and reliability of the GAA device. Furthermore, this present method also provides versatility allowing the designers to selectively optimize the capacitances of different regions of the GAA device according to design needs. As such, the present disclosure provides methods that improve the performance, functionality, and/or reliability of GAA devices. Stated differently, The GAA devices and methods of manufacture that are proposed in the present disclosure exhibit desirable properties, examples being: (1) a bottom-up epitaxial growth process that forms source/drain regions that are free from voids; (2) a large formation margin/window for gate dielectric and electrode in narrow channel-channel spaces; and (3) decreased capacitance between a source/drain region and an adjacent active gate structure.

An embodiment integrated circuit (IC) device includes: a semiconductor substrate having a top surface; a first source/drain feature and a second source/drain feature disposed on the semiconductor substrate; and a plurality of semiconductor layers including a first semiconductor layer and a second semiconductor layer. The first semiconductor layer extends longitudinally in a first direction and connects the first source/drain feature and the second source/drain feature, while the second semiconductor layer extends longitudinally in the first direction and connects the first source/drain feature and the second source/drain feature. The first semiconductor layer is stacked over and spaced apart the second semiconductor layer in a second direction perpendicular to the first direction, the second direction being normal to the top surface of the semiconductor substrate. The IC device further includes a gate structure engaging center portions of the first semiconductor layer and the second semiconductor layer, wherein a length of the first semiconductor layer along the first direction is less than a length of the second semiconductor layer along the first direction.

An embodiment integrated circuit (IC) device includes: a semiconductor substrate having a top surface; a first source/drain feature and a second source/drain feature disposed on the semiconductor substrate; and a plurality of semiconductor layers extending from the first source/drain feature to the second source/drain feature along a first direction. The plurality of semiconductor layers is stacked over each other along a second direction normal to the top surface and perpendicular to the first direction, wherein each of the semiconductor layers has a center portion and two side portions laterally adjacent to the center portion. The IC device also includes a gate electrode engaging the center portion of each of the semiconductor layers; a first spacer over the two side portions of a topmost semiconductor layer of the semiconductor layers; and a second spacer between vertically adjacent side portions of the semiconductor layers along the second direction. The first spacer includes a first dielectric material having a first dielectric constant, the second spacer includes a second dielectric material having a second dielectric constant different from the first dielectric constant, and a length of the topmost semiconductor layer, as measured along the first direction, is smaller than a length of each of the semiconductor layers vertically below the topmost semiconductor layer.

An embodiment method includes: forming a stack including first semiconductor layers and second semiconductor layers over a semiconductor substrate, wherein the first semiconductor layers and the second semiconductor layers

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have different material compositions and alternate with one another within the stack; forming a dummy gate structure over the stack, wherein the dummy gate structure wraps around top and sidewall surfaces of the stack; forming first spacers on sidewalls of the dummy gate structure, the first spacer being disposed on the top of the stack; forming a first tapered trench and a second tapered trench in the stack to expose a top surface of the semiconductor substrate; forming a first source/drain feature in the first tapered trench and a second source/drain feature in the second tapered trench; removing the dummy gate structure from the top and the sidewall surfaces of the stack; and removing the second semiconductor layers such that the first semiconductor layers remain and form semiconductor sheets connecting the first source/drain feature and the second source/drain feature to each other, wherein a length of a topmost first semiconductor layer is less than a length of a bottommost first semiconductor layer, as measured in a direction parallel to the top surface of the semiconductor substrate.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. An integrated circuit device, comprising:

- a semiconductor substrate having a top surface;
- a first source/drain feature and a second source/drain feature disposed on the semiconductor substrate;
- a plurality of semiconductor layers extending from the first source/drain feature to the second source/drain feature along a first direction, the plurality of semiconductor layers stacked over each other along a second direction normal to the top surface and perpendicular to the first direction; and
- a gate structure wrapping around each of the semiconductor layers, wherein the gate structure includes a tapered profile;

wherein

- the plurality of semiconductor layers includes a first semiconductor layer and a second semiconductor layer, the first semiconductor layer extending longitudinally in the first direction and connecting the first source/drain feature and the second source/drain feature;
- the second semiconductor layer extending longitudinally in the first direction and connecting the first source/drain feature and the second source/drain feature;
- the first semiconductor layer is stacked over and spaced apart the second semiconductor layer;
- the gate structure engages the first semiconductor layer and the second semiconductor layer between the first source/drain feature and the second source/drain feature; and
- a width of the first semiconductor layer is different from a width of the second semiconductor layer, each of the width of the first semiconductor layer and the width of the second semiconductor layer being measured in a third direction perpendicular to the first direction and the second direction.

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2. The IC device of claim 1, wherein the width of the first semiconductor layer is less than the width of the second semiconductor layer.

3. The IC device of claim 1, wherein a first portion of the gate structure engages the first semiconductor layer and a second portion of the gate structure engages the second semiconductor layer, wherein a length of the first portion of the gate structure along the first direction is less than a length of the second portion of the gate structure along the first direction.

4. The IC device of claim 3, wherein the length of the second portion of the gate structure is larger than the length of the first portion of the gate structure by at least 0.5 nanometers.

5. The IC device of claim 1, wherein
a length of the first semiconductor layer along the first direction is less than a length of the second semiconductor layer along the first direction; and
the width of the first semiconductor layer is less than the width of the second semiconductor layer.

6. The IC device of claim 5, wherein
a difference between the length of the second semiconductor layer and the length of the first semiconductor layer is at least 1 nanometer; and
a difference between the width of the first semiconductor layer and the width of the second semiconductor layer is in a range from about 0.5 nanometers to about 5 nanometers.

7. The IC device of claim 1, further comprising:

a first spacer disposed over side portions of the first semiconductor layer, the first semiconductor layer being a topmost layer of the plurality of semiconductor layers; and
a second spacer disposed between the side portions of the first semiconductor layer and side portions of the second semiconductor layer, wherein the first spacer and the second spacer have different material compositions.

8. The IC device of claim 7, wherein the first spacer spans a first dimension along the first direction and the second spacer spans a second dimension along the first direction, wherein the first dimension of the first spacer is less than the second dimension of the second spacer.

9. The IC device of claim 1, wherein each of the first and second source/drain features includes a tapered sectional profile.

10. The IC device of claim 1, wherein an acute angle subtended by a sidewall of the first source/drain feature and the top surface of the semiconductor substrate is in a range from about 80 degrees to about 88 degrees.

11. An integrated circuit device, comprising:

a semiconductor substrate having a top surface;
a first source/drain feature and a second source/drain feature disposed on the semiconductor substrate;
a plurality of semiconductor layers extending from the first source/drain feature to the second source/drain feature along a first direction, the plurality of semiconductor layers stacked over each other along a second direction normal to the top surface and perpendicular to the first direction; and
a gate structure wrapping around each of the semiconductor layers, wherein a width of a topmost semiconductor layer is different from a width of a bottommost semiconductor layer, as measured in a third direction perpendicular to the first and second directions.

12. The IC device of claim 11, wherein a length of the topmost semiconductor layer of the semiconductor layers is

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less than a length of the bottommost semiconductor layer of the semiconductor layers, as measured in the first direction.

13. The IC device of claim 11, wherein a first portion of the gate structure engages the topmost semiconductor layer and a second portion of the gate structure engages the bottommost semiconductor layer, wherein a length of the first portion of the gate structure along the first direction is less than a length of the second portion of the gate structure along the first direction.

14. The IC device of claim 11, wherein each of the first and second source/drain features includes a tapered profile.

15. The IC device of claim 14, wherein an acute angle subtended by a sidewall of the first source/drain feature and the top surface of the semiconductor substrate is in a range from about 80 degrees to about 88 degrees.

16. The IC device of claim 11, further comprising:

a first spacer over two side portions of the topmost semiconductor layer of the semiconductor layers; and
a second spacer between vertically adjacent side portions of the semiconductor layers along the second direction, wherein the first spacer includes a first dielectric material having a first dielectric constant, wherein the second spacer includes a second dielectric material having a second dielectric constant different from the first dielectric constant.

17. The IC device of claim 16, wherein the first spacer includes the first dielectric material selected from the group consisting of SiON, SiOC, SiOCN, and combinations thereof, and wherein the second spacer includes the second dielectric material selected from the group consisting of SiO₂, Si₃N₄, carbon doped oxide, nitrogen doped oxide, porous oxide, and combinations thereof.

18. The IC device of claim 16, further comprising a gate end dielectric layer contacting opposing ends of each of the plurality of semiconductor layers, wherein the gate end dielectric layer includes a third dielectric material having a third dielectric constant, the third dielectric constant being larger than the first dielectric constant and the second dielectric constant.

19. An integrated circuit device, comprising:

a semiconductor substrate having a top surface;
a first source/drain feature and a second source/drain feature disposed on the semiconductor substrate, wherein each of the first and second source/drain features includes a tapered profile;
a plurality of semiconductor layers extending from the first source/drain feature to the second source/drain feature along a first direction, the plurality of semiconductor layers stacked over each other along a second direction normal to the top surface and perpendicular to the first direction;
a gate structure wrapping around each of the semiconductor layers;
the gate structure comprising a top portion over the plurality of semiconductor layers; and
first and second top spacers disposed on opposing sidewalls of the top portion of the gate structure;
wherein a width of a topmost semiconductor layer is different from a width of a bottommost semiconductor layer, as measured in a third direction perpendicular to the first and second directions.

20. The IC device of claim 19, wherein

the semiconductor layers include a bottom semiconductor layer, a middle semiconductor layer stacked over the bottom semiconductor layer, and a top semiconductor layer stacked over the middle semiconductor layer;

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the bottom semiconductor layer includes a first width
along a third direction perpendicular to the first direc-
tion and the second direction;

the middle semiconductor layer includes a second width
along the third direction;

the top semiconductor layer includes a third width along
the third direction; and

the second width is less than the first width and is greater
than the third width.

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